Panasonic

Aquarea

Heat Pump Manager

PAW – HPM 1

PAW – HPM 2

Manual:

Part 2 - Library Programs

Revision A, February 2013

Software revision: 1.0-0-00

NOTICE:

Before removing the controller from the terminal block, be sure to switch off the flow voltage!

Table of Contents

Part I Module library

Table of contents

Chapter 1 The Timer modules

The HPM controller is equipped with a maximum of 4 timer modules (timers). One timer module is assigned to every heating circuit and domestic hot water circuit. Additionally, there is one free timer which can be used for multiple purposes.

All timer modules can be assigned to an output terminal. This switches On/Off in accordance with the specified operating times (OTx = On, NO = Off). In manual operation, the timer status and assigned output terminal can be selected manually.

Each timer module consists of a week program and a year program.

Week program

4 operating times are available for every day of the week. The operating times can be entered in a random manner. The parameter Block grouping can be used to group week days with the same operating times and to copy them to the remaining days.

Year program

The year program enables the system operator to define specific periods of operation or non-operation. During these specially defined times, the settings in the week program do not apply. The year program provides 15 non-operating times (e.g. public holidays or company holidays in a company building) and 10 operating times (e.g. special shifts at a specific weekend). Such non-operating times are referred to as Special Non-Operating Times (SNOT). Every operating time, referred to as Special Operating Time (SOT), has a day program with a maximum of 4 operating times, see chapter 2.1.2.n. There are 10 special operating times available.

1.1 Current values (2.1.1.n)

The calculated current values for the season and the day of the week are displayed in the menu “Current values”.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.1.1	season:	current season summer / winter					
2.1.1.3	day:	calculated from current date					

1.2 Timer selection / setpoints (2.1.2.n)

After selecting the control circuit, e.g. domestic hot water circuit or heating circuit (1 or 2) and/or the free timer, the relevant week and year programs are displayed. In the parameter numbers of the subordinate menu , “n” refers to the number of the selected control circuit.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.2.1	DHW	Domestic hot water					
2.1.2.2	heatCirc1	Heating Circuit 1					
2.1.2.3	heatCirc2	Heating Circuit 2					
2.1.2.4	fr.Channel	Free channel					

Week program (2.1.2.n.1.n)

In the week program of the selected controller circuit, the begin and end time for up to four operating times can be entered for every day of the week. Non-operation applies between individual operating times. The Block function can be used to copy the setting for Monday to other days of the week.

Example: Block = 2 (Mo-Fr) means that the settings for Monday are also valid Tuesday to Friday. Consequently, if on Monday operating time 1 starts at 8:13 AM, it will also start at 8:13 AM for the remaining days. In this example, operating times for Saturday and Sunday will have to be entered separately.

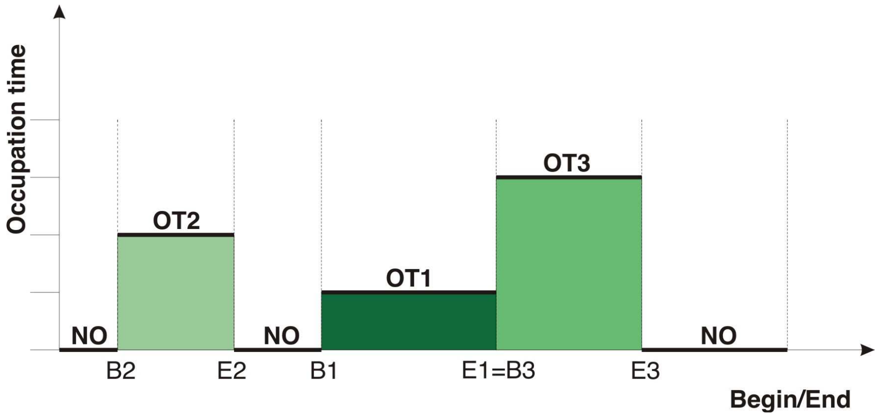


Fig. 1: Operating time state in the week program

The above figure (Fig. 1) shows a possible operating time (OT) state. As is visible, the end of an operating time is followed either by a non-operating time (NO) or by the beginning of a new operating time (see E1=B3). It is not possible for operating times to overlap each other.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.2.n.1.1	group	0:none 1:mo-th 2:mo-fr 3:mo-su		0	3	0	
2.1.2.n.1.2	mo no.OT	Monday numOf OT		0	4	1	
2.1.2.n.1.3	mo beg OT1	Monday begin OT1				06:00	
2.1.2.n.1.4	mo end OT1	Monday end OT1				22:00	
2.1.2.n.1.5	mo beg OT2	Monday begin OT2				--:--	
2.1.2.n.1.6	mo end OT2	Monday end OT2				--:--	
2.1.2.n.1.7	mo beg OT3	Monday begin OT3				--:--	
2.1.2.n.1.8	mo end OT3	Monday end OT3				--:--	
2.1.2.n.1.9	mo beg OT4	Monday begin OT4				--:--	
2.1.2.n.1.10	mo end OT4	Monday end OT4				--:--	
2.1.2.n.1.11	tu no.OT	Tuesday numOf OT		0	4	1	
2.1.2.n.1.12	tu beg OT1	Tuesday begin OT1				06:00	
2.1.2.n.1.13	tu end OT1	Tuesday end OT1				22:00	
2.1.2.n.1.14	tu beg OT2	Tuesday begin OT2				--:--	
2.1.2.n.1.15	tu end OT2	Tuesday end OT2				--:--	
2.1.2.n.1.16	tu beg OT3	Tuesday begin OT3				--:--	
2.1.2.n.1.17	tu end OT3	Tuesday end OT3				--:--	
2.1.2.n.1.18	tu beg OT4	Tuesday begin OT4				--:--	
2.1.2.n.1.19	tu end OT4	Tuesday end OT4				--:--	
…	…	…					
2.1.2.n.1.47	sa no.OT	Saturday numOf OT		0	4	1	
2.1.2.n.1.48	sa beg OT1	Saturday begin OT1				06:00	
2.1.2.n.1.49	sa end OT1	Saturday end OT1				22:00	
2.1.2.n.1.50	sa beg OT2	Saturday begin OT2				--:--	

2.1.2.n.1.51	sa end OT2	Saturday end OT2				--:--	
2.1.2.n.1.52	sa beg OT3	Saturday begin OT3				--:--	
2.1.2.n.1.53	sa end OT3	Saturday end OT3				--:--	
2.1.2.n.1.54	sa beg OT4	Saturday begin OT4				--:--	
2.1.2.n.1.55	sa end OT4	Saturday end OT4				--:--	
2.1.2.n.1.56	su no.OT	Sunday numOf OT		0	4	1	
2.1.2.n.1.57	su beg OT1	Sunday begin OT1				06:00	
2.1.2.n.1.58	su end OT1	Sunday end OT1				22:00	
2.1.2.n.1.59	su beg OT2	Sunday begin OT2				--:--	
2.1.2.n.1.60	su end OT2	Sunday end OT2				--:--	
2.1.2.n.1.61	su beg OT3	Sunday begin OT3				--:--	
2.1.2.n.1.62	su end OT3	Sunday end OT3				--:--	
2.1.2.n.1.63	su beg OT4	Sunday begin OT4				--:--	
2.1.2.n.1.64	su end OT4	Sunday end OT4				--:--	

Explanation:

group:		Block grouping over several week days
	= 0	No block grouping
	= 1	Monday to Thursday same operating times
	= 2	Monday to Friday same operating times
	= 3	Monday to Sunday same operating times
mo no. OT:		Number of operating times on Monday
	= 0	No operating time (→ non-operation)
	= 1	One operating time, begin and end OT1 have to be entered (→ Basis parameter setting)
	= 2	Two operating times, begin and end OT2 have to be entered
	= 3	Three operating times, begin and end OT3 have to be entered
	= 4	Four operating times, begin and end OT4 have to be entered
mo beg OT1:		Beginning of operating time 1 on Monday, enter time from 00.00 ... 23:59 hrs
mo end OT1:		End of operating time 1 on Monday, enter time from 00.01 ... 24:00 hours

Example: Continuous operation, Monday to Sunday - around the clock (e.g. for the domestic hot water circuit in an apartment building)

Block = 3, mo no. OT = 1, mo beg OT1 = 00:00, mo end OT1 = 24:00

Special non-occupation time (vacation) (2.1.2.n.2.n)

This menu permits defining special non-occupation times which have their own setpoint (see setpoints in chapter 6.2 (heating circuit) and 5.2 (domestic hot water circuit)). A special non-occupation time can be used to define a time when the building is not occupied; such as a vacation.

A special non-occupation time consists of:

- **begSNOx**: The date on which the non-occupation time begins.
- **endSNOx**: The date on which the non-occupation time ends.

Up to 15 special non-occupation times can be defined.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.2.n.2.1	no.SNO	number SNO		0	15	0	
2.1.2.n.2.2	begSNO1	beginning SNO 1				--.--.--	

2.1.2.n.2.3	endSNO1	end SNO 1				--.--.--	
2.1.2.n.2.4	begSNO2	beginning SNO 2				--.--.--	
2.1.2.n.2.5	endSNO2	end SNO 2				--.--.--	
…	…	…					
2.1.2.n.2.28	begSNO14	beginning SNO14				--.--.--	
2.1.2.n.2.29	endSNO14	end SNO14				--.--.--	
2.1.2.n.2.30	begSNO15	beginning SNO15				--.--.--	
2.1.2.n.2.31	endSNO15	end SNO15				--.--.--	

Explanation:

No.SNO: Number of special non-operating times

BegSNO1: Begin of special non-operating time 1

= 01.01.08 Begins on 1st January 2008

= 01.01.-- Begins on 1st January of every year

Special occupation time (2.1.2.n.3.n)

This menu permits defining special occupation times which use the same setpoints that have been defined for the week program (see chapter 6.2 (heating circuit) and 5.2 (domestic hot water circuit)). Special occupation time may be used either to define different occupation times during the normal week program, or to define a period of occupation during what would otherwise be a non-occupied period (e.g. a person staying at a house for the duration of one week while its regular owners are away on vacation).

A special occupation time consists of:

- **begSOTx**: The date on which the special occupation time begins.
- **endSOTx**: The date on which the special occupation time ends.
- **SOT1xx-SOT4xx**: Begin and end time of an occupation time, valid for each day during the special occupation time.

Up to 10 special non-occupation times can be defined.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.2.n.3.1	no.SOT	number SOT		0	10	0	
2.1.2.n.3.2	begSOT1	beginning SOT1				--.--.--	
2.1.2.n.3.3	endSOT1	end SOT1				--.--.--	
2.1.2.n.3.4	SOT1no.OT	SOT1: number OT		0	4	0	
2.1.2.n.3.5	SOT1beg1	SOT1: beginning OT1				--:--	
2.1.2.n.3.6	SOT1End1	SOT1: Ende OT1				--:--	
2.1.2.n.3.7	SOT1beg2	SOT1: beginning OT2				--:--	
2.1.2.n.3.8	SOT1End2	SOT1: Ende OT2				--:--	
2.1.2.n.3.9	SOT1beg3	SOT1: beginning OT3				--:--	
2.1.2.n.3.10	SOT1End3	SOT1: Ende OT3				--:--	
2.1.2.n.3.11	SOT1beg4	SOT1: beginning OT4				--:--	
2.1.2.n.3.12	SOT1End4	SOT1: Ende OT4				--:--	
…	…	…					
2.1.2.n.3.101	begSOT10	beginning SOT10				--.--.--	
2.1.2.n.3.102	endSOT10	end SOT10				--.--.--	
2.1.2.n.3.103	SOT10no.OT	SOT10: number OT		0	4	0	
2.1.2.n.3.104	SOT10beg1	SOT10: beginning OT1				--:--	
2.1.2.n.3.105	SOT10End1	SOT10: Ende OT1				--:--	
2.1.2.n.3.106	SOT10beg2	SOT10: beginning OT2				--:--	

2.1.2.n.3.107	SOT10End2	SOT10: Ende OT2				--:--	
2.1.2.n.3.108	SOT10beg3	SOT10: beginning OT3				--:--	
2.1.2.n.3.109	SOT10End3	SOT10: Ende OT3				--:--	
2.1.2.n.3.110	SOT10beg4	SOT10: beginning OT4				--:--	
2.1.2.n.3.111	SOT10End4	SOT10: Ende OT4				--:--	

Explanation:

no.SOT:		Number of special operating times
BegSOT1:		Beginning of special operating time 1
	= 01.01.08	Begins on 1st January 2008
	= 01.01.--	Begins on 1st January of every year
SOT1no.OT:		Number of operating times on each day of special operating time1
	= 0	No operating time on each day of special operating time 1 (→ non-operation)
	= 1	One operating time on each day of special operating time
SOT1beg1:		Begin of operating time 1 on each day of special operating time 1
SOT1end1:		End of operating time 1 on each day of special operating time 1

Priority (2.1.2.n.4.1)

The menu “Priority” can be used to determine whether the special operating time (SOT) or the special non-operating time (SNOT) should have priority in case several times of the year program should happen to overlap.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.2.n.4.1	typprior	0:SOT has priority 1:SNO has priority		0	1	0	

Explanation:

Typprior:	= 0	Special operating time has priority.
	= 1	Special non-operating time has priority.

1.3 Status (2.1.4.n)

After selecting the control circuit, e.g. domestic hot water circuit or heating circuit (1 or 2) and/or the free timer 1, the relevant status menu is displayed. In the subordinate menu, the item “n” in the parameter number refers to the number of the selected control circuit.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.4.1	DHW	Domestic hot water					
2.1.4.2	heatCirc1	Heating Circuit 1					
2.1.4.3	heatCirc2	Heating Circuit 2					
2.1.4.4	fr.Channel	Free channel					

Status of selected control circuit (2.1.4.n.n)

The timer status of the control circuits displays the current operating status of the timer program, the next status and the time difference between the two.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.4.n.1	timer status						
2.1.4.n.2	curStat	cur status					
2.1.4.n.3	nxtStat	next status					
2.1.4.n.4	timeDiff	time to end of cur. Status	min				

Explanation:

Timer status:		Displays operating status.
	= Time program	Normal operating mode, execution of time programs entered.
	= Manual [status]	Status of manual control example: Manual [OT1].
curSta:		Displays the current status.
	= NO	Week program non-operation.
	= OT1...OT4	Week program operating time 1 to 4.
	= SNO	Special non-operating time non-operation.
	= SOT1...4	Special operating time.
	= SNOT	Special non-operating time.
nxtStat:		Displays the next status.
	= NO	Week program non-operation.
	= OT1...OT4	Week program operating time 1 to 4.
	= SNO	Special non-operating time non-operation.
	= SOT1...4	Special operating time.
	= SNOT	Special non-operating time.

1.4 Manual control (2.1.5.n)

Manual control is used when commissioning the controller. Manual control of the timer allows actual testing of the required setpoint switch in a control circuit and to actually trigger the assigned outputs. After a successful test, the manual control should be switched back to *automatic* as otherwisethe week and year program will not be effective.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.5.1	DHW	manVal timerChan		0	11	11	
2.1.5.2	heatCirc1	manVal timerChan		0	11	11	
2.1.5.3	heatCirc2	manVal timerChan		0	11	11	
2.1.5.4	fr.chan.1	manVal timerChan		0	11	11	

Explanation:

Value for manual control of the selected control channel:	= 0	Non-operation (NO), output = OFF.
	= 1 ... 4	Operating time 1 ... 4 (OT1 ... OT4), output = ON.
	= 5	Special operating time non-operation SNO, output = OFF.
	= 6 ...9	Special operating time 1 ... 4 (SOT1 ... SOT4), output = ON.

= 10	Special non-operating time, output = OFF.
= 11	Automatic.

1.5 Service (2.1.6.n)

In the Service menu, the time, date and **System clock** mode can be set. Under Terminal assignment, the timers can be assigned output terminals.

Time (2.1.6.1.1)

The menu “time” permits viewing and adjusting the current time of the system clock.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.6.1.1	curTime	current time		00:00	23:59		

Date (2.1.6.2.1)

The menu “date” permits viewing and adjusting the current date of the system clock.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.6.2.1	curDate	current date		01.01.90	31.12.89		

Mode (2.1.6.3.n)

The menu “mode” permits entering special settings for the system clock.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.6.3.1	typSuWiSwitch	su/wi-switch at 0:date/time 1:by law		0	1	1	
2.1.6.3.2	dateSu	date summer		01.01.90	31.12.89	--.--.--	
2.1.6.3.4	dateWi	date winter		01.01.90	31.12.89	--.--.--	
2.1.6.3.7	operating mode	0:powerln. 1:crystal		0	1	1	
2.1.6.3.8	adjustment			-99	99		

Explanations:

typSuWiSwitch	= 0	1.5.1 Summer (daylight saving)/winter time switch according to date and time set.
	= 1	1.5.2 Automatic summer/winter time switch according to legal regulation:
		1.5.3 Winter/summer time switch on the last Sunday in March 02:00 am by +1h.
		1.5.4 Summer/winter time switch on the last Sunday in October 03:00 am by +1h.
dateSu:		The date when summer (daylight saving) time should begin. On this day, the clock will be put forward by 1h. at 2am.
dateWi:		The date when winter time should begin. On this day, the clock will be put forward by 1h. at 3am.
Operating mode	= 0	1.5.5 Operation synchronized by the frequency of the power flow.

= 1 1.5.6 Operation quartz synchronized.

Correction: Correction value for quartz clock in seconds/week.

Terminal assignment (2.1.6.4.n)

The menu “terminal assignment” permits assigning an output terminal or virtual terminal separately to each timer. This assignment is made by entering the terminal number.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.1.6.4.1	DHW	timer		0	255	0	
2.1.6.4.2	heatCirc1	timer		0	255	0	
2.1.6.4.3	heatCirc2	timer		0	255	0	
2.1.6.4.4	fr.chan.1	timer		0	255	0	

Assignment example:

1 Relay output terminal 1 assigned.

25 OC output terminal 25 assigned.

50 Virtual terminal 50 assigned.

Chapter 2 **The Heat pump module**

Depending on the loaded system diagram and the configuration of the parameters of the HPM you will have up to 3 heat pumps in the sup-menues 2.2.1, 2.2.2 and 2.2.3.

The control of the heat pump can be done via a contact or via a direct communication.

When using a **contact** a "heat demand"-signal is being issued to the heat pump. An actual temperature demand is not possible via the contact signal. In this case the calculation of the flow temperature is accomplished by the heat pump itself.

The HPM can **communicate directly** with the heat pump. To realize this, you have to connect the adapter cable, normally used for the operating unit, with the HPM. In the "direct communication" mode the HPM can distinguish between the demands to the heat pump for "heating", "cooling" and "domestic hot water production". All parameters of the operating unit of the heat pump are shown in the sub menues (current values, setpoints, additional functions, status, etc...) of the library program "heat pump" and can be adjusted there.

In connection with a second or third heat pump you will need a second or third HPM (without display) and one or two additional adapter cables. After loading the system diagrams in the slave controllers (SD.-No. 80000 > 2. heat pump with bus communication:Slave; SD.-No. 90000 > 3. heat pump with bus communication: Slave) and loading the master controller (SD.-No. 6xxxx > two heat pumps with bus communication Master; SD.-No. 7xxxx > three heat pumps with bus communication:Master) and realizing a LAN connection between the controllers, you can reach all parameters in each heat pump from the master controller in the su menues "heat pump" 1,2 and 3.

The HPM collects all temperature demands of the heating circuits 1 and 2 and adds the heat demand coming from the external demand (contact signal or VDV(0..10V)) to calculate the overall demand for "heating"

The external demands are only visible in the heat pump porgram, if there is only one heat pump in the system and no buffer tank is installed. A control of the heat pump to demand heat for domestic hot water production is only possible in "communication-mode". (Not in "contact-mode")

The following menues describe all functions of the heat pump program. (The x in the parameters stands for 1=heat pump 1; 2=heat pump 2; 3=heat pump 3)

2.1 Current values (2.2.x.1)

In the menu "current values" you will find the values of all inputs, f.i. all values read via direct communication from the heat pump and the values from the HPM (like external demand, operating mode switch, heat or electric consumption values etc. ...)

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.1.1	VDV	dem. VDV	°C				ext.demand.0-10V
2.2.x.1.2	ifc	dem. ifc	°C				ext.demand Interface
2.2.x.1.3	dem-cont.	dem. contact					ext.demand contact
2.2.x.1.4	oModLoc	oMod-local switch					Operating mode switch
2.2.x.1.5	mainten.	mainten.					Maintenance switch
2.2.x.1.6	aMntH	heat amount heating					
2.2.x.1.7	aMntE	energy electric					

2.2 Setpoint (2.2.x.2)

In the menue "setpoints" you wil find the calculated setpoint for "heating"t.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.2.1	setpoint	setpoint	°C				

2.3 Additional Functions (2.2.x.3)

Setpoint limitation (2.2.x.3.2)

The function “setpoint limitation” enables you to adjust the heat pump to the operating conditions specified by the manufacturer.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.3.2.1	maxSP	max. setpoint	°C	30	90	55	

Boost (2.2.x.3.3)

The additional function “boost” can be used to set boosts (e.g. HC1 10 %, DHW1 -100%) for the temperature demands of the internal heating and domestic hot water circuits and for the external demand (via VDV). This allows to compensate for transmission losses of the heating system.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.3.3.1	active	- look at op manual -		0	1	0	
2.2.x.3.3.2	boost HC1	boost HC1	%	-100	100	10	
2.2.x.3.3.3	boost HC2	boost HC2	%	-100	100	10	
2.2.x.3.3.4	boost DHW1	boost DHW1	%	-100	100	10	
2.2.x.3.3.5	boost VDV	boost VDV	%	-100	100	10	

Maintenance (2.2.x.3.4)

The function “maintenance” can be used by the service technician to start up the heat pump irrespective of a heat demand. For this purpose, the additional function “maintenance” has to be activated (**Active** = 1). In order to change the activation parameter the access code for level 4 has to be entered (see chapter “global – access code”). The maintenance function is started by pressing the “UP”-key.

The release time can be set in the menu "Heat Pump x - SF - maintenance (default value is 15 min)

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.3.3.1	active			0	1	0	

Signal (2.2.x.3.5)

The function “signal” gives you the possibility to configure the way a signal should be handled. When this function is activated, you can decide what should happen, if

a. an error occurred
b. a trouble indication occurred
c. a limit was exceeded
d. a timer has rundown (maintenance etc.)

TI-all	setting of the trouble indication outputs **troubleInd** .	indication as an alarm in the central **building management software**,	indication on the controller **display**
0			
1			x
2		x	
3		x	x
4	x		
5	x		x
6	x	x	
7	x	x	x

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.3.5.1	active			0	1	0	
2.2.x.3.5.2	TI-all	TI to BMS:2,3,6,7 TI-O:4-7 disp:1,3,5,7		0	7	1	

External demand (2.2.x.3.6)

The HPM offers the possibility to control the heat pump circuit in such a way, that a heat demand from an external heat consumer can be fulfilled. This heat consumer can send his demand in different ways:

- **Demand via contact**

A fixed temperature demand can be transmitted to the heat pump circuit via a potential-free contact which is connected to an input terminal of the controller. For this purpose, the input terminal dem-cont has to be assigned to a terminal (see chapter “SF – terminal assignment”). The activation is done in this menu at **dem-cont. = 1**. The required temperature demand can be specified in the parameter **dem-T-^**. If the contact is closed, the temperature demand is integrated in the setpoint calculation. The display of the status (On/Off) of the input terminal is shown in the menu “current values”.

- **Demand via VDV (variable demand via voltage)**

Variable temperature demands can be transmitted between HPM and other controllers via VDV (0 ... 10V signal). A VDV demand can also be received by “third-party controllers” with 0..10V signal. Up to 9 controllers can transmit their temperature demand to this controller. First, assign the input terminal for VDV (see chapter “SF – terminal assignment”). The activation is done in this menu at **VDV = 1**. The input signal 0 ... 10 V equals 0 ... 150 °C. The output signals of the demanding controllers have to send the signal accordingly in this format. The display of the setpoint temperature requested via VDV is shown in the menu “current values”.

- **Demand via interface**

A device with masterbus interface can send a setpoint temperature demand to the HPM via the bus (RS 485). The activation is done in this menu at **ifc = 1**. The display of the setpoint temperature requested via interface is shown in the menu “current values”. If the interface connection fails, a programable substitute value is active (see chapter “SF – external demand”).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.3.6.1	dem-cont.	dem. contact active		0	1	0	
2.2.x.3.6.2	dem-T-^	flow setpoint temp. contact	°C	2.0	160.0	50.0	
2.2.x.3.6.3	VDV	dem. VDV active		0	1	0	
2.2.x.3.6.4	ifc	dem. interface active		0	1	0	

Controller (2.2.x.3.7)

In the menu "controller" you have the possibility to give the library program "heat pump x" a project related name, to simplify the menu for the service personal.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.3.7.1	longDes	free adjustab. Prog.-longRef.					

2.4 Status

The menu “status” gives an overview of the current mode of operation of the heat pump circuit.

Every function affecting the operation of the heat pump is considered, when the main status **opStatus** or the auxiliary status **opStatCode** is generated. The **opStatCode** is a hexadecimal number. The meaning of the characters shown in the parameters **opStatCode** and **eStatCode** is explained in the following tables.

opStatCode:

The auxiliary operating status consists of six digits as various functions of the heat pump circuit can be effective at the same time. The display helps during analysis of the control. The example below, together with the translation table, explains how to decode the sequence of characters.

	1. digit	2. digit	3. digit	4. digit	5. digit	6. digit
1					Mainte- nance	setpoint limitation
2						OP-mode-switch
3						setpoint limitation OP-mode-switch
4						demand "heating"
5						demand "heating", setpoint limitation
6						demand "heating", OP-mode-switch
7						demand "heating", setpoint limitation, OP-mode-switch
8						cooling
9						cooling, setpoint limitation
A						Kühlung, OP-mode-switch
B						cooling, setpoint limitation, OP-mode-switch
C						cooling, demand "heating"
D						cooling, demand "heating", setpoint limitation
E						cooling, demand "heating", OP-mode-switch
F						cooling, demand "heating", setpoint limitation OP-mode-switch

Tab. 1: operating status code "heat pump"

Example:

Display: OPStatCode: = 05

Meaning: 6. digit = demand "heating", setpoint limitation is active

sStatCode:

The auxiliary trouble indication status consists of two digits as several trouble indications can occur at the same time. The example below, together with the translation table, explains how to decode the sequence of characters

	1. digit	2. digit
1		error Input

Tab. 2: trouble indication code "heat pump"

Example:

Display: sStatCode: = 1

Meaning: 2. digit = error input

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.4.1	opStatus						
2.2.x.4.2	trouble						
2.2.x.4.3	source	setpoint source and -compensation					
2.2.x.4.4	HP-rel	HP release					
2.2.x.4.5	troubInd	trouble indication					
2.2.x.4.6	opStatCode						
2.2.x.4.7	eStatCode						

Description:

opStatus: [ain status of the heat pump

= *Not active/Off* required inputs are not assigned or operating mode switch **oModRem** or **oModLoc** assigned and position Off

= *Frost protection* value at flow sensor **flow** is below frost limit

= *Maintenance* maintenance function started

= *Switch-off operation* system main switch **System** assigned and switched off (see chapter “terminal assignment”). Flow temperature demand = 2°C

= *Nominal operation* flow temperature demand >2°C

= *Manual operation* manual control of the output signals. Operating mode switch **oModRem** or

oModLoc assigned and in position “Manual”

source: setpoint source for heat pump circuit

= *dem. XXX* max. flow temperature demand by HC1, HC2, DHW, VDV or Ext (contact or ifc), strategy

= *no demand* flow temperature demand = 2°C, switch-off operation

= *xxxxxxxx **L*** setpoint compensation by setpoint limitation

2.5 Manual operation (2.2.x.5)

The manual control can be used to check the release of the heat pump during commissioning.

Note!

In case of improper handling, the operating mode “Manual operation” can cause damages to the system as the manual control overrides the limitation and the prevention of blocking and frost protection functions.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.5.1	HP-rel	0:OFF 1:ON 3:automatic		0	3	3	

2.6 Service Functions (2.2.x.6)

General values (2.2.x.6.1)

The service menu “general value” displays the operating hours of the heat pump (and/or the fan). During pump replacement or maintenance the counter can be adjusted to a required value. You will also find the number of starts of the heat pump.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.6.1.1	heatpump		h	0	999999	0	
2.2.x.6.1.2	numStartHP			0	999999	0	

Maintenance (2.2.x.6.4)

The maintenance function is starting the heat pump by pressing and holding the “UP” key (in the standard display). Pressing and holding the "UP" key again or when the timer has run down the time, set in this menu the heat pump will return into the normal operating mode.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.6.4.1	duration	duration meas.	min	1	60	15	
2.2.x.6.4.2	timer	curTimer	min				

Signal (2.2.x.6.5)

The last 10 errors detected are recorded and displayed in the service menu “signal”. The first parameter always shows the most recent error and the last parameter the oldest error. The parameter text displayed is the short text of the error detected. By pressing the “OK” key the info text appears, giving you more detailed description of the error. The date and time at which the error was detected is documented. A power failure or a controller cold start will delete the recorded errors!

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.6.5.1		TT.MM.JJ hh:mm					
2.2.x.6.5.2		TT.MM.JJ hh:mm					
2.2.x.6.5.3		TT.MM.JJ hh:mm					
2.2.x.6.5.4		TT.MM.JJ hh:mm					
2.2.x.6.5.5		TT.MM.JJ hh:mm					
2.2.x.6.5.6		TT.MM.JJ hh:mm					
2.2.x.6.5.7		TT.MM.JJ hh:mm					
2.2.x.6.5.8		TT.MM.JJ hh:mm					
2.2.x.6.5.9		TT.MM.JJ hh:mm					
2.2.x.6.5.10		TT.MM.JJ hh:mm					

Description:

parameter text:	TI-system	trouble indication in the system
	sensor defect	sensor fault was detected
Info-Text:	<date, time>	f.i..: at 18.02.06 at 13:57 an error was recorded

External demand (2.2.x.6.6)

In the service menu of the function “external demand” you can set the basic parameters for communication between the controller and external components that demand heat either by voltage signal (VDV) or via the interface. With **invertVDV** you can configure the conversion between the input voltage to the setpoint temperature: = *0*: 0V->0°C, 10V->100°C; = *1*: 0V->100°C, 10V-0°C. The parameter Interfac shows the setpoint temperature coming from an external controller via the interface.The target address for the external controllers is the parameter number of this parameter. If no valid value has been received after a waiting time **waitingT** the substitute value **^-subst** will become effective.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.6.6.1	invert VDV	0:0V-0°C,10V-100°C 1:0V-100°C,10V-0°C		0	1	0	
2.2.x.6.5.2	waitingT	idle time	Min	0.1	999.9	1.0	
2.2.x.6.5.3	interface	setpoint demand via interface	°C	0.0	160.0	0.0	
2.2.x.6.5.4	^-subst.	subst. value	°C	0.0	160.0	5.0	

Amount of heat (2.2.x.6.8)

After the assignment of a pulse counter to the appropreate input terminal (25 or 26) for the amount of heat and the amount of electric energy (see chapter SF/terminal assignment) you have the possibilty the view the consumption values in this menu. The consumption values for "heating" and "domestic hot water production" are counted sparately. In addition the total COP-value is calculated.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.6.8.1	HAmount-HC	heatAmount heatCirc	kWh	0	9999999	0	
2.2.x.6.8.2	HAmount-DHW	heatAmount domHWat	kWh	0	9999999	0	
2.2.x.6.8.3	EAmount	electric energie	kWh	0	9999999	0	
2.2.x.6.8.4	COP-T	Total-COP					

Terminal assignment (2.2.x.6.9)

In this service menu “terminal assignment” you are able to assign the components used in the library program “heat pump circuit” to an input or output terminal. Each input terminal has a substitute value, which is being used by the library program in case of a sensor error. The program will then signal the sensor error and use the substitute value. The substitute value will only be shown in the list, if you enter terminal number “99” as the terminal for the sensor. If you enter the terminal number 99 for the sensor, the library program will keep working with the substitute value, but will show no error anymore.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.x.6.9.1	VDV	dem. VDV		0	255	0	
2.2.x.6.9.2	^-subst.		°C	-40.0	160.0	2.0	
2.2.x.6.9.3	dem-cont.	dem. contact		0	255	0	
2.2.x.6.9.4	^-subst.			0	1	0	
2.2.x.6.9.5	OSwLoc	oMod-local switch		0	255	151	
2.2.x.6.9.6	^-subst.			0	5	1	
2.2.x.6.9.7	mainten.	mainten.		0	255	141	
2.2.x.6.9.8	^-subst.			0	1	0	
2.2.x.6.9.9	hAmount	pulse for amount of heat		0	255	0	
2.2.x.6.9.10	aMntE	pulse for amount of electricity		0	255	0	
2.2.x.6.9.11	HP-rel	HP release		0	255	0	
2.2.x.6.9.12	troubleInd	trouble indication		0	255	0	

Chapter 3 **Strategy Circuit Module**

IF the HPM was loaded with a system diagram, that needs to control more than one heat pump, the library program "strategy circuit" takes over the demand oriented control of the heat pumps. The change of the leading heat pump can be switched according to the running time (leading / running time) and via a switch due to an error in one of the heat producers.

There are a following possibilities to release or block a heat producer

change of the heat demand

running time dependend change of the leading heat producer

when a switch is made in the running order,

when a heat producer is in error or switched off.

Functionality of the strategy circuit:

Each heat pump is being controlled by his own library program. This library program receives the heat demand and the status in the running order from the strategy circuit. Using the setpoint deviation, the strategy circuit calculates the heat demand and decides which heat pump is released and which heat pump is blocked. If the calculated heat demand has reached 100% of the current available capacity, the start-up delay of the next heat pump is activated. If the timer has run down, the following heat pump is released.

If the heat demand falls to 0%, the switch-off delay for the last released heat pump is activated. If the time has run down, the last released heat pump is switched off.

3.1 Current values (2.2.4.1)

In the menu ”current values” all inputs assigned to the library program heat pump, such as temperatures, capacity and volume flow measuring values, operating mode selection and system trouble indications, are displayed

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.1.1	flowHC	flow temperature heatCirc.	°C				
2.2.4.1.2	flowDHW	flow temperature domHotWater	°C				
2.2.4.1.3	outdoor	outdoor temperature	°C				
2.2.4.1.4	VDV	dem. VDV	°C				
2.2.4.1.5	ifc	dem. ifc	°C				
2.2.4.1.6	dem-cont.	dem. contact					
2.2.4.1.7	oModLoc	oMod-local switch					
2.2.4.1.8	Ymin1	min. corrValue. HP1	%				
2.2.4.1.9	StatHS1	status HP1					
2.2.4.1.10	Ymin2	min. corrValue. HP2	%				
2.2.4.1.11	StatHS2	status HP2					
2.2.4.1.12	Ymin3	min. corrValue. HP3	%				
2.2.4.1.13	StatHS3	status HP3					

3.2 Setpoints (2.2.4.2)

In the menu “setpoints” you will see the currently calculated flow temperature setpoints for "heating" and for "domestic hot water production"

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.2.1	SP-FlowHC	setpoint flow.-temp	°C				
2.2.4.2.2	SP-FlowDHW	setpoint flow.-temp	°C				

3.3 Additional Functions (2.2.4.3)

Leading (2.2.4.3.2)

Changing the leading heat producer gives you the possibility to devide the running time evenly on all connected heat pumps. The results are equal idle times, equal maintenance intervals, and equal wearout of the components of the heat pumps. A switch of the leading heat pump takes place, if the operating hours of the leading heat pump is higher than the operating hours of the following heat pump.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.2.1	active			0	1	1	
2.2.4.3.2.2	running time	switching lead after running time	h	10	10000	100	

Sequence (2.2.4.3.3)

In the menu "sequence" you can define the sequence in which way the heat pumps should be released for "heating" and "domestic hot water production" separately. Switching the heat pumps of is done in the opposite way.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.3.2	SP-FlowHC	setpoint flow.-temp					
2.2.4.3.3.3	SP-FlowDHW	setpoint flow.-temp					

Boost (2.2.4.3.6)

The additional function “boost” can be used to set boosts (e.g. HC1 10 %, DHW1 -100%) for the temperature demands of the internal heating and domestic hot water circuits and for the external demand (via VDV). This allows to compensate for transmission losses of the heating system.

If a value of -100% is entered for the boost, the setpoint of the heat producer is not affected by this demand (e.g. boost DHW1 -100 %, for domestic hot water circuits with external supply and/or DHW with own heat source, e.g. - solar power -). If this function is activated with active = 1, the demands of the consumer circuits will be increased with the relevant boost factor.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.6.1	active			0	1	0	

2.2.4.3.6.2	boost HC1	boost HC1	%	-100	100	10	
2.2.4.3.6.3	boost HC2	boost HC2	%	-100	100	10	
2.2.4.3.6.4	boost DHW	boost DHW	%	-100	100	10	
2.2.4.3.6.5	boost VDV	boost VDV	%	-100	100	10	

Signal (2.2.4.3.7)

The function “signal” gives you the possibility to configure the way a signal should be handled. When this function is activated, you can decide what should happen, if

a) an error occurred
b) a trouble indication occurred
c) a limit was exceeded
d) a timer has rundown (maintenance etc.)

In the strategy circuit you have the possibility to handle the signals for "heating" and "domestic hot water production" separately.

TI-all	setting of the trouble indication outputs **troubleInd** .	indication as an alarm in the central **building management software**,	indication on the controller **display**
0			
1			x
2		x	
3		x	x
4	x		
5	x		x
6	x	x	
7	x	x	x

Par. no.	**Name**	**Info text**	**Unit**	**Min.**	**Max.**	**Default**	**Description**
2.2.4.3.7.1	active			0	1	0	
2.2.4.3.7.2	TI-all	TI to BMS:2,3,6,7 TI-O:4-7 disp:1,3,5,7		0	7	1	
2.2.4.3.7.3	Xw-HC	max. contrDev. flow temperature	K	0.5	50.0	50.0	
2.2.4.3.7.4	del-Xw-HC	delayed contrDev. flow temperature	min	0	600	600	
2.2.4.3.7.5	Xw-DHW	max. contrDev. flow temperature	K	0.5	50.0	50.0	
2.2.4.3.7.6	del-Xw-DHW	delayed contrDev. flow temperature	min	0	600	600	

,

Frost protection (2.2.4.3.8)

The frost protection function prevents the heating system from freezing by monitoring the flow and the outdoor temperature. The function is using an assigned pump.

Caution!

The frost protection function is not working in the status „NOT ACTIVE“ and in the status „manual operation“ of the pump.

Par. no.	**Name**	**Info text**	**Unit**	**Min.**	**Max.**	**Default**	**Description**
2.2.4.3.8.1	outdoorfrLi	outdoor-frost limit	°C	-50.0	50.0	2.0	

Pump (2.2.4.3.9)

The switch-off delay prevents the heat producers from overheating by dissipating the residual heat. For this purpose switching off of the pump is delayed for the duration of **SwOfDel**.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.9.1	active			0	1	1	
2.2.4.3.9.2	SwOfDel	delayed switchOff	s	0	600	300	

External demand (2.2.4.3.10)

If the system diagram loaded into the HPM has more than one heat pump, the strategy circuit can also receive external temperature demands (via VDV, contact and interface). After assignment of the input terminals (VDV, dem-cont.) the processing of the input signals can be activated in the menu AF - external demand.

- **Demand via contact**

A fixed temperature demand can be transmitted to the strategy circuit via a potential-free contact which is connected to an input terminal of the controller. For this purpose, the input terminal dem-cont has to be assigned to a terminal (see chapter “SF – terminal assignment”). The activation is done in this menu at **dem-cont. = 1**. The required temperature demand can be specified in the parameter **dem-T-^**. If the contact is closed, the temperature demand is integrated in the setpoint calculation. The display of the status (On/Off) of the input terminal is shown in the menu “current values”.

- **Demand via VDV (variable demand via voltage)**

Variable temperature demands can be transmitted between HPM and other controllers via VDV (0 ... 10V signal). A VDV demand can also be received by “third-party controllers” with 0..10V signal. Up to 9 controllers can transmit their temperature demand to this controller. First, assign the input terminal for VDV (see chapter “SF – terminal assignment”). The activation is done in this menu at **VDV = 1**. The input signal 0 ... 10 V equals 0 ... 150 °C. The output signals of the demanding controllers have to send the signal accordingly in this format. The display of the setpoint temperature requested via VDV is shown in the menu “current values”.

- **Demand via interface**

A device with masterbus interface can send a setpoint temperature demand to the HPM via the bus (RS 485). The activation is done in this menu at **ifc = 1**. The display of the setpoint temperature requested via interface is shown in the menu “current values”. If the interface connection fails, a programable substitute value is active (see chapter “SF – external demand”).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.10.1	dem-cont.	dem. contact active		0	1	0	
2.2.4.3.10.2	dem-T-^	flow setpoint temp. contact	°C	2.0	160.0	50.0	
2.2.4.3.10.3	VDV	dem. VDV active		0	1	0	
2.2.4.3.10.4	ifc	dem. interface active		0	1	0	

Controller (2.2.4.3.11)

During the initial installation, you have to set the parameters in this menu according to your system. You have to set system relevant parameters like

a) is a pump connected (yes/no)

b) you can define a system related name for the strategy circuit to simplify the structure for the service personal.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.11.1	pump	0:no 1:yes		0	1	0	
2.2.4.3.11.2	longDes	free adjustab. Prog.-longRef.					

Setpoint limitation (2.2.4.3.12)

An upper and lower limit for the flow temperature setpoint **SP-Flow** is set here.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.12.1	active			0	1	1	
2.2.4.3.12.2	minSP	min. flow-setpoint	°C	2.0	90.0	2.0	
2.2.4.3.12.3	maxSP	max. flow-setpoint	°C	50.0	160.0	85.0	

Preventation of blocking (2.2.4.3.13)

The "preventation of blocking" function detects automatically if the pump has moved due to performance of control tasks in the last 24 hours. If this is not the case, the pump is triggered for an adjustable time (duration) at 11 o'clock every day.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.3.13.1	duration	running time clogging protection	s	0	600	120	

3.4 Status (2.2.4.4)

The menu "status" gives an overview of the current mode of operation of the heat pump circuit.

Every function affecting the operation of the heat pump is considered, when the main status **opStatus** or the auxiliary status **opStatCode** is generated. The **opStatCode** is a hexadecimal number. The meaning of the characters shown in the parameters **opStatCode** and **eStatCode** is explained in the following tables.

opStatCode:

The auxiliary operating status consists of three digits as various functions of the heat pump circuit can be effective at the same time. The display helps during analysis of the control. The example below, together with the translation table, explains how to decode the sequence of characters.

	1. digit	2. digit	3. digit
1			
2			
3			
4			
5			
6			

7			
8			
9			
A			
B			
C			
D			
E			
F			

Tab. 3: operating mode status code, strategy circuit

Example:

Display: OPStatCode: = 05

Meaning: 6. digit = demand "heating", setpoint limitation is active

sStatCode:

The auxiliary trouble indication status consists of two digits as several trouble indications can occur at the same time. The example below, together with the translation table, explains how to decode the sequence of characters

	1. digit	2. digit
1		error input
2		maxim control deviation for flow temperature "heating"
3		error input maxim control deviation for flow temperature "heating"
4		maxim control deviation for flow temperature "DHW"
5		maxim control deviation for flow temperature "DHW", error input
6		maxim control deviation for flow temperature "DHW", maxim control deviation for flow temperature "heating"
7		maxim control deviation for flow temperature "DHW", error input maxim control deviation for flow temperature "heating"

Tab. 2: trouble indication code "strategy circuit"

Example:

Display: sStatCode: = 1

Meaning: 2. digit = error input

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.4.1	opStatus						
2.2.4.4.2	trouble						
2.2.4.4.3	source	setpoint source and -compensation					
2.2.4.4.4	curSequHC	cur sequence					
2.2.4.4.5	curSequDHW	cur sequence					

2.2.4.4.6	pump	pump					
2.2.4.4.7	fl-T-HP1	flow temp. HP1	°C				
2.2.4.4.8	seqStat1	sequence-status HP1					
2.2.4.4.9	fl-T-HP2	flow temp. HP2	°C				
2.2.4.4.10	seqStat2	sequence-status HP2					
2.2.4.4.11	fl-T-HP3	flow temp. HP3	°C				
2.2.4.4.12	seqStat3	sequence-status HP3					
2.2.4.4.13	troubInd	trouble indication					
2.2.4.4.14	opStatCode						
2.2.4.4.15	TIStatCode						

3.5 Manual operation (2.2.4.5)

The manual control can be used to check the release of the pump during commissioning.

Note!

In case of improper handling, the operating mode “Manual operation” can cause damages to the system as the manual control overrides the limitation and the prevention of blocking and frost protection functions.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.5.1	pump	0:off 1:on 3:automatic		0	3	3	

3.6 Service Functions (2.2.4.6)

In the Service menu of the strategy circuit, the basic configuration of the library program is made, inputs and outputs can be assigned and you can adjust the basic parameter settings for the additional functions.

General values (2.2.4.6.1)

The service menu “general values” displays the operating hours of the pump. During pump replacement or maintenance the counter can be adjusted to a required value.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.1.1	pump		h	0	999999	0	

Leading (2.2.4.6.2)

The running times of heat pumps 1 .. 3 are displayed.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.2.1	HP1	running time heat producer	h				
2.2.4.6.2.2	HP2	running time heat producer	h				
2.2.4.6.2.3	HP3	running time heat producer	h				

Release HC (2.2.4.6.4)

Using the deviation between the setpoint and the current value of the flow temperature of the heating circuit and the control parameters Xp and Tn a heat demand (control signal Y) is calculated. If the calculated heat demand has reached 100% of the current available capacity, the start-up delay of the next heat pump is activated. If the timer has run down, the following heat pump is released.

If the heat demand falls to 0%, the switch-off delay for the last released heat pump is activated. If the time has run down, the last released heat pump is switched off

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.4.1	Xp+	proportional band cur>SP	K	0.1	500.0	100.0	
2.2.4.6.4.2	Xp-	proportional range curVal<setpoint va-ope	K	0.1	500.0	100.0	
2.2.4.6.4.3	Tn+	integral action time curVal>setpoint va-c	min	0.1	121.0	2.0.	
2.2.4.6.4.4	Tn-	integral action time curVal<setpoint va-o	min	0.1	121.0	2.0	
2.2.4.6.4.5	Y	contr corr variable	%				
2.2.4.6.4.6	swOnDel	delayed switch-on	min	1.0	120.0	20.0	
2.2.4.6.4.7	timer	curTimer	min				
2.2.4.6.4.8	swOfDel	delayed switchOff	min	1.0	120.0	5.0	
2.2.4.6.4.9	timer	curTimer	min				

Description:

swOnDel: Delay for release of the next heat producer for current capacity = 100%

swOfDel: Delay for switching-off the last released heat producer for current capacity = 0%

Release DHW (2.2.4.6.5)

Using the deviation between the setpoint and the current value of the flow temperature of the DHW circuit and the control parameters Xp and Tn a heat demand (control signal Y) is calculated. If the calculated heat demand has reached 100% of the current available capacity, the start-up delay of the next heat pump is activated. If the timer has run down, the following heat pump is released.

If the heat demand falls to 0%, the switch-off delay for the last released heat pump is activated. If the time has run down, the last released heat pump is switched off

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.5.1	Xp+	proportional range curVal>setpoint va-clo	K	0.1	500.0	100.0	
2.2.4.6.5.2	Xp-	proportional range curVal<setpoint va-ope	K	0.1	500.0	100.0	
2.2.4.6.5.3	Tn+	integral action time curVal>setpoint va-c	min	0.1	121.0	2.0.	
2.2.4.6.5.4	Tn-	integral action time curVal<setpoint va-o	min	0.1	121.0	2.0	
2.2.4.6.5.5	Y	contr corr variable	%				
2.2.4.6.5.6	swOnDel	delayed switch-on	min	1.0	120.0	20.0	
2.2.4.6.5.7	timer	curTimer	min				
2.2.4.6.5.8	swOfDel	delayed switchOff	min	1.0	120.0	5.0	
2.2.4.6.5.9	timer	curTimer	min				

Signal (2.2.4.6.7)

The last 10 errors detected are recorded and displayed in the service menu “signal”. The first parameter always shows the most recent error and the last parameter the oldest error. The parameter text displayed is the short text of the error detected. By pressing the “OK” key the info text appears, giving you more detailed description of the error. The date and time at which the error was detected is documented. A power failure or a controller cold start will delete the recorded errors!

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.7.1		TT.MM.JJ hh:mm					
2.2.4.6.7.2		TT.MM.JJ hh:mm					
2.2.4.6.7.3		TT.MM.JJ hh:mm					
2.2.4.6.7.4		TT.MM.JJ hh:mm					
2.2.4.6.7.5		TT.MM.JJ hh:mm					
2.2.4.6.7.6		TT.MM.JJ hh:mm					
2.2.4.6.7.7		TT.MM.JJ hh:mm					
2.2.4.6.7.8		TT.MM.JJ hh:mm					
2.2.4.6.7.9		TT.MM.JJ hh:mm					
2.2.4.6.7.10		TT.MM.JJ hh:mm					

Pump (2.2.4.6.9)

The switch-off delay of the pump is used to prevent the heat pump to overheat by dissipating the heat. In the menu "AF - pump" you can adjust the duration of the extendet running time of the pump. The current value of this timer can be seen in this menu.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.9.1	swOffDel	curTimer	s				

External demand (2.2.4.6.10)

In the service menu of the function “external demand” you can set the basic parameters for communication between the controller and external components that demand heat either by voltage signal (VDV) or via the interface. With **invertVDV** you can configure the conversion between the input voltage to the setpoint temperature: = *0*: 0V->0°C, 10V->100°C; = *1*: 0V->100°C, 10V-0°C. The parameter Interfac shows the setpoint temperature coming from an external controller via the interface.The target address for the external controllers is the parameter number of this parameter. If no valid value has been received after a waiting time **waitingT** the substitute value **^-subst** will become effective.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.10.1	invert VDV	0:0V-0°C,10V-100°C 0:0V-100°C,10V-0°C		0	1	0	
2.2.4.6.10.2	waitingT	idle time	Min	0.1	999.9	1.0	
2.2.4.6.10.3	interface	setpoint demand via interface	°C	0.0	160.0	0.0	
2.2.4.6.10.4	^-subst.	subst. value	°C	0.0	160.0	5.0	

Sensor correction (2.2.4.6.14)

If the temperatures displayed in the menu "current values" differ from the current system values, a correction value can be entered to adjust the individual sensor values.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.14.1	flowHC		K	-10.0	10.0	0.0	
2.2.4.6.14.2	flowDHW		K	-10.0	10.0	0.0	
2.2.4.6.14.3	outdoor		K	-10.0	10.0	0.0	

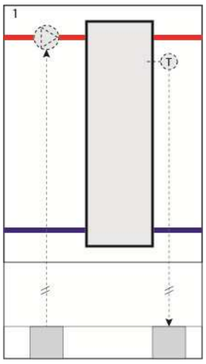
Terminal assignment (2.2.4.6.15)

In this service menu "terminal assignment" you are able to assign the components used in the library program "strategy circuit" to an input or output terminal. Each input terminal has a substitute value, which is being used by the library program in case of a sensor error. The program will then signal the sensor error and use the substitute value. The substitute value will only be shown in the list, if you enter terminal number "99" as the terminal for the sensor. If you enter the terminal number 99 for the sensor, the library program will keep working with the substitute value, but will show no error anymore.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.2.4.6.15.1	flowHC	flow temperature heatCirc.		0	255	0	
2.2.4.6.15.2	^-subst.	- look at op manual -	°C	-40.0	160.0	20.0	
2.2.4.6.15.3	flowDHW	flow temperature domHotWater		0	255	0	
2.2.4.6.15.4	^-subst.	- look at op manual -	°C	-40.0	160.0	20.0	
2.2.4.6.15.5	outdoor	outdoor temperature		0	255	0	
2.2.4.6.15.6	^-subst.	- look at op manual -	°C	-40.0	160.0	0.0	
2.2.4.6.15.7	VDV	dem. VDV		0	255	0	
2.2.4.6.15.8	^-subst.	- look at op manual -	°C	-40.0	160.0	2.0	
2.2.4.6.15.9	dem-cont.	dem. contact		0	255	0	
2.2.4.6.15.10	^-subst.	- look at op manual -		0	1	0	
2.2.4.6.15.11	OSwLoc	oMod-local switch		0	255	151	
2.2.4.6.15.12	^-subst.	- look at op manual -		0	6	0	
2.2.4.6.15.13	pump	pump		0	255	0	
2.2.4.6.15.14	troubleInd	trouble indication		0	255	207	

Chapter 4 The Buffer tank module

A buffer tank has many positive effects on a system, including reducing starts and stops from the heat pumps and generating cheaper heat production when available (solar, pellet, etc.). The temperature in the buffer tank is controlled according to the demands of the internal heat circuits and domestic hot water circuit, as well as external demand via 0…10V, interface or contact. The controller makes use of all available heat producers using the following priority: 1: Solar, 2: Additional heat source, 3: Internal heat pumps.



4.1 Current values (2.3.1.n)

The menu “Current values” displays an overview of the current values from temperature sensors, digital inputs, meters etc.

- **demCFlHeat**: Temperature demand via continuous signal for heating.
- **ifc**: Temperature demand via interface.
- **demContHeat**: Temperature demand via contact input.
- **buffer1**: Temperature from the sensor at the top of buffer tank.
- **buffer2**: Temperature from the sensor at the middle of buffer tank.
- **buffer3**: Temperature from the sensor at the bottom of buffer tank.
- **solarFl**: Temperature from flow temperature sensor in the solar circuit.
- **solarRt**: Temperature from return temperature sensor in the solar circuit.
- **aHS-fl**: Temperature from flow temperature sensor at additional heat source.
- **aHS-rt**: Temperature from return temperature sensor at additional heat source.
- **earthBuf**: Temperature from sensor in earth buffer.
- **oModLoc**: Position of local operating mode switch.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.1.1	demCFlHeat	tempDem. Cont. heat	°C				
2.3.1.2	ifc	dem. Ifc	°C				
2.3.1.3	demConHeat	dem. Contact heat					
2.3.1.4	buffer1	buffer temp top					
2.3.1.5	buffer2	buffer temp middle					
2.3.1.6	buffer3	buffer temp bottom					
2.3.1.7	solarFl	flTemp.solar	°C				

2.3.1.8	solarRt	rtTemp.solar	°C				
2.3.1.9	aHS-fl	flTemp.aHS	°C				
2.3.1.10	aHS-rt	rtTemp.aHS	°C				
2.3.1.11	earthBuf	earth buffer	°C				
2.3.1.12	oModLoc	oMod-local switch					

4.2 Setpoints (2.3.2.n)

HPM is capable of controlling normal buffer tanks with one zone. The menu “Setpoints” displays the currently calculated buffer temperature setpoint **SP-zone1** based on the demand from the internal heating circuits, the domestic hot water circuits and external heat consumers. These parameters can not be changed manually. A boost factor can also be set in order to compensate for heat loss due to poor insulation.

- **SP-zone1**: Currently calculated setpoint of buffer temperature for zone 1.
- **boostZ1**: Boost factor for zone 1.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.2.1	SP-zone1	current setpoint buffer zone 1	°C				
2.3.2.3	boostZ1	boost demand buffer zone 1		1.0	25.0	5.0	

4.3 Additional functions (2.3.3.n)

Boost (2.3.3.2.n)

The additional function “Boost” can be used to set boosts (e.g. HC1 10 %, DHW1 -100%) for the temperature demands of the internal heating and domestic hot water circuits and for external demand via 0…10V. This allows for compensation of transmission losses in the heating system.

If a value of -100% is entered for the boost, the setpoint of the boiler system is not affected by this demand (e.g. boost DHW1 -100 %, for domestic hot water circuits with external flow and/or DHW with own heat source, e.g. - heat pump or solar power).

If this function is activated, the demand of a heat consumer is multiplied with the boost value (in percentage 10=10%).

- **active**: This parameter activates (1) or deactivates (0) the function.
- **boost HC1**: Boost factor for the temperature demand of heating circuit 1.
- **boost HC2**: Boost factor for the temperature demand of heating circuit 2.
- **boost DHW**: Boost factor for the temperature demand of domestic hot water circuit.
- **boost VDV**: Boost factor for the temperature demand of external consumer.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.3.2.1	active			0	1	0	
2.3.3.2.2	boost HC1	boost HC1	%	-100	100	10	
2.3.3.2.3	boost HC2	boost HC2	%	-100	100	10	
2.3.3.2.4	4.3.7 boost	4.3.8 boost DHW1	4.3.9 %	4.3.10 -	4.3.11 1	4.3.12 1	
2.3.3.2.5	4.3.13 boost	4.3.14 boost VDV	4.3.15	4.3.16 -	4.3.17 1	4.3.18 1	

External demand (2.3.3.3.n)

The HPM offers the possibility to control the buffer tank in such a way that a heat demand from an external heat consumer can be easily fulfilled. This heat consumer can send a demand in different ways:

Demand via contact

A fixed temperature demand can be transmitted to the boiler circuit via a potential-free contact connected to an input terminal of the controller. For this purpose, the input terminal **dem-cont** has to be assigned to a terminal (see chapter “SF – terminal assignement”). Activation is performed in this menu at **dem-cont. = 1**. The required temperature demand can be specified by using the parameter **dem-T-^**. If the contact is closed, the temperature demand is integrated in the setpoint calculation. The status (On/Off) of the input terminal is displayed in the menu “Current values”.

Demand via voltage (VDV) (continuous signal)

Variable temperature demands can be transmitted between HPM and other controllers (like HPM, RU 6X and RU 9X³ controllers via a 0...10V signal. A heat demand can also be received by third-party controllers with a 0...10V signal. Up to 9 controllers can transmit their temperature demand to this controller. First, assign the input terminal for the heat demand (see chapter “SF – terminal assignment”). The activation is performed in this menu at **VDV = 1**. The input signal 0...10V equals 0...150 °C. The output signals of the demanding controllers have to send the signal in this format accordingly. The setpoint temperature requested via the 0…10V signal is displayed in the menu ”Current values”.

Demand via interface

A device with masterbus interface (e.g. CLEVER master) can transmit a setpoint temperature demand to the HPM via the R+S bus (RS 485). Activation is performed in this menu at **ifc = 1**. The setpoint temperature requested via interface is displayed in the menu “Current values”. If the interface connection fails, a programmable substitute value is active (see chapter “SF – external demand”).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.3.3.1	dem-cont.	dem. Contact active		0	1	0	
2.3.3.3.2	dem-T-^	flow setpoint temp. contact	°C	2.0	160.0	50.0	
2.3.3.3.3	VDV	dem. 0…10V active		0	1	0	
2.3.3.3.4	ifc	dem. Interface active		0	1	0	

Signal (2.3.3.4.n)

The “Signal” function permits configuring in what way a signal should be handled. When this function is activated, it is possible to decide what should happen if:

a) An error occurs.

b) A trouble indication occurs.

c) A limit is exceeded.

d) A timer has run down (maintenance, etc.).

- **active** : This parameter activates (1) or deactivates (0) the function.
- **TI-all**: This parameter permits configuring how the controller should react.

TI-all	setting of the trouble indication outputs troubleInd		
		indication as an alarm in the central building management software	
			indication on the controller display
0			
1			x
2		x	
3		x	x
4	x		
5	x		x
6	x	x	
7	x	x	x

- **uLim-aHS**: This parameter permits setting an upper limit for the flow temperature of the additional heat source, which, when violated, leads to a signal.
- **uLim-sol**: This parameter permits setting an upper limit for the flow temperature of the solar circuit, which, when violated, leads to a signal.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.3.4.1	active			0	1	0	
2.3.3.4.2	TI-all	TI to BMS:2,3,6,7 TI-A:4-7 disp:1,3,5		0	7	1	
2.3.3.4.3	uLim-aHS	upper limit temp. add. Heat source	°C	50.0	160.0	130.0	
2.3.3.4.4	uLim-sol	upper limit temp. solar	°C	50.0	160.0	130.0	

Buffer control (2.3.3.5.n)

The “Buffer control” menu permits to enter specific information for your buffer tank.

- **maxTStor**: Maximum possible buffer tank temperature.
- **longDes**: Freely adjustable name for the buffer tank.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.3.5.6	maxTStor	max. temperature	°C	20.0	90.0	70.0	
2.3.3.5.7	longDes	free adjustab. Prog.-longRef.					

Additional heat source (2.3.3.6.n)

The menu “Additional heat source” permits activating and configuring the control of an additional heat source, which can be used to push additional “cheap” heat into the buffer tank, when it is available. A maximum temperature and frost limit can also be defined.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **tDiffOn**: Temperature difference between flow temperature of additional heat source and the temperature in the buffer tank at which loading is possible.
- **tDiffOff**: Temperature difference between flow temperature of additional heat source and the temperature in the buffer tank at which loading is switched off.
- **maxTaHS**: Maximum flow temperature of additional heat source. If the flow temperature is higher than this limit, then the additional heat source will be cooled down, by activating the output **aHS** until the buffer tank temperature is lower than **maxTaHS – 10K.**
- **aHSFrLi**: With this parameter you can monitor the frost limit of the additional heat source. If the limit is violated, the output aHS is activated, to push warm water from the puffer tank into the pipes to prevent freezing. The frost protection is switched of when there is a switching difference of 1K.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.3.6.1	active			0	1	0	
2.3.3.6.2	tDiffOn	temp-difference add.heat source on	K	0.0	30.0	8.0	
2.3.3.6.3	tDiffOff	temp-difference add.heat source off	K	0.0	20.0	2.0	
2.3.3.6.4	maxTaHS	max. temperature add. Heat source	°C	50.0	160.0	90.0	
2.3.3.6.5	aHSFrLi	frost limit	°C	-60.0	10.0	-10.0	

Solar (2.3.3.7.n)

The menu “Solar” permits activating and configuring control of an integrated solar thermal collector, which can deliver additional heat to the buffer tank. A maximum temperature and frost limit can also be defined, and an earth buffer cam be integrated.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **tDiffOn**: Temperature difference between flow temperature of the solar circuit and the temperature in the buffer tank at which loading is possible.
- **tDiffOff**: Temperature difference between flow temperature of the solar circuit and the temperature in the buffer tank at which loading is switched off.
- **maxTsolar**: Maximum flow temperature of the solar circuit. If the flow temperature is higher than this limit, then the solar circuit will be cooled down, by activating the heating circuit until the buffer tank temperature is lower than **maxTsolar – 10K**.
- **solFrLi**: With this parameter you can monitor the frost limit of the solar circuit. If the limit is violated, the solar circuit is activated, to push warm water from the puffer tank into the pipes to prevent freezing. The frost protection is switched of when there is a switching difference of 1K.
- **actEarthB**: Loading of an earth buffer is possible (0=no, 1=yes).
- **earthBLim**: Maximum temperature of the earth buffer.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.3.7.1	active			0	1	0	
2.3.3.7.2	tDiffOn	temp-difference solar on	K	0.0	30.0	8.0	
2.3.3.7.3	tDiffOff	temp-difference solar off	K	0.0	20.0	2.0	
2.3.3.7.4	maxTsolar	max. temperature solar circuit	°C	50.0	160.0	90.0	

2.3.3.7.5	solFrLi	frost limit	°C	-60.0	10.0	-10.0	
2.3.3.7.6	actEarthB	loading earth buffer		0	1	0	
2.3.3.7.7	earthBLim	maximal temp. earth buffer	°C	10.0	30.0	20.0	

4.4 Status (2.3.4.n)

The “Status” menu provides an overview of the current operating mode of the buffer tank module.

Every function affecting the operation of the district heating circuit is considered when the main status **opStatus** or the auxiliary status **opStatCode** is generated. The **opStatCode** is a hexadecimal number. The meaning of the characters shown in the parameters **opStatCode** and **eStatCode** is explained in the following tables.

- **opStatus**: Main status of the buffer tank module:
  = *Not active/Off:* The required inputs are not assigned or the operating mode switch **oModRem** or **oModLoc** is assigned and in position Off.
  = *Switch-off operation*: System main switch **System** assigned and switched off. Demand = 2°C.
  = *Nominal operation*: Demand >2°C.
  = *Manual operation*: Manual control of the outputs. Operating mode switch **oModRem** or **oModLoc** assigned and in position “Manual”.
- **trouble:** Current trouble status (text).
- **aHS**: Current signal to additional heat source.
- **solPu**: Current signal to solar pump.
- **loadPu**: Current signal to loading pump.
- **earthBPu**: Current signal to earth buffer pump.
- **zone1**: Current signal for zone 1.
- **statAHS**: Current status of additional heat source.
- **oStatSol**: Current operationg status of solar system.
- **statSol1**: Current operating status solar, zone 1.
- **statEarthB:** Current status of earth buffer.
- **troubInd**: Pending trouble indication.
- **opStatCode**: The auxiliary operating status consists of two digits, since various functions of the buffer tank module can be effective at the same time. The example below, together with the translation table, explains how to decode the sequence of characters.

	1st digit	2nd digit
1	frostSolar	aHsCool
2	heatZone1	aHsFrost
3	heatZone1, frostSolar	aHsFrost, aHsCool
8	statEarthB	SolarCool
9	statEarthB,rostSolar	SolarCool, aHsCool
A	statEarthB,heatZone1	SolarCool, aHsFrost
B	statEarthB,heatZone1,frostSolar	SolarCool, aHsFrost, aHsCool
C		SolarCool, aHsLoad
D		SolarCool, aHsLoad, aHsCool
E		SolarCool, aHsLoad, aHsFrost
F		SolarCool, aHsLoad, aHsFrost, aHsCool

Tab. 4: Operating status code buffer tank

Example:

Display: opStatCode: = 10

Meaning: 1st digit = Heating zone 1.

- **eStatCode:** The auxiliary trouble indication status consists of two digits**.** The example below, together with the translation table, explains how to decode the sequence of characters.

	1st digit
1	TI-nput
2	uL-Solar,
3	uL-Solar, TI-nput
4	uL-aHs,
5	uL-aHs, TI-nput
6	uL-aHs, uL-Solar,
7	uL-aHs, uL-Solar, TI-nput

Tab. 5: Trouble indication status buffer tank

Example:

Display: sStatCode: = 10

Meaning: 1st digit = Trouble indication from input terminal.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.4.1	opStatus						
2.3.4.2	trouble						

2.3.4.4	aHS	add. Heat source					
2.3.4.5	solPu	solarPump					
2.3.4.6	loadPu	buffer load.pump					
2.3.4.7	earthBPu	earth buffer pump					
2.3.4.8	zone1	loading zone 1					
2.3.4.10	statAHS	operating status add. Heat source					
2.3.4.11	oStatSol	operating status solar					
2.3.4.12	statSol1	operating status solar zone 1					
2.3.4.14	statEarthB	oStat earth buffer					
2.3.4.15	oStatCode						
2.3.4.16	TIStatCode						

4.5 Manual operation (2.3.5.n)

Manual operatation can be used to check the field components connected to the buffer tank (e.g. pumps, etc.) for correct operation and sense of rotation during the commissioning phase.

NOTE:

In case of improper handling, the manual operating mode can cause damages to the system! Manual control overrides all functional limitations, including pump/valve exercise and frost protection functions.

- **aHS** : Signal to additional heat source: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **solPu**: Solar pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **loadPu**: Loading pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **earthBPu**: Earth buffer pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **zone1**: Signal zone 1: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.5.1	aHS	0:off 1:on 3:automatic		0	3	3	
2.3.5.2	solPu	0:off 1:on 3:automatic		0	3	3	
2.3.5.3	loadPu	0:off 1:on 3:automatic		0	3	3	
2.3.5.4	earthBPu	0:off 1:on 3:automatic		0	3	3	
2.3.5.5	zone1	0:off 1:on 3:automatic		0	3	3	

4.6 Service functions (2.3.6.n)

The service menu for the buffer tank circuit permits configuration of the module, assignment of inputs and outputs and setting of individual parameters for the additional functions. This section is reserved for skilled service personnel.

General values (2.3.6.1.n)

The service menu “General values” displays the operating hours of the pumps and/or given signals.

During pump replacement or maintenance, the counter can be adjusted to a required value.

- **aHS:** Operating hours of the signal to the external heat source.
- **solPu:** Operating hours of the solar pump.
- **loadPu:** Operating hours of the loading pump.
- **eartBPu:** Operating hours of the earth buffer pump.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.6.1.1	aHS		h	0	999999	0	
2.3.6.1.2	solPu		h	0	999999	0	
2.3.6.1.3	loadPu		h	0	999999	0	
2.3.6.1.4	earthBPu		h	0	999999	0	

External demand (2.3.6.3.n)

The service menu of the function "External demand" permits setting of basic parameters for communication between the controller and external components that demand heat either by voltage signal (0…10V) or via the interface.

- **Invert VDV**: This parameter configures the conversion between the input voltage to the setpoint temperature:

  = *0*: 0V→0°C, 10V→150°C

  = *1*: 0V→150°C, 10V-0°C
- **interface**: This parameter shows the setpoint temperature coming from an external controller via the interface. The parameter number of this parameter equals the target address for the external controllers.
- **waitingT**: If no valid value has been received within the waiting time **waitingT**…
- **^-subst.:** …the substitute value **^-subst** will take effect.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.6.3.1	4.6.19 invert	4.6.20 0:0V-0°C,10V-150°C 1:0V-150°C,10V-0°C		4.6.21	4.6.22 1	4.6.23 0	
2.3.6.3.2	waitingT	idle time	Min	0.1	999.9	1.0	
2.3.6.3.3	interface	setpoint demand via interface	°C	0.0	160.0	0.0	
2.3.6.3.4	^-subst.	subst. value	°C	0.0	160.0	2.0	

Sensor correction (2.3.6.8.n)

If the temperature values displayed in the menu "Current values" (see chapter "Buffer tank – current values") differ from those of the current system values, a correction factor can be entered to adjust the values of individual sensors.

- **demCFlHeat**: Correction factor for temperature demand via continuous signal for heating.
- **buffer1**: Correction factor for the temperature from the sensor at the top of the buffer tank.
- **buffer3**: Correction factor for the temperature from sensor at the bottom of the buffer tank.
- **solarFl**: Correction factor for the temperature from flow temperature sensor in the solar circuit.

- **solarRt**: Correction factor for the temperature from return temperature sensor in the solar circuit.
- **aHS-Fl**: Correction factor for the temperature from flow temperature sensor of additional heat source.
- **aHS-Rt**: Correction factor for the temperature from return temperature sensor of additional heat source.
- **earthBuf**: Correction factor for the temperature from sensor in earth buffer.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.6.8.1	4.6.24 demC	4.6.25	4.6.26 K	4.6.27 -	4.6.28 10.	4.6.29 0.0	
2.3.6.8.2	4.6.30 buffe	4.6.31	4.6.32 K	4.6.33 -	4.6.34 10.	4.6.35 0.0	
2.3.6.8.4	4.6.36 buffe	4.6.37	4.6.38 K	4.6.39 -	4.6.40 10.	4.6.41 0.0	
2.3.6.8.5	4.6.42 solFl	4.6.43	4.6.44 K	4.6.45 -	4.6.46 10.	4.6.47 0.0	
2.3.6.8.6	4.6.48 solRt	4.6.49	4.6.50 K	4.6.51 -	4.6.52 10.	4.6.53 0.0	
2.3.6.8.7	4.6.54 aHS-	4.6.55	4.6.56 K	4.6.57 -	4.6.58 10.	4.6.59 0.0	
2.3.6.8.8	4.6.60 aHS-	4.6.61	4.6.62 K	4.6.63 -	4.6.64 10.	4.6.65 0.0	
2.3.6.8.9	4.6.66 earth	4.6.67	4.6.68 K	4.6.69 -	4.6.70 10.	4.6.71 0.0	

Terminal assignment (2.3.6.9.n)

The “Terminal assignment” service menu permits assigning the components used in the Heat pump circuit modules to an input or output terminal. Each input terminal has a substitute value which is used by the modules in case of a sensor error. The modules will then signal the sensor error and use the substitute value. The substitute value will only be shown in the list if the terminal number “99” is entered as the terminal for the sensor. If this number is entered for the sensor, the modules will keep working with the substitute value but will no longer show any error.

- **demCFlHeat**: The terminal number for temperature demand via continuous signal for heating.
- **^subst:** The substitute value for the above mentioned sensor (terminal number= 99).

- **demConHeat:** The terminal number for demand via contact input.
- **buffer1:** The terminal number for the buffer temperature sensor at the top of the buffer tank.
- **buffer3:** The terminal number for the buffer temperature sensor at the bottom of the buffer tank.
- **solarFl:** The terminal number for the flow temperature sensor in the solar circuit.
- **solarRt**: The terminal number of the return temperature sensor in the solar circuit.
- **aHSFl:** The terminal number for the flow temperature sensor from the additional heat source.
- **aHSRt:** The terminal number of the return temperature sensor from the additional heat source.
- **earthB:** The terminal number of the temperature sensor in the earth buffer.
- **oModLoc:** The terminal number for the local operating mode switch (via remote control).
- **flow-T-heat:** The terminal number for temperature demand to heat source.
- **aHS**: The terminal number for release of additional heat source.
- **solPu**: The terminal number for the solar pump.
- **loadPu**: The terminal number for the loading pump.
- **earthBPu:** The terminal number for the earth buffer pump.
- **zone1**: The terminal number for loading pump or switch-over valve zone 1.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.3.6.9.1	demCFlHeat	tempDem cont heat		0	255	0	
2.3.6.9.2	^-subst.		°C	2.0	160.0	2.0	
2.3.6.9.3	demConHeat	dem. Contact heat		0	255	0	
2.3.6.9.4	^-subst.			0	1	0	
2.3.6.9.5	buffer1	buffer temp top		0	255	0	
2.3.6.9.6	^-subst.		°C	-40.0	160.0	20.0	
2.3.6.9.9	buffer3	buffer temp bottom		0	255	0	
2.3.6.9.10	^-subst.		°C	-40.0	160.0	20.0	
2.3.6.9.11	solarFl	flTemp.solar		0	255	0	
2.3.6.9.12	^-subst.		°C	-40.0	160.0	20.0	
2.3.6.9.13	solarRt	rtTemp.solar		0	255	0	
2.3.6.9.14	^-subst.		°C	-40.0	160.0	20.0	
2.3.6.9.15	aHS-Fl	flTemp.aHS		0	255	0	
2.3.6.9.16	^-subst.		°C	-40.0	160.0	20.0	
2.3.6.9.17	aHS-Ret	rtTemp.aHS		0	255	0	
2.3.6.9.18	^-subst.		°C	-40.0	160.0	20.0	
2.3.6.9.19	earthB	earth buffer		0	255	0	
2.3.6.9.20	^-subst.		°C	-40.0	160.0	20.0	
2.3.6.9.21	oModLoc	oMod-local switch		0	255	151	
2.3.6.9.22	^-subst.			0	5	0	
2.3.6.9.23	4.6.72 flow-	flow temp. Heat		0	255	0	
2.3.6.9.24	aHS	add. Heat source		0	255	0	
2.3.6.9.25	solPu	solarPump		0	255	0	
2.3.6.9.26	loadPu	buffer load.pump		0	255	0	

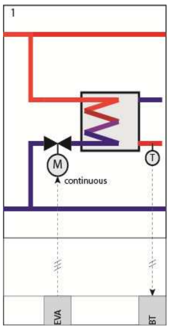
2.3.6.9.27	earthBPu	earth buffer pump		0	255	0	
2.3.6.9.28	zone1	loading zone 1		0	255	0	

Chapter 5 The Domestic hot water circuit modules

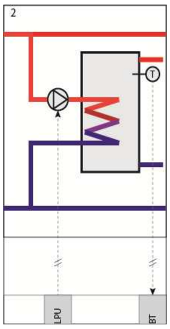
Depending on the selected system diagram, HPM can control 4 different types of domestic hot water. Naturally, the controller is capable of fulfilling many more configurations if the modules are configured manually.

The Domestic hot water circuit modules have the following preprogrammed system diagram configurations.

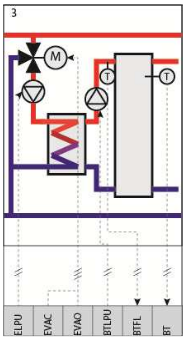
Domestic hot water production in a flow-through system using a fast 0...10V actuator



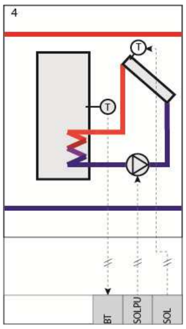
Domestic hot water production in a buffer tank using 2-point control



Domestic hot water production in a buffer tank using a loading system with 3-point actuator (control of loading temperature)



Domestic hot water production with a stand-alone solar system



For all system types, it is possible to activate additional functions such as “thermal disinfection” (anti-legionella circuit). After a cold start or initial installation, an automatic configuration of terminals and parameters is initiated

5.1 Current values (2.4.1.n)

The menu ”Current values” provides an overview of the current values from temperature sensors, digital inputs, meters etc.

- **storage**: Current value of the storage temperature sensor (in the middle of the tank).
- **storage2**: Current value of the additional storage temperature sensor (at the bottom of the tank).
- **system:** Status of the system main switch of this heating circuit module, status of the components of a connected remote control unit for this heating circuit (see below).
- **stoFlow**: Current value of a flow temperature sensor in systems where the storage temperature can be controlled.
- **loadFlow**: Current value of a flow temperature sensor in systems where the loading temperature can be controlled.
- **release**: Current value of the release of the loading pump.
- **solStor**: Current value of the storage temperature sensor in systems with solar thermal collectors.
- **solColl**: Current value of the collector temperature sensor in systems with solar thermal collectors.
- **solRet**: Current value of the return temperature sensor in systems with solar thermal collectors.
- **outdoor**: Current value of the outdoor temperature sensor.
- **limit**: Current value of the limit temperature sensor.
- **shift**: Current value of the shift temperature sensor.
- **volFlow**: Current volumetric flow (measured by the heat meter).
- **heatCapa**: Heat capacity consumed by the domestic hot water circuit (measured by the heat meter).
- **HeAmount**: Heat capacity consumed by the domestic hot water circuit using solar energy.
- **system:** The system main switch (**system)** of the module is responsible for whether the module is running or not. This switch can by used by other modules or the BMS system.
- **oModLoc**: Status of the local system operating switch, which has an influence of the functionality of the heating circuit module.

By connecting a remote control unit, it is possible to change the setpoint from domestic hot water, press an overtime key or change operating mode.

- **setp-poti**: A potentiometer can be used to change the setpoint of the water temperature.
- **key**: Overtime key on the remote control unit for extending the occupation time.

- **oModRC**: Switch for changing the operating mode (can be set to ON/OFF or AUTO/Manual/Off).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.1.1	5.1.73 stor	5.1.74 storage temperature	5.1.75 °				
2.4.1.2	5.1.76 stor	5.1.77 storage temperature2	5.1.78 °				
2.4.1.3	5.1.79 stoF	5.1.80 storage flow temp	5.1.81 °				
2.4.1.4	5.1.82 load	5.1.83 loadFlow temp.	5.1.84 °				
2.4.1.5	5.1.85 rele	5.1.86 releaseTemperature	5.1.87 °				
2.4.1.6	5.1.88 solS	5.1.89 solar-storage temp.	5.1.90 °				
2.4.1.7	5.1.91 solC	5.1.92 solar-collector temp.	5.1.93 °				
2.4.1.8	5.1.94 solR	5.1.95 solar-return temp.	5.1.96 °				
2.4.1.9	5.1.97 outd	5.1.98 outdoor temperature	5.1.99 °				
2.4.1.10	5.1.100 lim	5.1.101 limitation sensor	5.1.102				
2.4.1.11	5.1.103 shi	5.1.104 shift sensor	5.1.105				
2.4.1.12	5.1.106 set	5.1.107 remote setting unit	5.1.108				
2.4.1.13	5.1.109 vol	5.1.110 volumetric flow	5.1.111				
2.4.1.14	5.1.112 he	5.1.113 heat capacity	5.1.114				
2.4.1.15	5.1.115 He	5.1.116 amount of heat					

2.4.1.16	5.1.117 sys	5.1.118 sys.-main switch					
2.4.1.17	5.1.119 ke	5.1.120 RC-key					
2.4.1.18	5.1.121 o	5.1.122 RC-opMode-switch					
2.4.1.19	5.1.123 o	5.1.124 oMod-local switch					

5.2 Setpoint (2.4.2.n)

In the “Setpoints” menu, the storage temperature setpoints for the relevant occupation and non-occupation times are defined. The parameter **boostLoadFl** can be used to replace the setpoint of the storage temperature with the setpoint of the load flow temperature. This ensures that a sufficient amount of hot medium is available at the heat exchanger during loading of the storage. Such boost is effective for both the setpoint for control of the load flow temperature **SP-loadFl** and the current heat demand.

In case of storage loading systems with control sensor on the secondary side of the heat exchanger **boostStorFl** is used to increase the setpoint for the storage flow temperature **SP-storFl**. This accelerates the storage loading process. The loading process is terminated after the storage temperature has reached the valid setpoint. In addition, the current setpoints of the domestic hot water circuit **SP-stor**, **SP-storFl**, **SP-loadFl** are displayed.

- **SP-stor**: Setpoint for the storage tank temperature.
- **SP-storFl:** Setpoint for the flow temperature of the storage tank.
- **SP-loadFl:** Setpoint of the load flow temperature.
- **SP-OTx**: Setpoint of the domestic hot water temperature in occupation time x.
- **SP-NO**: Setpoint of the domestic hot water temperature in non-occupation time.
- **SP-SNOT**: Setpoint of the domestic hot water temperature in special non-occupation time 1.
- **boostStorFL**: Boost for the setpoint of the storage flow temperature.
- **boostLoadFL**: Boost for the setpoint of the loading flow temperature.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.2.1	5.2.125 SP-	5.2.126 setpoint storage	5.2.12	5.2.12	5.2.129		
2.4.2.2	5.2.130 SP-	5.2.131 setpoint controller	5.2.13	5.2.13	5.2.134		

2.4.2.3	5.2.135 SP-	5.2.136 setpoint controller	5.2.13	5.2.13	5.2.139		
2.4.2.4	5.2.140 SP-	5.2.141 setpoint OT1	5.2.14	5.2.14	5.2.144	5.2.145	
2.4.2.5	5.2.146 SP-	5.2.147 setpoint OT2	5.2.14	5.2.14	5.2.150	5.2.151	
2.4.2.6	5.2.152 SP-	5.2.153 setpoint OT3	5.2.15	5.2.15	5.2.156	5.2.157	
2.4.2.7	5.2.158 SP-	5.2.159 setpoint OT4	5.2.16	5.2.16	5.2.162	5.2.163	
2.4.2.8	5.2.164 SP-	5.2.165 setpoint NO	5.2.16	5.2.16	5.2.168	5.2.169	
2.4.2.9	5.2.170 SP-	5.2.171 setpoint SNOT	5.2.17	5.2.17	5.2.174	5.2.175	
2.4.2.10	5.2.176 boos	5.2.177 boost setpoint storage flow	5.2.17	5.2.17	5.2.180	5.2.181	
2.4.2.11	5.2.182 boos	5.2.183 boost setpoint load flow	5.2.18	5.2.18	5.2.186	5.2.187	
2.4.2.12	5.2.188 boos	5.2.189 boost setpoint load flow	5.2.19	5.2.19	5.2.192	5.2.193	

5.3 Additional functions (2.4.3.n)

Priority (2.4.3.2.n)

The “Priority” function permits defining if and how the DHW circuit should have priority over the other heat consumers (e.g. heating circuits), when heat is required. This function is needed when the heat producer is unable to simultaneously flow heat to all heat consumers. Therefore, it is possible to define a reduction of the heating circuits during the loading phase of the storage tank or they can even be switched off during this time.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type**: Type of reduction.

  =1: Heating circuits are switched off completely (switch-off operation)

  =2: Heating circuits are switched off after a delay time

  =3: Heating circuits are reduced to the setpoint defined for the non-occupation time **SP-NO** (see chapter “setpoints”)
- **dest.:** This parameter enables the user to define for which heating circuits the reduction should be effective.
- **timer:** If type=2, a delay time can be defined, after which the heating circuits are switched off.
- **max-red-D:** This parameter is used to set the maximum reduction and/or switch-off time of the heating circuits. If the loading of the storage tank is not completed within this period, the heating circuits return to their normal operation, parallel operation of domestic hot water and heating circuits takes place. After this period has expired a second time the priority function is repeated.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.2.1	active			0	1	0	
2.4.3.2.2	type	1:absolute 2:time dependent 3:reduced NOT		1	3	1	
2.4.3.2.3	dest.	HC21 = 11		0	11	11	
2.4.3.2.4	timer	reducing after hh:mm		00:00	23:59	02:00	
2.4.3.2.5	max-red-D	max duration of reduction		00:00	23:59	02:00	

Controller release (2.4.3.3.n)

When using a solar system in the DHW circuit (solar = 1 in the menu “AF – controller”) the criteria for **hot water loading release** can be defined in this menu. The solar loading is always released. The hot water loading can be blocked, released or reserved according to the current operating and/or non-operating time. In reduced operation the hot water loading is not released until the specified storage temperature difference is exceeded and the delay expired. If the hot water loading is blocked within an operating time, the domestic hot water circuit can only be loaded via the solar system.

- **relHLoadOTx**: Release of hot water loading in occupation time x

  =0: prohibited: No hot water loading release within operating/non-operating time.

  =1: released: Hot water loading released with highest priority within the operating/non-operating time.

  =2: reserved: Release of hot water loading within the operating/non-operating time only if control deviation **Xw-relHload** is exceeded and after expiration of delay time **delRelHload**.
- **relHLoadNO**: Release of hot water loading in non-occupation time.
- **relHLoadSNOT**: Release of hot water loading in special non-occupation time.

- **Xw-relHLoad**: Max. negative control deviation (actual<setpoint) for which hot water loading in reserve operation is released, if exceeded at sensor **storage** and **storage2**.
- **delRelHLoad**: Delay for hot water loading release in reserve operation.
- **timer:** Delay time.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.3.1	relHLoadOT1	0:prohibited 1:release 2:reserve		0	2	1	
2.4.3.3.2	relHLoadOT2	0:prohibited 1:release 2:reserve		0	2	1	
2.4.3.3.3	relHLoadOT3	0:prohibited 1:release 2:reserve		0	2	1	
2.4.3.3.4	relHLoadOT4	0:prohibited 1:release 2:reserve		0	2	1	
2.4.3.3.5	relHLoadNO	0:prohibited 1:release 2:reserve		0	2	1	
2.4.3.3.6	relHLoadSNOT	0:prohibited 1:release 2:reserve		0	2	1	
2.4.3.3.7	Xw-relHLoad	control deviation for backup operation	K	0.5	50.0	10.0	
2.4.3.3.8	delRelHLoad	delay for backup operation	min	1	600	10	
2.4.3.3.9	timer	curTimer	min				

Universal limitation (2.4.3.4.n)

The menu “Universal limitation” is used to limit the temperature at the limitation sensor **lim** to a maximum or minimum value.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type**: Depending on this parameter, the function can be used optionally for maximum or minimum limitation and the actuator can either open or close in case of the violation of the limit.

  *=1***:** Maximum limitation, open valve.

  *=2***:** Maximum limitation, close valve.

  *=3***:** Minimum limitation, close valve.

  *=4***:** Minimum limitation, open valve.

- **shift**: The limit can be defined as fixed value or as shifting value with a variable shifting curve according to the value of the sensor input **shift**. (see chapter 0 for the definition of the shifting sensor **shift**)

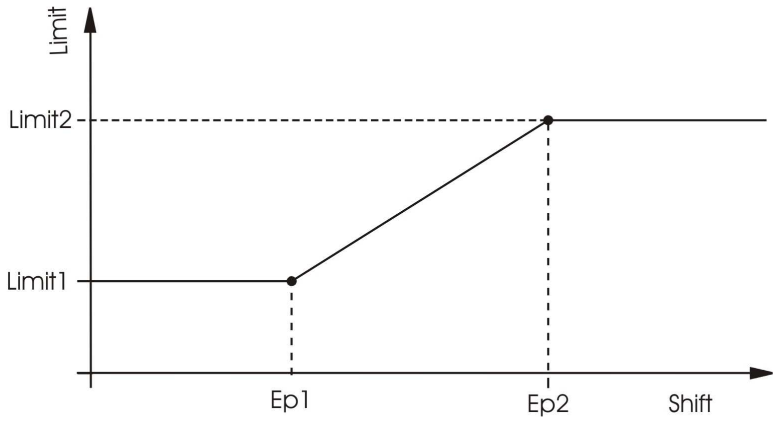
  *=0***:** Fixed value (constant).

  *=1***:** Shifting value from shifting sensor.

- **limit1, limit2, EP1, EP2:**

  shift=0: If the current limit **limit1** at the limitation sensor is violated, the return limitation takes over the control of the actuator from the flow temperature controller.

  shift=1: If the curve between **limit1/EP1** and **limit2/EP2** is violated, the return limitation takes over the control of the actuator from the flow temperature controller.



Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.4.1	active			0	1	0	
2.4.3.4.2	type	1:max open 2:max close 3:min close 4:min		1	4	2	
2.4.3.4.3	shift	shift with shift sensor		0	1	0	
2.4.3.4.4	limit1	limit value 1	°C	-999999.9	999999.9	0.0	
2.4.3.4.5	EP1	entry point 1	°C	-999999.9	999999.9	0.0	
2.4.3.4.6	limit2	limit value 2	°C	-999999.9	999999.9	0.0	
2.4.3.4.7	EP2	entry point 2	°C	-999999.9	999999.9	0.0	

Remote control (2.4.3.5.n)

In this menu, the functionality of a connected remote control unit can be defined.

- **activeOTx**: This parameter activates (1) or deactivates (0) the remote control unit for occupation time x.
- **keyEff**: Function of the key.
  =1: Overtime function: The current occupation time is prolonged by the time defined in the parameter **duratOvertime** or an additional occupation time with the setpoint of OT1 is actuvated with the same duration.
  =2: The storage tank is loaded with the setpoint of the current occupation time or OT1.
- **duratOvertime:** Overtime duration.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.5.1	active OT1	activation poti OT1		0	1	1	
2.4.3.5.2	active OT2	activation poti OT2		0	1	1	
2.4.3.5.3	active OT3	activation poti OT3		0	1	1	
2.4.3.5.4	active OT4	activation poti OT4		0	1	1	
2.4.3.5.5	keyEff	1:ovTiFct 2:loadStor		1	2	2	
2.4.3.5.6	duratOvertime	duration for overtime	h	0.0	24.0	1.0	

Signal (2.4.3.6.n)

The “Signal” function permits configuring in what way a signal should be handled. When this function is activated, it is possible to decide what should happen if:

a) An error occurs.
b) A trouble indication occurs.
c) A limit is exceeded.
d) A timer has run down (maintenance, etc.).

- **active**: This parameter activates (1) or deactivates (0) the function.
- **TI-all**: This parameter permits configuring how the controller should react.

TI-all	setting of the trouble indication outputs troubleInd (see menu “SF – terminal assignement”).		
		indication as an alarm in the central building management software (EXOscada or IRMA),	
			indication on the controller display
0			
1			x
2		x	
3		x	x
4	x		
5	x		x
6	x	x	
7	x	x	x

- **Xw-stor**: This parameter permits setting the maximum deviation of the storage tank temperature between setpoint and current value before a trouble indication signal is sent.
- **Xw-storFl**: This parameter permits setting the maximum deviation of the flow temperature of the storage tank between setpoint and current value before a trouble indication signal is sent.
- **Xw-loadFl**: This parameter permits setting the maximum loading temperature deviation between setpoint and current value before a trouble indication signal is sent.
- **uLimStor**: This parameter permits setting the upper limit of the storage temperature. When violated, a trouble indication signal is sent.
- **uLimColl**: This parameter permits setting the upper limit of the solar collector temperature. When violated, a trouble indication signal is sent.
- **Xw-stor**: This parameter permits delaying the trouble indication signal for the storage tank temperature deviation.
- **Xw-storFl**: This parameter permits delaying the trouble indication signal for the flow temperature deviation for the storage tank.
- **Xw-stor**: This parameter permits delaying the trouble indication signal for the loading temperature deviation for the storage tank.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.6.1	active			0	1	0	
2.4.3.6.2	TI-all	TI to BMS:2,3,6,7 TI-A:4-7 disp:1,3,5		0	7	1	
2.4.3.6.3	Xw-stor	max. contrDev. Storage	K	0.5	50.0	50.0	
2.4.3.6.4	Xw-storFl	max. contrDev. Storage flowT	K	0.5	50.0	50.0	
2.4.3.6.5	Xw-loadFl	max. contrDev. loadFlow	K	0.5	50.0	50.0	
2.4.3.6.6	uLimStor	temp-upper limit storage	°C	50.0	160.0	80.0	
2.4.3.6.7	uLimColl	temp-upper limit solar-collector	°C	50.0	160.0	130.0	
2.4.3.6.8	Xw-stor	delayed contrDev. Storage	min	0	600	600	
2.4.3.6.9	Xw-storFl	delayed contrDev. storageFlow	min	0	600	600	
2.4.3.6.10	Xw-loadFl	delayed contrDev. loadFlow	min	0	600	600	

Frost protection (2.4.3.7.1)

In order to prevent damage to the domestic hot water circuit, the “Frost protection” function is always active (unless the controller is in “non-active” or “manual” mode).

- **outdoorfrLi:** This parameter permits setting the outdoor temperature at which the pump of the domestic hot water circuit will be started. The water in the pipes is kept in motion even if the domestic hot water circuit does not require any heat, thereby preventing freezing.

This should prevent freezing of pipes running at the outdoor wall of the building. The frost protection mode stops when the outdoor temperature is 1 K above the value set in the parameter **outdoorfrLi**.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.7.1	outdoorfrLi	outdoor-frost limit	°C	-50.0	50.0	2.0	

Setpoint limitation (2.4.3.8.n)

The “Setpoint limitation” function can be used to specify a maximum limit for the *calculated* storage tank temperature setpoint and/or the maximum flow temperature requested by the domestic hot water circuit.

- **maxSP**: Maximum limit for the calculated storage tank temperature.
- **maxDemFl**-T: Maximum flow temperature requested by the domestic hot water circuit.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.8.1	maxSP	max. setpoint storage temperature	°C	2.0	160.0	55.0	
2.4.3.8.2	maxDemFl-T	max. flow temp.demand	°C	0.0	160.0	90.0	

Controller (2.4.3.9.n)

The parameters in the menu “Controller” are set automatically, depending on the selected system diagram of the domestic hot water circuit (storage with internal or external heat exchanger, instant water heater, solar). Naturally, these parameters can also be manually adjusted as desired.

- **sys-type**: Type of domestic hot water control.

  =2: DHW system with buffer loading pump.

  =5: DHW system with external loading pump + valve + buffer loading pump.

  =7: DHW system with external loading pump.
- **va-outp**: If **sys-type**=5, the control signal can be set to the valve.

  =1: Continuous signal (0..10 V).

  =3: 3-point control.
- **solar**: Solar system included.

  =0: No.

  =1: Yes, solar pump included.
- **lonDes**: Freely adjustable name for the DHW circuit.

If a system diagram without any valve or solar system is selected, only for these components later to be added manually, the required input and output terminals have to be assigned in the menu “service – terminal assignment”.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.9.1	sys-type	2:buffLPu 7:exLPu 5:exLPu+Va+buffLPu		1	7	2	
2.4.3.9.2	va-outp	1:continuous 3:3pnt		1	3	3	
2.4.3.9.3	solar	0:none 1:solar pump		0	1	0	
2.4.3.9.4	longDes	free adjustab. Prog.-longRef.					

Forced load (2.4.3.10.n)

The “Forced load” function was developed to prevent loading of the DHW tank from failing during the heat-up phase of the heating circuits, thereby preventing the rooms from warming up (e.g. in large apartment blocks). This function forces the system to heat up the DHW tank immediately before an occupation time of the heating circuits is initiated.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **HCx**: These parameters permit deciding for which heating circuit the function is enabled.

  =0: No forced load before heat-up phase of heating circuit x.

  =1: Forced load of DHW storage tank before heat-up phase of heating circuit x.
- **duration**: This parameter permits setting how long before the start of the heat-up phase of the heating circuits the forced loading should take place.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.10.1	active			0	1	0	
2.4.3.10.2	HC1			0	1	1	
2.4.3.10.3	HC2			0	1	1	
2.4.3.10.4	duration		min	0	60	60	

Thermal disinfection (2.4.3.15.n)

The “Thermal disinfection” function prevents the build-up of bacteria (Legionella) in drinking water by heating up the temperature in the storage tank to at least 60°C.

If required, the storage setpoint **SP-stor**, the storage flow setpoint **SP-storFl** and the load flow setpoint **SP-loadFl** can be adjusted in order to achieve shorter heat-up times or higher storage temperatures.

When activated, thermal disinfection will be performed at regular intervals, according to the configuration of the parameters **Day** (day of the week, 0 = daily) and **Time** (start time).

During thermal disinfection, the switch output **thermDOn** (see chapter “service – terminal assignment”) is triggered in order to, for example, switch an additional heat source on (electric heating, magnet valve for additional volume). If the storage temperature reaches the specified setpoint, all fittings (taps, shower heads, etc.) should be cleaned. For this purpose, the thermal disinfection triggers the switch output **thermDbuf** for the running time set in the parameter **Duration**. By opening a magnetic valve, automatic cleaning is possible. Alternatively, a warning device (lamp, horn, etc.) can request cleaning. During the cleaning process, the storage setpoint of the thermal disinfection remains effective. If required, the storage tank will be reloaded.

If the circulation pump is supposed to run during thermal disinfection, the circulation pump has to be released (see chapter “AF – circulation pump”) with the parameter **relCircThD** = 1.

The current storage temperature **cur-stor** and the related **time** and **date** are displayed in the menu “SF – thermal disinfection”.

If the specified storage setpoint **SP-stor** is not reached within 2 hours after start of this function, the trouble indication switch is triggered with the status “thermal disinfection” and the function is interrupted. The display message “Fault DHW-circuit” can be reset by means of parameter **Reset-mess** (see chapter “SF – thermal disinfection”).

If the power flow of the controller fails during the thermal disinfection (power failure, voltage fluctuations, reset/hot start), the controller automatically repeats the function after restart, if it has not been completed or if a time of 2 hours after start (basis: 01:00am + 2h = until 03:00am) has been exceeded.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **SP-stor**: Setpoint for storage tank temperature during thermal disinfection.
- **SP-storFl**: Setpoint for flow temperature for the storage tank during thermal disinfection.

- **SP-loadFl**: Setpoint for loading temperature for the storage tank during thermal disinfection.
- **day**: Day(s) on which thermal disinfection should take place.

  =0: Everyday.

  =1: Monday.

  =2: Tuesday.

  …

  =7: Sunday.
- **time**: Time of the day when thermal disinfection should start.
- **duration**: Running time of thermal disinfection.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.15.1	active			0	1	0	
2.4.3.15.2	SP-stor	storage-setpoint therm.desinfection	°C	60.0	160.0	70.0	
2.4.3.15.3	SP-storFl	storage flow-SP therm.desinfection	°C	60.0	100.0	75.0	
2.4.3.15.4	SP-loadFl	loadFlow-SP Therm.desinfection	°C	60.0	160.0	85.0	
2.4.3.15.5	day	0:daily 1-7:Monday-Sunday		0	7	1	
2.4.3.15.6	time	time beginning therm. Desinfection		00:00	23:59	01:00	
2.4.3.15.7	duration	running time/cleaning therm. Desinfektion	min	0	240	0	

Circulation pump (2.4.3.16.n)

This menu permits defining at what point an installed circulation pump should run. This function can only be used if a terminal is assigned in the parameter **circPu** (see chapter “SF – terminal assignment”).

- **relCircOTx**: Release of circulation pump in occupation time x.
- **relCircNO**: Release of circulation pump in non-occupation time.
- **relCircSNOT**: Release of circulation pump in the special non-occupation time.
- **relCircThD**: Release of circulation pump during thermal disinfection.
- **relCircLoad**: Release of circulation pump while loading of the storage tank.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.16.1	5.3.194 relCir	5.3.195 circulation OT1		5.3.19	5.3.19	5.3.198	
2.4.3.16.2	5.3.199 relCir	5.3.200 circulation OT2		5.3.20	5.3.20	5.3.203	
2.4.3.16.3	5.3.204 relCir	5.3.205 circulation OT3		5.3.20	5.3.20	5.3.208	
2.4.3.16.4	5.3.209 relCir	5.3.210 circulation OT4		5.3.21	5.3.21	5.3.213	

2.4.3.16.5	5.3.214 relCir	5.3.215 circulation NO		5.3.21	5.3.21	5.3.218	
2.4.3.16.6	5.3.219 relCir	5.3.220 circulation SNOT		5.3.22	5.3.22	5.3.223	
2.4.3.16.7	5.3.224 relCir	5.3.225 circ. Therm. Des.		5.3.22	5.3.22	5.3.228	
2.4.3.16.8	5.3.229 relCir	5.3.230 circ. When loading		5.3.23	5.3.23	5.3.233	

Pump/valve exercise (2.4.3.17.1)

The "Pump/valve exercise" function automatically detects if the actuators (pump, valve) have moved due to having undertaken any control tasks in the last 24 hours. If this is not the case the pump, followed by the valve, is triggered for an adjustable interval at 11 o'clock each day, thereby preventing blocking in the actuator and the pump.

- **duration:** This parameter permits setting how long the pump and actuator should run. This time should be at least equal to the running time of the valve actuator motor in order to ensure that the district heating valve is moved over the total lift.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.17.1	5.3.234 d	running time pump/valve exercise	5.3.23	5.3.23	5.3.23	5.3.238	

Capacity limitation (2.4.3.18.n)

The "capacity limitation" function permits limitation of either the capacity output to the heating circuit or of the volumetric flow in the heating circuit. This function requires a heat meter to be installed and connected, providing the currently used capacity or the current volumetric flow. Limitation affects the signal to the valve of the heating circuit.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type**: This parameter permits setting the type of limitation.

  =1: Capacity limitation.

  =2: Volumetric flow limitation.
- **limCapa**: This parameter permits setting the limit of the capacity.
- **lim1olFl:** This parameter permits setting the limit for the volumetric flow.
- **Kp**: This parameter represents the amplifier gain for the actuator.

- **maxCorr**: This parameter permits setting the maximum allowed setpoint correction.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.18.1	active			0	1	0	
2.4.3.18.2	type	1:capa 2:volFlow		1	2	1	
2.4.3.18.3	limCapa	limit1 capacity	kW	0.0	3200.0	3200.0	
2.4.3.18.4	limVolFl	limit1 volumet.	l/h	0	999999	999999	
2.4.3.18.5	Kp	amplifier gain	K/%	0.1	99.9	1.0	
2.4.3.18.6	maxCorr	max. SP correction	K	-100.0	0.0	50.0	

Solar statistics (2.4.3.21.1)

If a solar system is used in the DHW production, it is usually desirable to know what amount of energy is produced by the sun. HPM is capable of calculating this figure with precision. The “Solar statistics” menu permits activation of this function. Both the values needed for calculation as well as the results can be found in the service section of this function (see chapter “SF – solar statistics”)

- **active**: This parameter activates (1) or deactivates (0) the function.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.3.21.1	active			0	1	0	

5.4 Status (2.4.4.n)

The “status” menu provides an overview of the current operating mode of the DHW circuit.

When the main status **opStatus** or the auxiliary status **opStatCode** is generated, every function affecting the operation of the DHW circuit is taken into consideration. The **opStatCode** is a hexadecimal number. The meaning of the characters shown in the parameters **opStatCode** and **eStatCode** is explained in the following tables.

- **opStatus**: Main status of the DHW circuit, controller status + timer status, e.g. “nominal operation OT1”.
  = *Not active/Off:* Required inputs are not assigned or operating mode switch **oModRem** or **oModLoc** is assigned and in position Off.
  = *Frost protection*: Value at sensors **storage** or **storage2** are below frost limit.
  = *Switch-off operation*: System main switch **system** is assigned and switched off.
  = *Therm. Des.:* Thermal disinfection active.
  = *Heating-up operation*: Storage loading via remote control by means of **key** and **keyEff** = 2.
  = *Nominal operation*: Status of timer channel **OT1...OT4** or **SOT1...SOT4**.
  = *Manual operation*: Manual control for an output or operating mode switch **oModRem** or **oModLoc** assigned and in position *manual.*
- **trouble**: Trouble indication status of the DHW circuit.
- **source:** Setpoint source for the DHW circuit.
  = timer OT1…4: Setpoint for operating time **SP-OT1…4**, **status OT1…4**.
  *= timer NO***:** Setpoint for non-occupation time **SP-NO**.
  *= timer SNOT***:** Setpoint for special non occupation time **SP-SNOT**.
  *= xxxxxxxx R***:** Setpoint compensated by remote control / overtime key.
  *= xxxxxxxx S***:** Setpoint compensated by operating mode switch.
  *= xxxxxxxx L***:** Setpoint compensated by setpoint limitation.
- **va-cont**: Current position of the valve in percent (0..100 %).
- **va-open:** Current open signal to 3-point actuator.
- **va-close:** Current close signal to 3-point actuator.
- **loadPu**: Current signal to loading pump.
- **exchPu**: Current signal to exchanger pump.

- **stoPu**: Current signal to storage tank loading pump.
- **stoUnloa**: Signal if storage tank is being unloaded.
- **solPu**: Current signal to the solar pump.
- **circPu**: Current signal to circulation pump.
- **thermDon**: Signal indicating if thermal disinfection is active.
- **thermDBuf**: Signal if tabs should be cleaned.
- **troubInd**: Pending trouble indication.
- **Hea-dem**: Current variable demand via voltage input.
- **opStatCode**: The auxiliary operating status consists of seven digits, as various functions of the domestic hot water circuit can be effective at the same time. The example below, together with the translation table, explains how to decode the sequence of characters.

	1st digit	2nd digit	3rd digit	4th digit	5th digit	6th digit	7th digit
1	RC-ovKey4h	setpoint limitation	throughput	solarHold	Outdoor-frost limit	switch-on delay release controller	capacity limitation
2	RC-button	oMod-remote switch	cooling	solar loading	flow frost limit	switch-on delay loading	circulation pump
3	RC-ovKey4h RC-button	setpoint limitation, oMod-remote switch	throughput, cooling	solarHold, solar loading	outdoor frost limit, flow frost limit	switch-on delay release controller, switch-on delay loading	capacity limitation, circulation pump
4	sys.-main switch	oMod-local switch	hold	2Pnt-switch-off exchanger load pump	frost protection collector	pump/valve exercise	switch-off delay and storage pump
5	sys.-main switch, RC-ovKey4h	oMod-local switch, setpoint limitation	hold, throughput	2Pnt-switch-off exchanger load pump, solarHold	frost protection collector, outdoor frost limit	pump/valve exercise, switch-on delay release controller	switch-off delay storage pump capacity limitation
6	sys.-main switch, RC-button	oMod-local switch, oMod-remote switch	hold, cooling	2Pnt-switch-off exchanger load pump, solar loading	frost protection collector, flow frost limit	pump/valve exercise, switch-on delay loading	switch-off delay storage pump, circulation pump
7	sys.-main switch, RC-ovKey4h, RC-button	oMod-local switch, setpoint limitation, oMod-remote switch	hold throughput, cooling	2Pnt-switch-off exchanger load pump, solarHold, solar loading	outdoor frost limit, flow frost limit	pump/valve exercise, switch-on delay release controller, switch-on delay loading	switch-off delay storage pump, capacity limitation, circulation pump
8		RC-oMod-switch	loading	unloading	cooling collector	universal limitation	switch-off delay loading
9		RC-oMod-switch, setpoint limitation	loading, throughput	unloading, solarHold	cooling collector, outdoor frost limit	universal limitation, switch-on delay release controller	switch-off delay loading, storage pump capacity limitation
A		RC-oMod-switch, oMod-remote switch	loading, cooling	unloading, solar loading	cooling collector, flow frost limit	universal limitation, switch-on delay loading	switch-off delay loading, circulation pump
B		RC-oMod-switch, setpoint limitation, oMod-remote	loading, throughput, cooling	unloading, solarHold, solar loading	cooling collector, outdoor frost limit, flow frost limit	universal limitation, switch-on delay release controller,	switch-off delay loading, capacity limitation,

		switch				switch-on delay loading	circulation pump
C		RC-oMod-switch, oMod-local switch	loading, hold	unloading, 2Pnt-switch-off exchanger load pump	cooling collector, frost protection collector	universal limitation, pump/valve exercise	switch-off delay loading, switch-off delay storage pump
D		RC-oMod-switch, oMod-local switch, setpoint limitation	loading, hold, throughput	unloading, 2Pnt-switch-off exchanger load pump, solarHold	cooling collector, frost protection collector, outdoor frost limit	universal limitation, pump/valve exercise, switch-on delay release controller	switch-off delay and loading, switch-off delay storage pump, capacity limitation
E		RC-oMod-switch, oMod-local switch, oMod-remote switch	loading, hold, cooling	unloading, 2Pnt-switch-off exchanger load pump, solar loading	cooling collector, frost protection collector, flow frost limit	universal limitation, pump/valve exercise, switch-on delay loading	switch-off delay loading, switch-off delay storage pump, circulation pump
F		RC-oMod-switch, oMod-local switch, setpoint limitation, oMod-remote switch	loading, hold, throughput, cooling	unloading, 2Pnt-switch-off exchanger load pump, solarHold, solar loading	cooling collector, frost protection collector, outdoor frost limit, flow frost limit	universal limitation, pump/valve exercise, switch-on delay release controller, switch-on delay loading	switch-off delay loading, switch-off delay storage pump, capacity limitation, circulation pump

Tab. 10: Operating status code domestic hot water circuit

Example: Display: opStatCode: = 0080100

Meaning: 3rd digit = loading of storage active, 5th digit = below frost limit outdoor

- **sStatCode**: The auxiliary trouble indication status consists of three digits, as several trouble indications can occur at the same time. The example below, together with the translation table, explains how to decode the sequence of characters.

	1st digit	**2nd digit**	**3rd digit**
1	Frost protection storage	Therm. disinfection failed	error input
2		max. contrDev. load flow	input terminal
3		Therm. disinfection failed, max. contrDev. load flow	error input, input terminal
4		max. contrDev. storage flow	temp. upper limit solar collector
5		max. contrDev. storage flow, Therm. disinfection failed	temp. upper limit solar collector, error input
6		max. contrDev. storage flow, max. contrDev. load flow	temp. upper limit solar collector, input terminal
7		max. contrDev. storage flow, Therm. disinfection failed, max. contrDev. load flow	temp. upper limit solar collector, error input, input terminal
8		max. contrDev. storage	temp.-upper limit storage
9		max. contrDev. storage, Therm. disinfection failed	temp.-upper limit storage, error input
A		max. contrDev. storage, max. contrDev. load flow	temp.-upper limit storage, input terminal

B		max. contrDev. storage, Therm. disinfection failed, max. contrDev. load flow	temp.-upper limit storage, error input, input terminal
C		max. contrDev. storage, max. contrDev. storage flow	temp.-upper limit storage, temp. upper limit solar collector
D		max. contrDev. storage, max. contrDev. storage flow, Therm. disinfection failed	temp.-upper limit storage, temp. upper limit solar collector, error input
		max. contrDev. storage, max. contrDev. storage flow, max. contrDev. load flow	temp.-upper limit storage, temp. upper limit solar collector, input terminal
		max. contrDev. storage, max. contrDev. storage flow, Therm. disinfection failed, max. contrDev. load flow	temp.-upper limit storage, temp. upper limit solar collector, error input, input terminal

Tab. 11: Trouble indication code domestic hot water circuit

Example: Display: sStatCode: = 0B0,

Meaning: 2nd digit = Maximum admissible control deviation of storage temperature and load flow temperature exceeded, therm. disinfection failed.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.4.1	5.4.239 opStatus	5.4.240 OpStatus					
2.4.4.2	5.4.241 trouble	5.4.242					
5.4.243 2.4.4.3	5.4.244 source	5.4.245 setpoint source and – compensation					
2.4.4.4	5.4.246 va-cont	5.4.247 valve continuous	5.4.24				
2.4.4.5	5.4.249 val-open	5.4.250 valve open					
2.4.4.6	5.4.251 val-close	5.4.252 valve close					
2.4.4.7	5.4.253 loadPu	5.4.254 loadPump					
2.4.4.8	5.4.255 exchPu	5.4.256 exchanger load pump					
5.4.257 2.4.4.9	5.4.258 stoPu	5.4.259 storage load pump					
2.4.4.10	5.4.260 stoUnloa	5.4.261 storage unloading					
2.4.4.11	5.4.262 solPu	5.4.263 solarPump					
2.4.4.12	5.4.264 circPu	5.4.265 circulation pump					
2.4.4.13	5.4.266 thermDOn	5.4.267 therm. Desinfection					
2.4.4.14	5.4.268 thermDBuf	5.4.269 tab cleaning					
2.4.4.15	5.4.270 troubInd	5.4.271 trouble indication					
2.4.4.16	Hea-dem	Hea-dem	5.4.27				
2.4.4.17	5.4.273 opStatCo	5.4.274					

	d e						
5.4.275 2.4.4.18	5.4.276 eStatCod e	5.4.277					

5.5 Manual operation (2.4.5.n)

Manual operation is used during commissioning in order to check the heating circuit pump and the valve or mixer drive for proper operation, control direction and sense of rotation

NOTE:

In case of improper handling, the manual operating mode can cause damages to the system! Manual control overrides all functional limitations, including pump/valve exercise, frost protection and monitoring and signal functions.

- **valve**: Control of valve (continuous): 0..100=0..100%, 101=AUTOMATIC (operated by program).
- **valve:** Control of valve (3-point): 0=close, 1=open, 2=stop, 3=AUTOMATIC (operated by program).
- **loadPu**: Loading pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **exchPu**: Heat exchanger pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **stoTanPu**: Storage tank pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **stoTanUnl**: Unloading storage tank signal (could be a magnetic valve): 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **solPu**: Solar pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **circPu**: Circulation pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **thermDOn**: Signal indicating that thermal disinfection is active: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **thermDBuf**: Signal, that the tabs should be opened during thermal disinfection: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.5.1	valve	0-100:man 101:automatic	%	0	101	101	
2.4.5.2	valve	0:close 1:open 2:stop 3:automatic		0	3	3	
2.4.5.3	loadPu	0:off 1:on 3:automatic		0	3	3	
2.4.5.4	exchPu	0:off 1:on 3:automatic		0	3	3	
2.4.5.5	storTanPu	0:off 1:on 3:automatic		0	3	3	
2.4.5.6	storTanUnl	0:off 1:on 3:automatic		0	3	3	
2.4.5.7	solPu	0:off 1:on 3:automatic		0	3	3	
2.4.5.8	circPu	0:off 1:on 3:automatic		0	3	3	
2.4.5.9	thermDOn	0:off 1:on 3:automatic		0	3	3	
2.4.5.10	thermDBuf	0:off 1:on 3:automatic		0	3	3	

5.6 Service function (2.4.6.n)

The domestic hot water circuit service menu permits trained personnel to configure modules, assign inputs and outputs and to set individual parameters for additional functions. This menu also displays data of interest to maintenance and service personnel.

General values (2.4.6.1.n)

The “General values” menu offers data of interest to maintenance and service personnel.

- **loadPu:** Running time of the loading pump since installation.
- **exchPu:** Running time of the lheat exchanger pump since installation.
- **storPu:** Running time of the storage tank pump since installation.
- **solPu:** Running time of the solar pump since installation.
- **circPu:** Running time of the circulation pump since installation.

All these parameters can also be set manually in case the controller or a pump has to be exchanged.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.1.1	loadPu		h	0	999999	0	
2.4.6.1.2	exchPu		h	0	999999	0	
2.4.6.1.3	storPu		h	0	999999	0	
2.4.6.1.4	solPu		h	0	999999	0	
2.4.6.1.5	circPu		h	0	999999	0	

Priority (2.4.6.2.n)

The “Priority” service function displays information of interest when the priority of the domestic hot water circuit is active.

- **timer:** This parameter shows the running time since the loading of the DHW circuit has started and priority is set to “time dependent priority” **type**=2 (see chapter “AF – priority”).
- **max-red-D**: This parameter shows the actual value of the duration, the heating circuits have been switched off.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.2.1	timer						
5.6.278 2.4.6.2.2	5.6.279 max-red-D	5.6.280					

Universal limitation (2.4.6.4.n)

If the current limit **curLim** is violated by the value from the limitation sensor **limit** (see chapter “SF – terminl assignment”), the universal limitation takes over control of the actuator from the flow temperature controller. The universal limitation control behaviour can be adjusted to the control system characteristics by means of the controller parameters

- **Xp:** PID-proportional band. An increase will reduce the step response of the P proportion.
- **Tn:** PID-integral action time. An increase will extend the I-rate in time and thus reduce the impact. Tn=121.0min will deactivate the I-rate.

The following parameters show the current status of the universal limitation:

- **curLim**: This parameter shows the currently valid limit of the universal limitation (result from the parameters set in chapter “AF – universal limitation”)

- **Y-uniLim:** This parameter shows the current correcting variable (signal to the actuator).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.4.1	Xp	proportional band		0.1	500.0	100.0	
2.4.6.4.2	Tn	integral action time	min	0.1	121.0	1.0	
2.4.6.4.3	curLim	cur limit value	°C	-999999.9	999999.9		
2.4.6.4.4	Y-uniLim	correcting variable universal limitation	%	0.0	100.0		

Signal (2.4.6.6.n)

The last 10 errors detected by the DHW circuit module are recorded and displayed in the "Signal" service menu. The first parameter always shows the most recent error while the last parameter shows the oldest error. The parameter text displayed constitutes the short text of the detected error. When pressing "OK", the info text appears, providing a more detailed description of the error. The date and time at which the error was detected is documented. **Please note that a power failure or controller cold start will delete all recorded errors!**

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.6.1		--.--.-- --:--					
2.4.6.6.2		--.--.-- --:--					
2.4.6.6.3		--.--.-- --:--					
2.4.6.6.4		--.--.-- --:--					
2.4.6.6.5		--.--.-- --:--					
2.4.6.6.6		--.--.-- --:--					
2.4.6.6.7		--.--.-- --:--					
2.4.6.6.8		--.--.-- --:--					
2.4.6.6.9		--.--.-- --:--					
2.4.6.6.10		--.--.-- --:--					

Explanation:

Parameter text:	FP system	Frost protection system, flow temperature below frost limit
	FP-stor	Frost protection storage, storage temperature below frost limit
	uLimColl	Maximum limit solar collector temperature exceeded
	uLimStor	Maximum limit storage temperature exceeded
	uLimFlow	Maximum limit flow temperature exceeded
	TI system	Trouble indication system detected (input TI-system = 1)
	Defective sensor	Sensor fault detected
	ThermDes	Thermal disinfection failed
	Xw-loadFl	max. control deviation load flow temperature exceeded
	Xw-stor	max. control deviation storage temperature exceeded
	Xw-storFl	max. control deviation storage flow temperature exceeded
Info text	<date, time>	e.g.: the error was recorded on 18.02.06 at 13:57

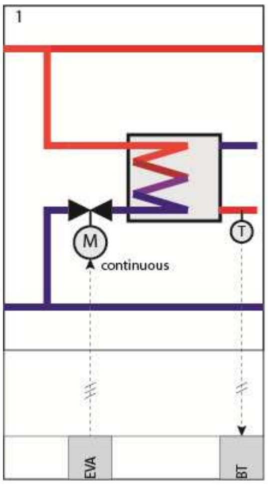
Controller (2.4.6.9.1)

The control behaviour of the temperature control can be changed by adjusting the parameters.

- **va-TMot:** The actual running time of the actuator should be set accordingly.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.9.1	va-TMot	actuator running time	s	10	600	120	

Storage tank (2.4.6.11.n)



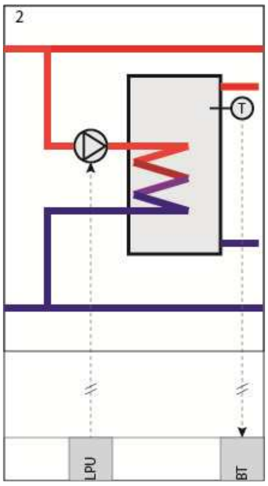
This function generally applies to all system types in which the storage tank temperature is controlled by means of a two-point control. A higher switch difference **switchD** can prevent frequent reloading.

For **sys-typ** = 1 the storage temperature is controlled continuously during loading by triggering the valve in the loading circuit. The behaviour of the PI controller can be compensated by adjusting parameters **Xp, Tn+, Tn-**.

- **switchD:** Switching difference for storage loading.
- **sys-typ**=1...3: Pos. switch difference (loading On for storage and storage2 < SP-stor, loading Off for storage and storage2 >= SP-stor + switchD).
- **sys-typ**=4...7: Neg. switch difference (loading On for storage and storage2 < SP-stor - switchD, loading Off for storage and storage2 >= SP-stor).
- **XP:** Proportional band, an increase will reduce the step response of the P proportion.
- **TN+:** Integral action time in case of a positive control deviation, an increase will extend the I rate in time and thus reduce the impact on the **closing** of the valve (Actual value > setpoint).
- **TN-**: Integral action time in case of a negative control deviation, an increase will extend the I rate in time and thus reduce the impact on the **opening** of the valve (Actual value < setpoint).
- **Y-stor**: Control correcting variable.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.11.1	switchD	switch difference storage load	K	0.1	50.0	5.0	
2.4.6.11.2	Xp	proportional band	K	0.1	500.0	25.0	
2.4.6.11.3	Tn+	integral action time curVal>setpoint va-c	min	0.1	121.0	1.0	
2.4.6.11.4	Tn-	integral action time curVal<setpoint va-o	min	0.1	121.0	1.0	
2.4.6.11.5	Y-stor	contr corr variable	%				

Storage flow (2.4.6.12.n)

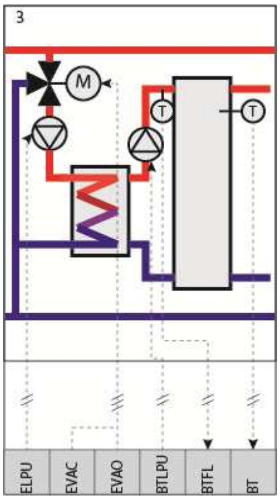


For **sys-typ** = 2, the storage flow temperature **storFl** is controlled by means of setpoint **SP-storFl** during storage loading. This setpoint is the sum of **SP-stor** and the boost factor **boost StorFl**. The controller performs a two-point control with internal heat exchanger and loading pump **exPu**

- **switchD:** Switch difference for exchanger load pump for sys-typ=7 (exPu On for storFl < SP-storFl, exPu Off for storFl >= SP-storFl + switchD).
- **XP:** Proportional band, an increase will reduce the step response of the P proportion.
- **TN+:** Integral action time in case of a positive control deviation, an increase will extend the I rate in time and thus reduce the impact on the **closing** of the valve (Actual value > setpoint).
- **TN-**: Integral action time in case of a negative control deviation, an increase will extend the I rate in time and thus reduce the impact on the **opening** of the valve (Actual value < setpoint).
- **Y-storFL**: Control correcting variable.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.12.1	switchD	switch difference exchanger loadpump	K	0.0	50.0	10.0	
2.4.6.12.2	Xp	proportional band	K	0.1	500.0	100.0	
2.4.6.12.3	Tn+	integral action time (+xw)	min	0.1	121.0	1.0	
2.4.6.12.4	Tn-	integral action time (-xw)	min	0.1	121.0	1.0	
2.4.6.12.5	Y-storFl	contr corr variable	%				

Load flow (2.4.6.13.n)



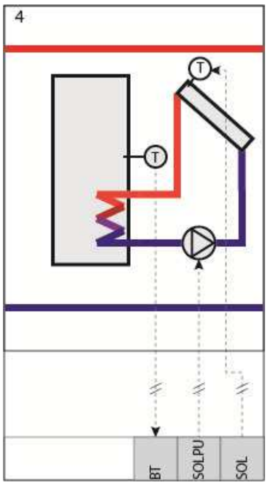
Buffer tank controlled via heat excahnger. See above

For **sys-typ** = 5, the load flow temperature **loadFl** is controlled by means of setpoint **SP-loadFl** during storage loading. This setpoint is the sum of **SP-stor** and the boost **boost StorFl**. By changing the parameters **Xp, Tn+, Tn-**, control of the control valve can be compensated.

- **XP:** Proportional band, an increase will reduce the step response of the P proportion.
- **TN+:** Integral action time in case of a positive control deviation. An increase will extend the I rate in time and thus reduce the impact on the **closing** of the valve (Actual value > setpoint).
- **TN-**: Integral action time in case of a negative control deviation. An increase will extend the I rate in time and thus reduce the impact on the **opening** of the valve (Actual value < setpoint).
- **Y-loadFL**: Control correcting variable.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.13.1	Xp	proportional band	K	0.1	500.0	100.0	
2.4.6.13.2	Tn+	integral action time (+xw)	min	0.1	121.0	1.0	
2.4.6.13.3	Tn-	integral action time (-xw)	min	0.1	121.0	1.0	
2.4.6.13.4	Y-loadFl	contr corr variable	%				

Solar circuit (2.4.6.14.n)



The solar circuit can be used either for heating the domestic hot water storage tank or in order to heat up the buffer tank. If solar is used to heat up the buffer tank, it is possible to use tanks that have 2 zones or to use 2 buffer tanks for different purposes (e.g. heating or DHW production). If a buffer tank with 2 zones is being used, its lower zone is normally used to heat the heating circuits while the higher zone is used for heating of domestic hot water.

The basic function of the solar circuit is to control the temperature difference between the collector sensor **solColl** and the storage sensor **solStor** (see chapter "SF – terminal assignment"). If the temperature difference rises above **TDiffSolOn**, the solar pump **solPu** is switched on. This pump is switched off when the temperature difference decreases below **TDiffSolOff**.

If the solar circuit contains a return temperature sensor **solReturn**, the solar pump is not switched off until a temperature difference smaller than 1K is reached. If the temperature difference is smaller, the solar system is inefficient.

The parameters **maxTstor** and **maxTSolStor** permits setting the maximum temperatures for the DHW storage tank and the buffer tank. If a maximum value is exceeded, solar heating will be interrupted.

If the temperature limit **maxTColl** at the collector is exceeded, the solar pump is switched on and the collector cooled down using the storage or buffer tank. If in addition **maxTStor** is exceeded, the excessive heat will be dissipated by the heating circuits being switched on.

Frost protection in the solar circuit is normally realised by the addition of frost protection fluid. Furthermore, it is monitored if the temperature falls below the frost limit **collFrLi,** in which case the solar pump is switched on. The solar collector is flushed with water from the buffer tank until the danger of freezing no longer persists. These functions can be deactivaded using the settings **maxTColl** = 160°C and **collFrLi** = -50°.

- **TdiffSolOn:** If the temperature difference between the solar collector temperature and storage or buffer tank temperature rises above this value, the solar pump is switched on.
- **TdiffSolOff:** If the temperature difference between the solar collector temperature and storage or buffer tank temperature falls below this value, the solar pump is switched off.
- **maxTStor:** Maximum temperature in DHW storage tank.
- **maxTStorSt:** Maximum temperature in solar buffer
- **maxTcoll:** Maximum temperature in collector.
- **collFrLi:** The frost protection limit at which the solar pump is switched on.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.14.1	TdiffSolOn	temp-difference solar pump on	K	0.0	30.0	8.0	
2.4.6.14.2	TdiffSolOff	temp-difference solar pump off	K	0.0	20.0	2.0	
2.4.6.14.3	maxTStor	max. temperature storage sensor	°C	20.0	160.0	70.0	
2.4.6.14.4	maxTSolSt	max. temperature Sol.-storage sensor	°C	20.0	160.0	70.0	
2.4.6.14.5	maxTcoll	max. temperature collector sensor	°C	70.0	160.0	110.0	
2.4.6.14.6	collFrLi	frostLim collector	°C	-50.0	10.0	-10.0	

Thermal disinfection (2.4.6.15.n)

During the thermal disinfection period, the actual value reached is monitored and stored in the parameter **cur-stor** and the related **time** and **date** are displayed. The parameter **timer** shows the remaining time during which the function is active.

If the selected storage setpoint is not reached within 2 hours after the function “thermal disinfection” was started, the trouble indication “**Therm. disinfection**” is triggered. The display message **“Fault DHW circuit”** can be reset using **reset-mess**=1.

- **cur-stor:** Actual storage of buffer tank temperature that has been reached presently or at the last thermal disinfection.
- **time:** Time of the last thermal disinfection.
- **date:** Date of the last thermal disinfection.
- **timer:** Time until current thermal disinfection is finished.
- **reset-mess**: =1: Resets the trouble indication “thermal disinfection” (setpoint temperature was not reached).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.15.1	cur-stor	achieved actual value storage Tank temper	°C				
2.4.6.15.2	time	time buff.-temp.					
2.4.6.15.3	date	date buff.-temp.					
2.4.6.15.4	timer	curTim running time/cleaning	min				
2.4.6.15.5	Reset-mess	reset trouble indication		0	1	0	

Capacity limitation (2.4.6.18.n)

The “capacity limitation” service menu contains:

- **flowCorrAct:** Current setpoint correction due to capacity limitation.
- **Tn:** The integral action time exerting an influence on the signal to the actuator. An increase will extend the I-rate in time and thus reduce the impact. Tn=121.0min will deactivate the I-rate.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.18.1	flowCorrAct	correction setpoint with capacity limitat	K	-100.0	0.0		
2.4.6.18.2	Tn	integral action time	min	0.1	121.0	5.0	

Switch-on-delay (2.4.6.19.n)

The“switch-on-delay” function permits loading the storage tank or delaying the release of the DHW program.

- **swOnDelLoad:** Delay start of loading of the storage tank.
- **swOnDelContr.**: Delay of switch-on release of the control circuit.

timer: Displays the time before the delay time has run out.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.19.1	swOnDelLoad	delayed switch-on storage loading	s	0	600	0	
2.4.6.19.2	timer	curTimer	s				
2.4.6.19.3	swOnDelContr.	Delayed switch-on release controller	s	0	600	300	
2.4.6.19.4	timer	curTimer	s				

Switch-off-delay (2.4.6.20.n)

The “switch-off-delay” function permits delaying storage tank loading switch-off or the release of the heat exchanger.

- **swOfDelLoad**: Delay of the start of the loading of the storage tank.
- **swOfDelContr.**: Delay of the switch-on release of the heat exchanger.
- **timer:** Shows the current time before the delay time has run out.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.20.1	swOfDelLoad	delayed switchOff loading	min	0.0	60.0	3.0	
2.4.6.20.2	timer	curTimer	min				
2.4.6.20.3	swOfDelHEx	delayed switchOff circulation Hex	s	0	600	120	
2.4.6.20.4	timer	curTimer	s				

Solar statistic (2.4.6.21.n)

If a return temperature sensor (**solReturn**) is installed in the solar circuit, the current heat capacity (**curCapa**) is calculated using the measured differential temperature (**solColl** - **solReturn**) and the given volumetric flow of the solar pump (**volFlowPu**). The volumetric flow can be found in the data sheet for the solar pump. If glycol is being used in the solar circuit, the higher viscosity of this media must be taken into consideration and **percGlycol** adjusted, if required. The heat amount **HeAmount** is calculated based on the heat capacity and solar pump running time. This provides information on the heat energy saved by the solar system.

It is possible to define when counting should begin (day, month) by using the **begin** parameter. If this date is reached again, the current calculated value is stored in the parameter for the previous year **hAmount-LY**.

The maximum heat capacity reached (**maxCapa**) and the related **date** are displayed. A reset will delete the **maxCapa** and **date**.

- **volFlowPu:** The volumetric flow of the solar pump (must set by a technician).
- **percGlykol:** The percentage of glycol in the media (must be set by a technician).
- **curCapa:** Current capacity.
- **HeAmount:** The amount of heat produced by the solar system. Reset after each year.
- **Begin:** Start of recording of the amount of heat.
- **hAmount**-LY: Amount of heat produced during last year.
- **maxCapa**: Maximum heat output achieved (heat capacity).
- **date:** Date when the maximum amount was achieved.
- **reset:** =1: Resets parameters **maxCapa** and **date**.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.21.1	volFlowPu	volumetric flow solar pump	m3/h	0.00	10.00	0.00	
2.4.6.21.2	percGlykol	volume percent Glykol	%	0	100	60	
2.4.6.21.3	curCapa	cur capacity	W				
2.4.6.21.4	HeAmount	amount of heat	kWh				
2.4.6.21.5	begin	countBegin				01.01	
2.4.6.21.6	hAmount-LY	amount of heat last year	kWh				
2.4.6.21.7	maxCapa	maximum achieved heat output	W				
2.4.6.21.8	date	date of maxCapacity					

2.4.6.21.9	reset	reset maxCapa		0	1	0	

Sensor correction (2.4.6.22.n)

If the temperature values displayed in the "current values" menu differ from current system values, a correction factor can be entered to adjust the individual sensor values.

- **DHWTank:** Correction factor (offset) for the storage tank temperature sensor (top).
- **DHWTank2:** Correction factor (offset) for the storage tank temperature sensor (middle).
- **DHWFlow**: Correction factor (offset) for the flow temperature sensor.
- **loadFlow**: Correction factor (offset) for the loading temperature sensor.
- **release**: Correction factor (offset) for the release temperature sensor.
- **solTank**: Correction factor (offset) for the buffer tank temperature sensor.
- **solColl**: Correction factor (offset) for the flow collector temperature sensor.
- **solReturn**: Correction factor (offset) for the return temperature sensor.
- **outdoor**: Correction factor (offset) for the outdoor temperature sensor.
- **limit**: Correction factor (offset) for the limitation temperature sensor.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.22.1	DHWtank		K	-10.0	10.0	0.0	
2.4.6.22.2	DHWTanl2		K	-10.0	10.0	0.0	
2.4.6.22.3	DHWFlow		K	-10.0	10.0	0.0	
2.4.6.22.4	loadFlow		K	-10.0	10.0	0.0	
2.4.6.22.5	release		K	-10.0	10.0	0.0	
2.4.6.22.6	solTank		K	-10.0	10.0	0.0	
2.4.6.22.7	solColl		K	-10.0	10.0	0.0	
2.4.6.22.8	solReturn		K	-10.0	10.0	0.0	
2.4.6.22.9	outdoor		K	-10.0	10.0	0.0	
2.4.6.22.10	limit		K	-10.0	10.0	0.0	
2.4.6.22.11	limit		K	-10.0	10.0	0.0	

Terminal assignment (2.4.6.23.n)

The "Terminal assignment" service menu permits assigning the components used in the DHW circuit module to an input or output terminal. Each input terminal has a substitute value which is used by the module in case of a sensor error. The module will then signal the sensor error and use the substitute value. The substitute value will only be shown in the list if the terminal number "99" is entered as the terminal for the sensor. If this number is entered for the sensor, the module will keep working with the substitute value but will no longer show any error.

- **storage**: The terminal number for the storage temperature sensor (top).
- **^subst:** The substitute value for the above mentioned sensor (terminal number=99).
- **storage2**: The terminal number for the storage temperature sensor (middle).
- **storFlow**: The terminal number for the flow temperature sensor.
- **loadFlow**: The terminal number for the loading temperature sensor.
- **release**: The terminal number for the release temperature sensor.
- **solStor**: The terminal number for the solar storage temperature sensor.
- **solColl**: The terminal number for the collector temperature sensor.
- **solRet**: The terminal number for the return temperature sensor.

- **outdoor**: The terminal number for the outdoor temperature sensor.
- **limit**: The terminal number for the limitation temperature sensor.
- **shift**: The terminal number for the shifting temperature sensor.
- **setp-poti**: The terminal number for the external setpoint potentiometer.
- **volFlow**: The terminal number for the volumetric flow meter (pulse).
- **HeatCapa**: The terminal number for the heat capacity meter (pulse).
- **hAmount**: The terminal number for the heat amount.
- **system:** The terminal number for the system main switch (DI).
- **key:** The terminal number for the overtime key (DI).
- **oModRC:** The terminal number for the external operation mode switch at the remote control unit.
- **oModLoc:** The terminal number for the local operating mode switch.
- **valve-Cont**: The terminal number for control of valve (0…10V, continuous).
- **val-open**: The terminal number for control of valve (3-point, open).
- **val-close**: The terminal number for control of valve (3-point, close).
- **loadPu**: The terminal number for the loading pump.
- **exchPu**: The terminal number for the heat exchanger pump.
- **storPu**: The terminal number for the storage tank pump.
- **storUnload**: The terminal number of the magnetic valve to unload the storage tank.
- **solPu**: The terminal number for the solar pump.
- **circPu**: The terminal number for the circulation pump.
- **thermDon**: Terminal for signal when thermal disinfection starts.
- **thermDbuff**: Terminal for signal when tabs should be opened during thermal disinfection.
- **troublelnd**: The terminal number of the alarm signal (contact output).
- **VDV**: The terminal number for external demand (0...10V input).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.4.6.23.1	5.6.281 storage	5.6.282 storage tempe rature		5.6.28	5.6.284	5.6.285	
2.4.6.23.2	5.6.286 ^-subst.	5.6.287	5.6.28	5.6.28	5.6.290	5.6.291	
2.4.6.23.3	5.6.292 storage2	5.6.293 storage temperature2		5.6.29	5.6.295	5.6.296	
5.6.297 2.4.6.	5.6.298 ^-subst.	5.6.299	5.6.30	5.6.30	5.6.302	5.6.303	
2.4.6.23.5	5.6.304 storFlow	5.6.305 storage flow temp		5.6.30	5.6.307	5.6.308	
2.4.6.23.6	5.6.309 ^-subst.	5.6.310	5.6.31	5.6.31	5.6.313	5.6.314	

2.4.6.23.7	5.6.315 loadFlow	loadFlow temp.		5.6.31	5.6.317	5.6.318	
2.4.6.23.8	5.6.319 ^-subst.	5.6.320	5.6.32	5.6.32	5.6.323	5.6.324	
2.4.6.23.9	5.6.325 release	5.6.326 releaseTemperature		5.6.32	5.6.328	5.6.329	
2.4.6.23.10	5.6.330 ^-subst.	5.6.331	5.6.33	5.6.33	5.6.334	5.6.335	
2.4.6.23.11	5.6.336 solStor	5.6.337 solar-storage temp.		5.6.33	5.6.339	5.6.340	
2.4.6.23.12	5.6.341 ^-subst.	5.6.342	5.6.34	5.6.34	5.6.345	5.6.346	
2.4.6.23.13	5.6.347 solColl	5.6.348 solar-collector temp.		5.6.34	5.6.350	5.6.351	
2.4.6.23.14	5.6.352 ^-subst.	5.6.353	5.6.35	5.6.35	5.6.356	5.6.357	
2.4.6.23.15	5.6.358 solRet	5.6.359 solar-return temp.		5.6.36	5.6.361	5.6.362	
2.4.6.23.16	5.6.363 ^-subst.	5.6.364	5.6.36	5.6.36	5.6.367	5.6.368	
2.4.6.23.17	outdoor	outdoor temperature		0	255	0	
2.4.6.23.18	5.6.369 ^-subst.	5.6.370	5.6.37	5.6.37	5.6.373	5.6.374	
2.4.6.23.19	5.6.375 limit	5.6.376 limitation sensor		5.6.37	5.6.378	5.6.379	
2.4.6.23.20	5.6.380 ^-subst.	5.6.381	5.6.38	5.6.38	5.6.384	5.6.385	
2.4.6.23.21	5.6.386 shift	5.6.387 shift sensor		5.6.38	5.6.389	5.6.390	
2.4.6.23.22	5.6.391 ^-subst.	5.6.392	5.6.39	5.6.39	5.6.395	5.6.396	
2.4.6.23.23	5.6.397 setp-poti	5.6.398 setpoint-poti		5.6.39	5.6.400	5.6.401	
2.4.6.23.24	5.6.402 ^-subst.	5.6.403	5.6.40	5.6.40	5.6.406	5.6.407	
2.4.6.23.25	5.6.408 volFlow	5.6.409 volumetric flow		5.6.41	5.6.411	5.6.412	

2.4.6.23.26	5.6.413 ^-subst.	5.6.414	5.6.41	5.6.41	5.6.417	5.6.418	
2.4.6.23.27	5.6.419 heatCapa	5.6.420 heat capacity		5.6.42	5.6.422	5.6.423	
2.4.6.23.28	5.6.424 ^-subst.	5.6.425	5.6.42	5.6.42	5.6.428	5.6.429	
2.4.6.23.29	5.6.430 hAmount	5.6.431 amount of heat		5.6.43	5.6.433	5.6.434	
2.4.6.23.30	5.6.435 system	5.6.436 sys.-main switch		5.6.43	5.6.438	5.6.439	
2.4.6.23.31	5.6.440 ^-subst.	5.6.441		5.6.44	5.6.443	5.6.444	
5.6.445 2.4.6.	5.6.446 key	5.6.447 RC-key		5.6.44	5.6.449	5.6.450	
2.4.6.23.33	5.6.451 oModRC	5.6.452 RC-opMode-switch		5.6.45	5.6.454	5.6.455	
2.4.6.23.34	5.6.456 oModLoc	5.6.457 oMod-local switch		5.6.45	5.6.459	5.6.460	
2.4.6.23.35	5.6.461 ^-subst.	5.6.462		5.6.46	5.6.464	5.6.465	
5.6.466 2.4.6.	5.6.467 valve-cont	5.6.468 valve continuous		5.6.46	5.6.470	5.6.471	
2.4.6.23.37	5.6.472 val-open	5.6.473 valve open		5.6.47	5.6.475	5.6.476	
2.4.6.23.38	5.6.477 val-close	5.6.478 valve close		5.6.47	5.6.480	5.6.481	
2.4.6.23.39	5.6.482 loadPu	5.6.483 loadPump		5.6.48	5.6.485	5.6.486	
2.4.6.23.40	5.6.487 exchPu	5.6.488 exchanger load pump		5.6.48	5.6.490	5.6.491	
5.6.492 2.4.6.	5.6.493 storPu	5.6.494 storage load pump		5.6.49	5.6.496	5.6.497	
2.4.6.23.42	5.6.498 stoUnloa	5.6.499 storage unloading		5.6.50	5.6.501	5.6.502	
2.4.6.23.43	5.6.503 solPu	5.6.504 solarPump		5.6.50	5.6.506	5.6.507	
2.4.6.23.44	5.6.508 circPu	5.6.509 circulation pump		5.6.51	5.6.511	5.6.512	
2.4.6.23.45	5.6.513 thermDOn	5.6.514 therm.		5.6.51	5.6.516	5.6.517	

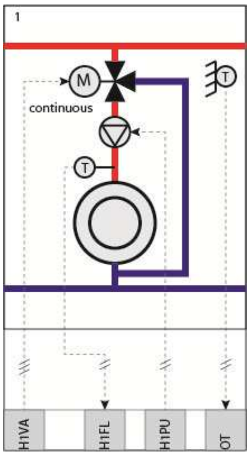
		Desin fectio n					
2.4.6.23.46	5.6.518 thermDbuf	5.6.519 tab cleaning		5.6.52	5.6.521	5.6.522	
2.4.6.23.47	5.6.523 troubInd	5.6.524 trouble indica tion		5.6.52	5.6.526	5.6.527	
2.4.6.23.48	VDV	Hea-dem		5.6.52	5.6.529	5.6.530	

Chapter 6 The Heating Circuit modules

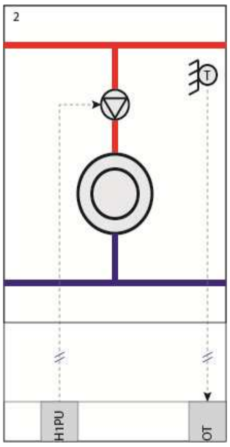
Depending on the type of controller, HPM can be equipped with up to 2 heating circuits. Each heating circuit is controlled by a module. The basic function of the Heating circuit module is outdoor temperature and/or room temperature-dependent control of the flow temperature. To control flow temperature to the radiators, underfloor heating or any other heat consumer, the HPM controls a valve with an actuator or a pump.

There are 3 different heating circuit diagrams available:

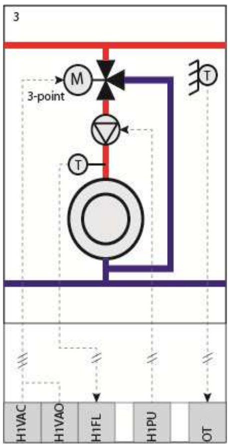
Heating circuit with outdoor temperature-dependent flow temperature control, using a valve with continuous signal:



Unmixed heating circuit:



Heating circuit with outdoor temperature-dependent flow temperature control, using a valve with 3-point control:



6.1 Current values (2.5.x.1.n)

The menu ”Current values” displays an overview of values from temperature sensors, digital inputs, meters, etc.

- **room, outdoor…**: Values of all sensors connected to this heating circuit.
- **volFlow, heatCapa**: Values of a connected heat or flow meter related to this heating circuit.
- **delOT**: The delayed outdoor temperature. This constitutes a calculcated outdoor temperature coming from the function “delayed outdoor temperature” (see chapter “AF – delayed outdoor temperature”).
- **oModLoc**: Status of the local system operating switch which has an influence of the functionalty of the heating circuit module.
- **begHeUp, endHeUp**: Provides relevant information on the beginning and end of the heating-up phase (e.g. such as when the room temperature is brought back to a desired level after a temperature reduction during the night).
- **fl-corr:** Correction of the flow temperature value by a potentiometer or other functions in this module.
- **system:** The system main switch (**system**) of the module is responsible for the module running or not running. This switch can by used by other modules or the BMS system.

Depending on the HPM unit in question, a large number of different remote control units may also be connected:

- **key**: Overtime key on the remote control unit for extending the occupation time.
- **oModRC**: Switch for changing the operating mode (can be set to ON/OFF or AUTO/Manual/Off).
- **setp-poti**: A potentiometer can also be connected for changing the room temperature setpoint.
- **room**: Each remote control unit is equipped with a room sensor which temperature is shown in parameter 2.5.x.1.1, where x stands for the number of the heating circuit (1 or 2).

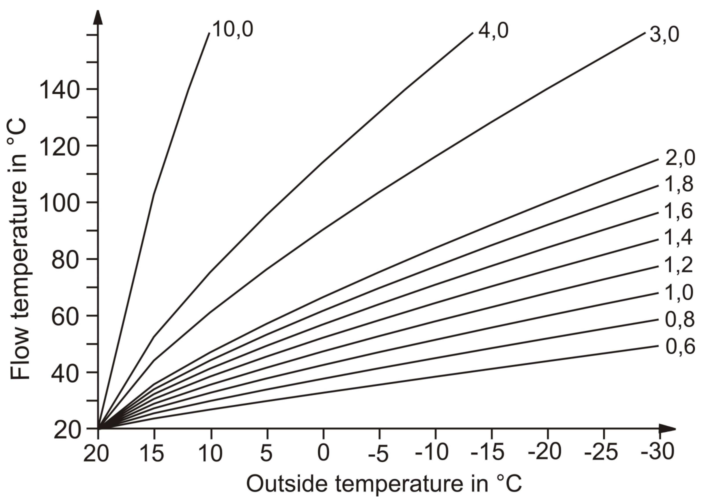
Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.1.1	room	room temperature	°C				
2.5.x.1.2	outdoor	outdoor temperature	°C				

2.5.x.1.3	flow	flow temperature	°C				
2.5.x.1.4	delOT	delayed outdoor temperature	°C				
2.5.x.1.5	return	return temperature	°C				
2.5.x.1.6	limit	limitation sensor	°C				
2.5.x.1.7	shift	shift sensor	°C				
2.5.x.1.8	setp-poti	setpoint poti	°C				
2.5.x.1.9	fl-corr	flow setpoint corr.	°C				
2.5.x.1.10	volFlow	volumetric flow	l/h				
2.5.x.1.11	heatCapa	heat capacity	kW				
2.5.x.1.12	HeAmount	amount of heat					
2.5.x.1.13	system	sys.-main switch					
2.5.x.1.14	key	RC-key					
2.5.x.1.15	oModRC	RC-opMode-switch					
2.5.x.1.16	oModLoc	oMod-local switch					
2.5.x.1.17	begHeUp	beginning heating up		00:00	23:59		
2.5.x.1.18	endHeUp	end heating up		00:00	23:59		

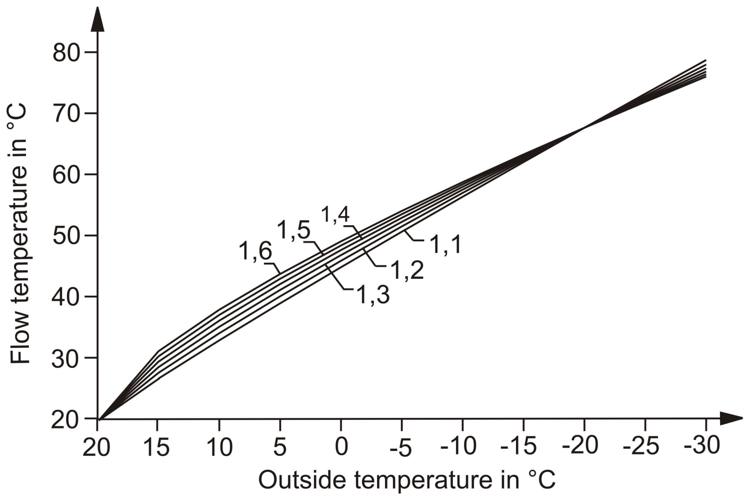
6.2 Setpoints (2.5.x.2.n)

The “Setpoints” menu contains all the normal parameters responsible for the basic functionality of the heating circuit module.

- **SP-room**: This is the calculated setpoint for the room temperature. It is a result of the setpoint choosen for the current timer status (occupation time 1, 2, 3, 4, non-occupation time or special non-occupation time) and is influenced by a remote setpoint potentiometer or the overtime key. This parameter can not be changed directly, but does show the current valid setpoint. A room sensor must be connected.
- **SP-Flow**: This is the currently calculated setpoint for the flow temperature. This is a result from the heating curve and some influence of the functions described below. This parameter can not be changed directly.
- **hCu-slope**: This parameter describes the slope of the heating curve:



- **hCU-exp**: This parameter describes how the curve is bent. This exponent is normally choosen based on the type of heating system used. The non-linear dependency between the flow temperature and heat output to the room is represented in this value.



Common radiator exponents are:

a) 1,10 – underfloor heating
b) 1,2 – radiators
c) 1,33 – DIN – radiators
d) 1,25 …1,40 – plate radiator
e) 1,40 … 1,60 – convectors

- **roomOT1..OT4** : Temperature setpoint for the operating time x (1..4).
- **roomNO**: Temperature setpoint for the non-occupation time.
- **roomSNOT**: Temperature setpoint for the special non-occupation time.
- **floReNO**: Setpoint for flow temperature reduction during the non-occupation time.
- **flowReSNOT:** Flow temperature reduction in the special non-occupation time.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.2.1	6.2.531 SP-	6.2.532 setpoint room temp.	6.2.53	6.2.53	6.2.535		
2.5.x.2.2	6.2.536 SP-	6.2.537 setpoint flow.-temp	6.2.53	6.2.53	6.2.540		
2.5.x.2.3	6.2.541 hCu-	6.2.542 characteristic curve slope		6.2.54	6.2.544	6.2.545	
2.5.x.2.4	6.2.546 hCu-	6.2.547 radiator-exponent		6.2.54	6.2.549	6.2.550	
2.5.x.2.5	6.2.551 room	6.2.552 setpoint OT1	6.2.55	6.2.55	6.2.555	6.2.556	
2.5.x.2.6	6.2.557 room	6.2.558 setpoint OT2	6.2.55	6.2.56	6.2.561	6.2.562	

2.5.x.2.7	6.2.563 room	6.2.564 setpoint OT3	6.2.56	6.2.56	6.2.567	6.2.568	
2.5.x.2.8	6.2.569 room	6.2.570 setpoint OT4	6.2.57	6.2.57	6.2.573	6.2.574	
2.5.x.2.9	6.2.575 room	6.2.576 setpoint NO	6.2.57	6.2.57	6.2.579	6.2.580	
2.5.x.2.10	6.2.581 room	6.2.582 setpoint SNOT	6.2.58	6.2.58	6.2.585	6.2.586	
2.5.x.2.11	6.2.587 floRe	6.2.588 flow-reduct. NO	6.2.58	6.2.59	6.2.591	6.2.592	
2.5.x.2.12	6.2.593 flowR	6.2.594 flow-setback SNOT	6.2.59	6.2.59	6.2.597	6.2.598	

6.3 Additional functions (2.5.x.3.n)

The heating circuit module is equipped with a number of additional functions which can be activated and configured as required. Some of these functions have parameters which should only be changed by trained personnel.

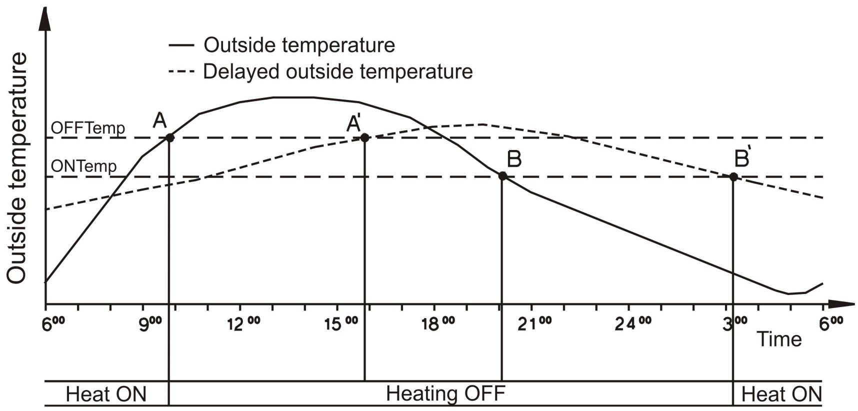
Caretaker (2.5.x.3.2.n)

The caretaker function performs exactly the same function that a building caretaker would perform; monitoring outdoor temperature and switching the heating circuit module into summer or winter mode. When in summer mode, the module only attempts to protect the building by preventing clogging or blocking of pumps and valves and watching for frost. In winter mode, it also controls the room or flow temperature and works according to other activated functions.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type**: There are 2 types of caretaker function; "comfort caretaker" (1) or "3-day-caretaker" (2).

  =1*: comfort caretaker*: If the outdoor temperature exceeds the switch-off temperature (**swOfTempOT** for the operating time and **swOfTempNO** for non-occupation time), the heating circuit module is switched into summer mode. If the outdoor temperate falls below the switch-on temperature (**swOnTempOT** for operating time and **swOnTempNO** for non-occupation time) the heating circuit module is switched into winter mode and heating is initiated.

  =2*: 3-day-caretaker*: The outdoor temperature is measured daily at a given measuring time (**meas.time**). If the outdoor temperature exceeds the switch-off temperature (**swOfTempOT** for the operating time and **swOfTempNO** for non-occupation time) on 3 consecutive days, the heating circuit module is switched into summer mode. If the outdoor temperate falls below the switch-on temperature (**swOnTempOT** for operating time and **swOnTempNO** for non-occupation time) on 3 consecutive days, the heating circuit module is switched into winter mode and heating is initiated.



Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.2.1	6.3.599 active	6.3.600		6.3.601	6.3.602	6.3.603	
2.5.x.3.2.2	6.3.604 type	6.3.605 1:comfort-CT 2:3-day-CT		6.3.606	6.3.607	6.3.608	
2.5.x.3.2.3	6.3.609 swOf	6.3.610 switchOff-temp OT	6.3.61	6.3.612	6.3.613	6.3.614	
2.5.x.3.2.4	6.3.615 swOf	6.3.616 switchOff-temp NO	6.3.61	6.3.618	6.3.619	6.3.620	
2.5.x.3.2.5	6.3.621 swOn	6.3.622 switchOn-temp OT	6.3.62	6.3.624	6.3.625	6.3.626	
2.5.x.3.2.6	6.3.627 swOn	6.3.628 switchOn-temp NO	6.3.62	6.3.630	6.3.631	6.3.632	
2.5.x.3.2.7	6.3.633 meas.t	6.3.634		6.3.635	6.3.636	6.3.637	

Switch-on optimization (2.5.x.3.3.n)

HPM is capable of postponing the start of the heat-up phase so that the room temperature will reach the given setpoint exactly when the given occupation time begins. The start is established based on a “heating-up” curve, which can be found in the service section for this function in the “SF – switch-on optimization” menu. This curve contains a time for each outdoor temperature that is needed by the heating system to heat up the room temperature by 5 K.

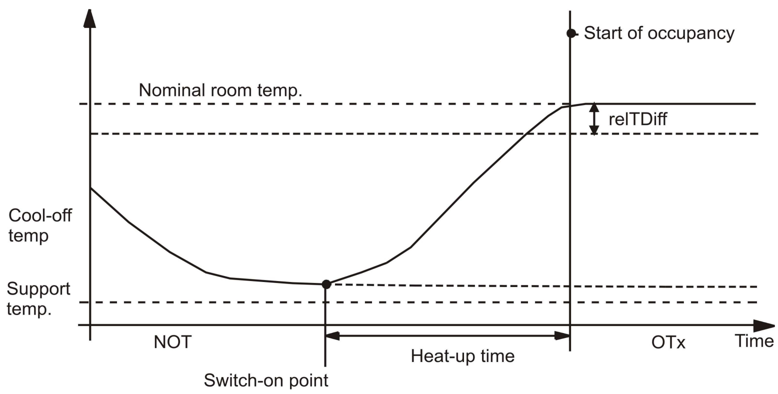
- **active**: This parameter activates (1) or deactivates (0) the function.
- **type**: There are 3 different types of optimization:

=1: *with adaptation* (using a room sensor): After each heating-up phase, the controller will check if the temperature was reached at the right time, not reached at all or reached earlier than expected. Depending on the result, the heating-up curve in the menu “SF – switch-on optimization” is optimized so that the system will get closer to the desired result the next time.

=2: *without adaptation* (using a room sensor): In this mode, the function works with a fixed heating curve. This can be set manually in the service section of the function in the “SF – switch-on optimization” menu. The room sensor is only used to calculate the temperature difference between the current temperature and the upcoming setpoint.

=3: *without a room sensor*: In this mode, the start of the heating-up phase is established according to the outdoor temperature and the heating-up curve in the menu “SF – switch-on optimization”

- **relTdiff:** The heating-up phase will only start if the temperature difference between the current room temperature and the upcoming setpoint is bigger than the “release temperature difference”:



Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.3.1	active			0	1	0	
2.5.x.3.3.2	type	1:with 2:without adapt. 3:without soomS		1	3	2	
2.5.x.3.3.4	relTdiff	release tempdiff.	K	0.0	5.0	1.0	

Switch-off optimization (2.5.x.3.4.n)

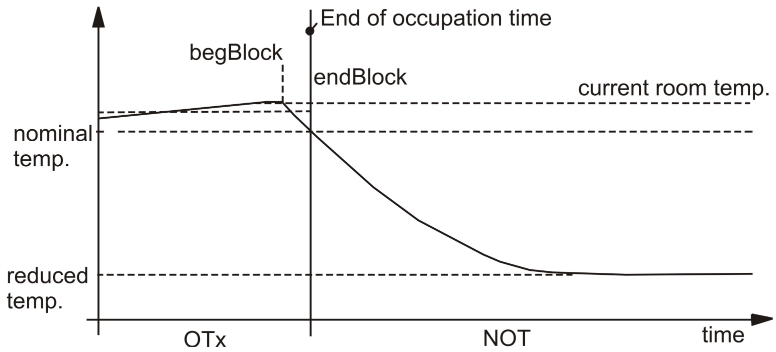
The “switch-off optimization” function enables postponing the end of the heating phase if the controller can establish that the room still has the desired room temperature at the end of the occupation time. The end of the heating phase is established by using the building time constant. This building time constant represents a value that indicates how fast the building is cooling down when no additional heat is coming from the heating system. A room sensor is required for this function to be activated.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type:** There are 2 different types of optimization:

  *=1***:** *automatic*: Each time a switch-off optimization is established, the controller checks if the room temperature is too high, too low or adequate at the end of the occupation time. The building time constant is adapted accordingly.

  *=2***:** *manual*: In this mode, the function operates using a fixed building time constant which can be set manually in the menu “SF – switch-off optimization”.

- **buildTiC**: This parameter shows the building time constant currently used.



Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.4.1	active			0	1	0	
2.5.x.3.4.2	type	building-timeconst. 1:autom. 2:manual		1	2	2	
2.5.x.3.4.3	buildTiC	building-timeconst.	H	0.0	48.0	12.0	

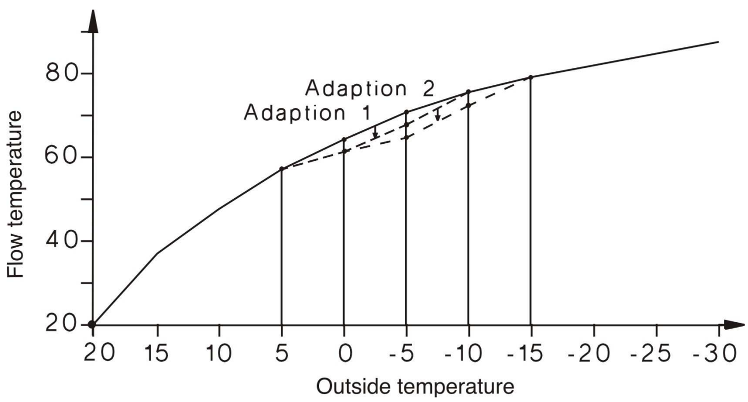
Heating curve adaption (2.5.x.3.5.n)

The heating curve represents the dependency between the outdoor temperature and the flow temperature. The lower the outdoor temperature, the higher the flow temperature as the area that is to be kept warm requires more heat. The heating curve is used by many functions in this module and it is therefore essential that it is as accurate as possible. The heating curve can be found in the service section concerning this function in the chapter "SF – heating curve adaption". The function requires a room sensor to be connected.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type:** There are 2 different types of heating curve adaptions:

  *=1***:** *adaption*: Each time a change in setpoint occurs, the program establishes the time needed to reach the new desired setpoint. This time is used to adapt the heating curve and obtain a perfect flow temperature for each outdoor temperature.

  =2: *manual*: This mode permits manual adaptation of the heating curve in the menu "SF – heating curve adaption" to the desire of the individual user. The controller will work accordingly.



Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.5.1	active			0	1	0	
2.5.x.3.5.2	type	1:adapt. 2:manual		1	2	1	

Setpoint limitation (2.5.x.3.6.n)

The “setpoint limitation” function can be used to specify a minimum and maximum limit for the *calculated* flow setpoint. By adapting the limits and activating the function, the flow temperature of, for instance, an underfloor heating system can be limited to a maximum permittable value. The ramp function is used to set the speed of the setpoint change. This setting can for instance be used to avoid power peaks during heating up of the pipe system.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **minFl:** Minimum limit of the flow temperature.
- **maxFl:** Maximum limit of the flow temperature.
- **posLim**: The setpoint limitation function makes it possible to limit the change of the calculated setpoint during the heating-up phase, to prevent noise in the system and extreme stress on the piping.
- **negLim:** This parameter has the same function as **posLim**, with the exception that the limitation is working during the cooling down phase.
- **maxDemFl-T**: This parameter represents the maximum allowed flow temperature demand requested *directly* from the heat producer.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.6.1	6.3.638 active	6.3.639		6.3.64	6.3.641	6.3.642	
2.5.x.3.6.2	6.3.643 minFl	6.3.644 min. flow temp.	6.3.64	6.3.64	6.3.647	6.3.648	
2.5.x.3.6.3	6.3.649 maxFl	6.3.650 max. flow temp.	6.3.65	6.3.65	6.3.653	6.3.654	
2.5.x.3.6.4	6.3.655 posLi	6.3.656 SP-ramp heating up	6.3.65	6.3.65	6.3.659	6.3.660	
2.5.x.3.6.5	6.3.661 negLi	6.3.662 SP-ramp coolDown	6.3.66	6.3.66	6.3.665	6.3.666	
2.5.x.3.6.6	6.3.667 maxDe	6.3.668 max. flow temp.demand	6.3.66	6.3.67	6.3.671	6.3.672	

Universal limitation (2.5.x.3.7.n)

This function requires a limitation sensor to be defined. A completely new sensor can either be connected or the universal limitation used for any sensor already connected (e.g. a return sensor, exhaust sensor etc.). See chapter “SF – terminal assignment” for the definition of the limitation sensor **limit**.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type:** Depending on this parameter, the function can be used optionally for maximum or minimum limitation and the actuator can either open or close in case of limit violation.

  *=1***:** Maximum limitation, open valve.

=2: Maximum limitation, close valve.

=3: Minimum limitation, close valve.

=4: Minimum limitation, open valve.

- **typLimit:** The limit can be defined as fixed value or as shifting value with a variable shifting curve according to the value of the sensor input **shift** (see chapter “SF – terminal assignment” for definition).

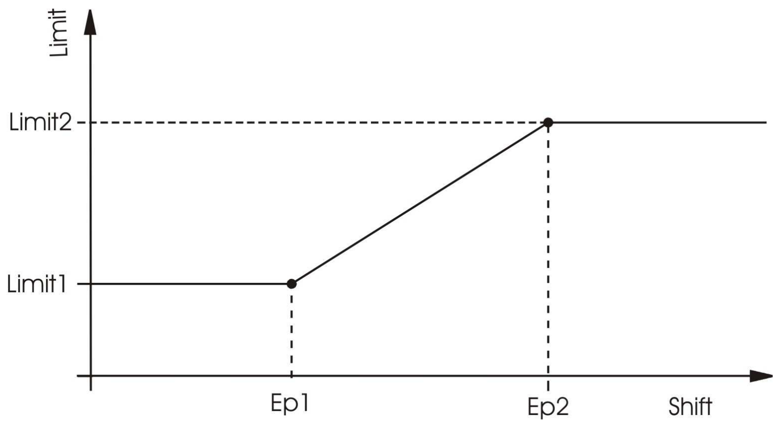
  =0: Fixed value (constant).

  =1: Shifting value.

- **limit1, limit2, EP1, EP2**:

  If **typLimit=0**: If the value measured at the limitation sensor violates the value in **limit1**, the universal limitation takes over control of the actuator.

  If **typLimit=1**: If the curve between **limit1/EP1** and **limit2/EP2** is violated, the universal limitation takes over control of the actuator. (See graphic below)



The control behaviour of the universal limitation can be adjusted to the characteristics of the control system by means of the controller parameters **Xp**, **Tn**, **nZone** and **switchD.** These control parameters can be configured in the menu “SF – universal limitation”.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.7.1	active			0	1	0	
2.5.x.3.7.2	type	1:max open 2:max close 3:min close 4:min		1	4	1	
2.5.x.3.7.3	typLimit	0:const. value 1:shift		0	1	0	
2.5.x.3.7.4	limit1	limit value 1	°C	-999999.9	999999.9	0.0	
2.5.x.3.7.5	EP1	entry point 1	°C	-999999.9	999999.9	0.0	
2.5.x.3.7.6	limit2	limit value 2	°C	-999999.9	999999.9	0.0	
2.5.x.3.7.7	EP2	entry point 2	°C	-999999.9	999999.9	0.0	

Room compensation (2.5.x.3.8.n)

If the function “room compensation” is activated, the calculated flow temperature setpoint **SP-flow** (see menu “Setpoints”) is corrected by means of a Pl controller according to the control deviation of the room temperature (difference between current room temperature and setpoint).

- **active**: This parameter activates (1) or deactivates (0) the function.
- **maxposCorr, minposCorr**: The rom compensation can be limited by these two parameters.

This function requires the assignment of the room sensor **Room** (see menu “SF – terminal assignment”). The room sensor has to be installed in a reference room of the heating circuit. The reference room can be the living room of a single-family home or the classroom of a school. The thermostat valves of the radiators installed in the reference room have to be fully opened so that they do not affect the heat emission. If the room compensation is active, the additional function “Heating curve adaptation” is not effective and the menu for this function is not displayed.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.8.1	6.3.673 acti	6.3.674		6.3.675	6.3.676	6.3.677	
2.5.x.3.8.2	6.3.678 max	6.3.679 max. pos. correction	6.3.68	6.3.681	6.3.682	6.3.683	
2.5.x.3.8.3	6.3.684 max	6.3.685 max. neg. correction	6.3.68	6.3.687	6.3.688	6.3.689	

Delayed outdoor temperature (2.5.x.3.11.1)

Every building has the capacity to store heat. The amount of heat that a building can store is described in the “building time constant”. This constant is dependent on the construction of the building (e.g. the thickness of its outdoor walls, insulation, type of windows, etc.). If this function is activated (1), the measured outdoor temperature is dampened using the “building time constant” of the building and therefore delaying functions, that are using the outdoor temperature. This “delayed” outdoor temperature can be seen in the menu ”Current values”.

The “delayed” outdoor temperature is used for the following functions:

a) Calculation of the flow temperature from the heating curve

b) Caretaker function

c) Heating curve adaption

d) Design temperature

The value for the “building time constant” can be found in the menu “SF – switch-off optimization”.

- **activeDelOT**: This parameter activates (1) or deactivates (0) the function

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.11.1	6.3.690 activ	6.3.691 release delayed outdoor temperature		6.3.69	6.3.69	6.3.694	

Design temperature (2.5.x.3.12.n)

The design temperature is the lowest outdoor temperature for which the heating system is just able to provide enough heat to hold a room temperature. If a room sensor is installed, this function is invalid.

- **designOT:** This parameter represents the minimum outdoor temperature for which the system was designed. The controller will use this parameter to establish how quickly the rooms will cool down and will use it to lessen the reduction of the flow temperature according to the measured outdoor temperature. If the outdoor temperature reaches the design temperature, the flow temperature will not be reduced during the non-occupation time.
- **limOutdoor:** This parameter permits limiting the reduction of the flow temperature during non-occupation time.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.12.2	designOT	design-outdoortemp.	°C	-50.0	0.0	-12.0	
2.5.x.3.12.3	limOutdoor	limit of shifted flow reduction	°C	-60.0	20.0	-12.0	

Remote control unit (2.5.x.3.13.n)

The “Remote control unit” function permits configuring the influence of the remote control unit. If the remote control unit is equipped with a setpoint potentiometer, the setpoint for the room temperature may be shifted remotely.

- **active OT1 … OT4**: These parameters give you the possibility to define, when the setpoint potentiometer at the remote control unit is active (1) or not active (0) for this heating cricuit
- **duratOvertime**: This parameter permits setting the duration in which the heating circuit should be in occupation time mode, once the overtime key has been pressed.

The name of the overtime key stems from offering the possibility to prolong the normal occupation time set in the timer. If the overtime key is pressed during an occupation time, this occupation time is prolonged for the duration set in this parameter. If the overtime key is pressed during non-occupation time, an extra occupation time is started using the setpoint set for occupation time 1. If the overtime is activated, a LED will light up at the remote control unit. The overtime can be stopped by pressing the overtime key again.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.13.1	6.3.695 active	6.3.696 activation poti OT1		6.3.697	6.3.69	6.3.699	
2.5.x.3.13.2	6.3.700 active	6.3.701 activation poti OT2		6.3.702	6.3.70	6.3.704	
2.5.x.3.13.3	6.3.705 active	6.3.706 activation poti OT3		6.3.707	6.3.70	6.3.709	
2.5.x.3.13.4	6.3.710 active	6.3.711 activation poti OT4		6.3.712	6.3.71	6.3.714	
2.5.x.3.13.5	6.3.715 duratO	6.3.716 duration for overtime	6.3.71	6.3.718	6.3.71	6.3.720	

Signal (2.5.x.3.14.n)

The "Signal" function permits configuring in what way a signal should be handled. When this function is activated it is possible to decide what should happen if:

e) An error occurs.

f) A trouble indication occurs.

g) A limit is exceeded.

h) A timer has run down (maintenance, etc.).

- **active** : This parameter activates (1) or deactivates (0) the function.
- **TI-all**: This parameter permits configuring how the controller should react.

TI-all	setting of the trouble indication outputs troubleInd (see menu "SF – terminal assignement").	indication as an alarm in the central building management software (EXOscada or IRMA),	indication on the controller display
0			
1			x
2		x	
3		x	x
4	x		
5	x		x
6	x	x	
7	x	x	x

- **Xw-Flow:** This parameter permits setting the maximum flow temperature deviation between setpoint and current value before a trouble indication signal is sent.
- **Xw-room**: This parameter permits setting the maximum room temperature deviation between setpoint and current value before a trouble indication signal is sent.
- **del-Xw-flow**: This parameter permits delaying the trouble indication signal for the flow temperature deviation.
- **del-Xw-room**: This parameter permits delaying the trouble indication signal for the room temperature deviation.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.14.1	active			0	1	0	
2.5.x.3.14.2	TI-all	TI to BMS:2,3,6,7 TI-O:4-7 disp:1,3,5		0	7	1	
2.5.x.3.14.3	Xw-Flow	max. contrDev. Flow temperature	K	0.5	50.0	50.0	
2.5.x.3.14.4	Xw-room	max. contrDev. Room temperature	K	0.5	20.0	20.0	
2.5.x.3.14.5	del-Xw-flow	delayed contrDev. Flow temperature	min	0	600	600	
2.5.x.3.14.6	del-Xw-room	delayed contrDev. Room temperature	min	0	600	600	

Frost protection (2.5.x.3.15.1)

In order to prevent damage to the heating circuit, the "Frost protection" function is always active (unless the controller is in "non-active" or "manual" mode).

- **outdoorfrLi:** This parameter permits setting the outdoor temperature at which the pump of the heating circuit will be started. The water in the pipes is kept in motion even if the heating circuit is does not require any heat, thereby preventing freezing.

This should prevent freezing of pipes running at the outdoor wall of the building. The frost protection mode stops when the outdoor temperature is 1 K above the value set in the parameter outdoorfrLi.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.15.1	outdoorfrLi	outdoor-frost limit	°C	-30.0	50.0	2.0	

Pump (2.5.x.3.16.n)

The “Pump” function permits configuring control of a pump in the heating circuit.

- **typePump**: With this parameter you can configure the type and the existence of the pump

  = 0: no pump

  = 1: pump with one speed rate. The terminal for the connection of the pump has to be assigned in the menu “SF – terminal assignment”.

- **typPuSwOff**: With this parameter you can configure, when the pump should be switched off. In general, the pump will switch off, when the outdoor or room temperature exceeds the setpoint of the flow temperature.

  =0: Pump is never switched off.

  =1: Pump is switched off if outdoor temperature is above the setpoint of the flow temperature.

  =2: Pump is switched off if room temperature is above the setpoint of the flow temperature.

  =3: Pump is switched off if outdoor temperature or room temperature are above the demanded flow temperature.

- **swOfDel:** This parameter permits setting an extended running time for the pump.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.16.1	typePump	0:none 1:1 rotRate		0	1	1	
2.5.x.3.16.2	typPuSwOff	pump switch-off 1:outsT 2:roomT		0	3	1	
2.5.x.3.16.3	swOfDel	delayed switchOff	min	0.0	60.0	0.0	

Controller (2.5.x.3.18.n)

The “controller” function permits configuring the type of output signal fed to the pump or the actuator to control the valve of the heating circuit.

- **contrOutp**: This parameter permits setting the type of output signal.

  =0: No control valve is installed (unmixed heating circuit).

  =1: Continuous output signal 0...10 V. The terminals must be assigned accordingly.

  =2: 2-point output. This type of signal is used for soleonit valves or actuators with spring return.

  =3: 3-point output (default). The actuator is controlled by an open signal and a close signal.

- **longDes**: This parameter permits assigning a name with max. 19 characters to the heating circuit that can be used to describe the heating circuit. This name can later be used in the display and by the BMS system.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.18.1	6.3.721 c	6.3.722 1:cont 2:2pnt 3:3pnt		6.3.72	6.3.72	6.3.725	
2.5.x.3.18.2	6.3.726 l	6.3.727 free adjustab. Prog.-longRef.					

Pump/valve exercise (2.5.x.3.19.1)

The “pump/valve exercise” function automatically detects if the actuators (pump, valve) have moved due to having undertaken any control tasks in the last 24 hours. If this is not the case the pump, followed by the valve, is triggered for an adjustable interval at 11 o’clock each day, thereby preventing blocking in the actuator and the pump.

- **duration:** This parameter permits setting how long the pump and actuator should run. This time should be at least equal to the running time of the valve actuator motor in order to ensure that during the “pump/valve exercise funjtion” the heating valve is moved over the total lift.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.19.1	duration	running time pump/valve exercise	s	0	600	120	

Capacity limitation (2.5.x.3.20.n)

The “capacity limitation” function permits limitation of either the capacity output to the heating circuit or of the volumetric flow in the heating circuit. This function requires a heat meter to be installed and connected, providing the currently used capacity or the current volumetric flow. Limitation affects the signal to the valve of the heating circuit.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **type**: This parameter permits setting the type of limitation.

  =1: Capacity limitation.

  =2: Volumetric flow limitation.
- **Lim1Capa**: This parameter permits setting the limit of the capacity.
- **Lim1VolFl:** This parameter permits setting the limit for the volumetric flow.
- **Kp**: This parameter represents the amplifier gain for the actuator.
- **maxCorr**: This parameter permits setting the maximum permitted setpoint correction.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.20.1	6.3.728 act	6.3.729		6.3.730	6.3.731	6.3.732	
2.5.x.3.20.2	6.3.733 ty	6.3.734 1:capa 2:volFlow		6.3.735	6.3.736	6.3.737	
2.5.x.3.20.3	6.3.738 Li	6.3.739 limit1 capacity	6.3.74	6.3.741	6.3.742	6.3.743	
2.5.x.3.20.4	6.3.744 Li	6.3.745 limit1 volumet.	6.3.74	6.3.747	6.3.748	6.3.749	
2.5.x.3.20.5	6.3.750 Kp	6.3.751 amplifier gain	6.3.75	6.3.753	6.3.754	6.3.755	
2.5.x.3.20.6	6.3.756 ma	6.3.757 max. SP correction	6.3.75	6.3.759	6.3.760	6.3.761	

Support operation (2.5.x.3.21.n)

If a room temperature sensor **room** has been assigned (see menu “SF – terminal assignment”), the additional function “support operation” can be used to switch off the pump and close the valve during the non-occupation time. In this case the room temperature is controlled like a 2-point control.

If the temperature is below the current room temperature setpoint **roomNO** and/or **roomSNOT** (see menu “setpoints”), the heating circuit is operated at the maximum flow temperature until the current room temperature exceeds the room temperature setpoint plus the switch-off difference **swOfDiff**. This reduces the running time of the heating circuit pump and saves electrical energy.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **swOfDiff**: Switch-off difference for the support operation.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.21.1	6.3.762 a	6.3.763		6.3.76	6.3.76	6.3.766	
2.5.x.3.21.2	6.3.767 s	6.3.768 switchOffdifference for support Oper.	6.3.76	6.3.77	6.3.77	6.3.772	

Screed drying (2.5.x.3.22.n)

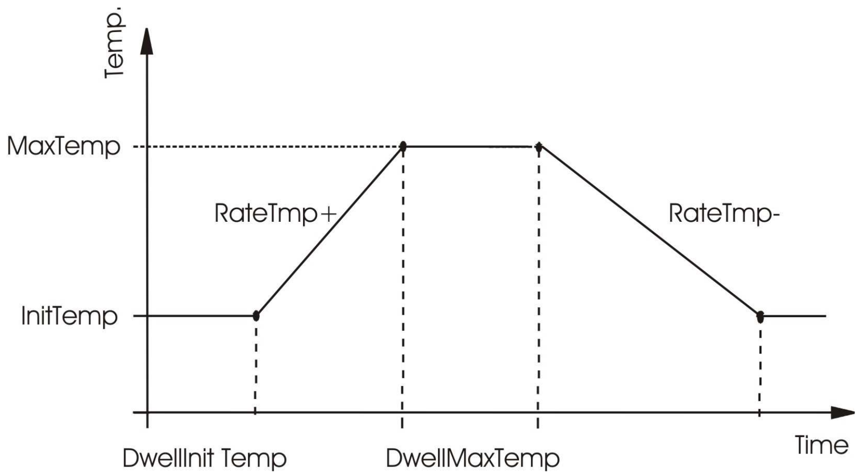
The “screed drying” function is used in order to support the drying of cement floors in which an underfloor heating system is installed. The function influences the flow temperature to optimize drying time and prevent the possibility of cracks in the floor.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **initTemp**: The flow temperature with which the screed drying phase will start.
- **dwellInitTemp**: The duration that the controller will hold the initial temperature
- **rateTemp (+/-):** The degree with which the flow temperature will rise/fall each day, once **dwellInitTemp/dwellMaxTemp** has run out.
- **maxTemp**: The maximum flow temperature during the function.
- **dwellMaxTemp**: The time during which the controller will hold the maximum temperature, after which the temperature will slowly decrease.
- **optPowerFail**: This parameter permits establishing how the controller should react if a power failure occurs during the screed drying period:

  =0: Restart the current step.

  =1: Restart completely.

  =2: Stop completely.
- **max-Xw**: This parameter permits setting a limit for the control deviation, which, when exceeded will result in a trouble indication signal.
- **dur-Xw**: This parameter permits setting a limit for the duration of the control deviation violation, which, when exceeded will result in a trouble indication signal.



Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.3.22.1	active			0	1	0	
2.5.x.3.22.2	initTemp	initial temperature		2.0	50.0	25.0	
2.5.x.3.22.3	dwellInitTemp	dwell initial temperature	d	0	10	1	
2.5.x.3.22.4	rateTmp+	rate of heat-up	K/d	1.0	50.0	5.0	
2.5.x.3.22.5	rateTmp-	rate of cool-down	K/d	1.0	50.0	5.0	
2.5.x.3.22.6	maxTemp	maximum temperature	°C	2.0	50.0	45.0	
2.5.x.3.22.7	dwellMaxTemp	Dwell maximum temperature	d	0	10	3	
2.5.x.3.22.8	optPowerFail	0:restart curr. Step 1:restart 2:stop		0	2	0	
2.5.x.3.22.9	max-Xw	max control deviat.	K	0.0	50.0	5.0	
2.5.x.3.22.10	dur-Xw	max duration of max control deviat.	H	0.0	240.0	0.5	

6.4 Status (2.5.x.4.n)

The “Status” menu provides an overview of the current operating mode of the heating circuit.

Every function affecting the operation of the heating circuit is considered, when the main status **opStatus** or the auxiliary status **opStatCode** is generated. The **opStatCode** is a hexadecimal number. The meaning of the characters shown in the parameters **opStatCode** and **eStatCode** is explained in the following tables.

- **opStatus:** Main status of heating circuit, controller status + timer status + compensation by:

  D = domestic hot water priority, **CT** = caretaker, **R** = remote control unit

  = Not active/Off: The required inputs are not assigned or the operating mode switch **oModRem** or **oModLoc** is assigned and in position “Off”.

  = Frost protection: The value at flow sensor is below the frost limit.

  = Building protection: The temperature at the room sensor is below the building protection limit.

  = Switch-off operation: The system main switch **System** assigned and switched off due to **D**omestic hot water priority or summer switch-off caused by caretaker function (**CT**).

  = Nominal operation: The status of timer channel **OT1...OT4** or **SOT1...SOT4.**

  = Manual operation: Manual control for an output is active. Operating mode switch **oModRem** or **oModLoc** is assigned and in position “Manual”

  = Red. Operation: Reduced operation (NO or SNOT) due to night reduction, **D**omestic hot water priority or **R**emote setting unit.

  = Support operation: Support operation activated (NO or SNOT) due to night switch-off, **D**omestic hot water priority or **R**emote setting unit.

  = Heating-up operation: Switch-on optimisation is effective.

- **trouble**: Trouble indication status of heating circuit (e.g. normal, outdoor too low)
- **source:** Composed of the setpoint source and the identification characters of the setpoint compensation e.g.: TIMER OT1 FTSR--BO—.
- **contr-**continuous: Current position of the valve in percent (continuous actuator).
- **contr2pnt:** Current signal to 2-point actuator.
- **contrOpen:** Current open signal to 3-point actuator.
- **contrClos:** Current close signal to 3-point actuator.
- **pump**: Current signal to pump.
- **troubInd**: Pending trouble indication.
- **Hea-dem**: Current variable demand via voltage input (external demand)

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.4.1	6.4.773 opStatus	6.4.774 OpStatus					
2.5.x.4.2	6.4.775 trouble	6.4.776					
6.4.777 2.5.x.	6.4.778 source	6.4.779 setpoint source and –compensation					
2.5.x.4.4	6.4.780 contr- cont	6.4.781 controller continuous	6.4.782				
2.5.x.4.5	6.4.783 contr2pnt	6.4.784 controller 2-point					

2.5.x.4.6	6.4.785 contrOpen	6.4.786 controller 3-pnt open					
2.5.x.4.7	6.4.787 contrClos	6.4.788 controller 3-pnt close					
2.5.x.4.8	6.4.789 pump	6.4.790 pump					
2.5.x.4.9	6.4.791 troubInd	6.4.792 trouble indication					
2.5.x.4.10	VDV	VDV	6.4.793				
2.5.x.4.11	6.4.794 opStatCode	6.4.795					
6.4.796 2.5.x.	6.4.797 eStatCode	6.4.798					

- **opStatCode:** The auxiliary operating status consists of six digits, as various functions of the heating circuit can be effective at the same time. The example below, together with the translation table, explains how to decode the sequence of characters.

	1st digit	2nd digit	3rd digit	4th digit	5th digit	6th digit
1	RC-button	oMod-remote switch	Universal limitation	pump extended running time	switch-on optimisation	flow correction
2	room occupancy	oMod-local switch	flow temperature limitation	pump switch-off	disconnection of loads	Caretaker
3	RC-button, room occupancy	oMod-remote switch, oMod-local switch	universal limitation, flow temperature limitation	Pump extended running, pump switch-off	switch-on optimisation, disconnection of loads	flow correction, caretaker
4	sys.-main switch	FC-oMod switch	Frost protection	pump/valve exercise	energy manager - limitation	heating up
5	sys.-main switch, RC-button	RC oMod switch, oMod-remote switch	frost protection, universal limitation	pump/valve exercise, pump extended running	energy manager – limitation, switch-on optimisation	heating up, flow correction
6	sys.-main switch, room occupancy	RC oMod switch, oMod-local switch	frost protection, flow temperature limitation	pump/valve exercise, pump switch-off	energy manager – limitation, disconnection of loads	heating up, caretaker
7	sys.-main switch, RC-button, room occupancy	RC oMod switch, oMod-remote switch, oMod-	frost protection, universal limitation, flow temperature	pump/valve exercise, pump extended running, switch-off	energy manager – limitation, switch-on	heating up, flow correction,

		local switch	limitation		optimisation, disconnection of loads	caretaker
8		RC-overtime-key 4h	adaptation 3Pnt	capacity limitation	pump speed reduction	switch-off optimisation
9		RC-overtime-key 4h, oMod-remote switch	adaptation 3Pnt, universal limitation	capacity limitation, pump extended running	pump speed reduction, switch-on optimisation	switch-off optimisation , flow correction
A		RC-overtime-key 4h, oMod-local switch	adaptation 3Pnt, flow temperature limitation	capacity limitation, pump switch-off	pump speed reduction, disconnection of loads	switch-off optimisation , caretaker
B		RC-overtime-key 4h, oMod-remote switch, oMod-local switch	adaptation 3Pnt, universal limitation, flow temperature limitation	capacity limitation, pump extended running, switch-off	pump speed reduction, switch-on optimisation, disconnection of loads	switch-off optimisation , flow correction, caretaker
C		RC-overtime-key 4h, RC oMod switch	adaptation 3Pnt, frost protection	capacity limitation, pump/valve exercise	pump speed reduction, energy manager - limitation	switch-off optimisation , heating up
D		RC-overtime-key 4h, RC oMod switch, oMod-remote switch	adaptation 3Pnt, frost protection, universal limitation	capacity limitation, pump/valve exercise, pump extended running	pump speed reduction, energy manager – limitation, switch-on optimisation	switch-off optimisation , heating up, flow correction
E		RC-overtime-key 4h, RC oMod switch, oMod-local switch	adaptation 3Pnt, frost protection, flow temperature limitation	capacity limitation, pump/valve exercise, pump switch-off	pump speed reduction, energy manager – limitation, disconnection of loads	switch-off optimisation , heating up, caretaker
F		RC-overtime-key 4h, RC oMod switch, oMod-local switch	adaptation 3Pnt, frost protection, universal limitation, flow temperature limitation	capacity limitation, pump/valve exercise, pump extended running, pump switch-off	pump speed reduction, energy manager – limitation, switch-on optimisation, disconnection of loads	switch-off optimisation , heating up, flow correction, caretaker

Tab. 12: Operating status code heating circuit

Example:

Display: opStatCode: = 083000

6.4.799 Meaning: 6.4.800 2nd digit = RC-overtime-key 4h effective.

6.4.801 3rd digit = Universal limitation and flow temperature limitation effective.

- **eStatCode:** The auxiliary trouble indication status consists of two digits, as several trouble indications can occur at the same time. The example below, together with the translation table, explains how to decode the sequence of characters:

	1st digit	2nd digit
1	Frost protection room	Error input
2	Frost protection system	Input terminal
3	Frost protection room, Frost protection system	Error input, Input terminal
4		max. contrDev. room temperature
5		max. contrDev. room temperature, error input

6		max. contrDev. room temperature, input terminal
7		max. contrDev. room temperature, Error input, input terminal
8		max. contrDev. flow temperature
9		max. contrDev. flow temperature, error input
A		max. contrDev. flow temperature, input terminal
B		max. contrDev. flow temperature, error input, input terminal
C		max. contrDev. flow temperature, max. contrDev. room temperature
D		max. contrDev. flow temperature, max. contrDev. room temperature, error input
E		max. contrDev. flow temperature, max. contrDev. room temperature, input terminal
F		max. contrDev. flow temperature, max. contrDev. room temperature, error input, input terminal

Tab. 13: Trouble indication status code heating circuit

6.4.802 **Example:**

Display: sStatCode: = 08

6.4.803 Meaning: 6.4.804 2nd digit = Maximum control deviation of flow temperature exceeded.

6.5 Manual operation (2.5.x.5.n)

Manual control is used to check the heating circuit pump and the valve or mixer drive for proper operation as well as to control direction and sense of rotation during commissioning.

NOTE:

In case of improper handling, the manual operating mode can cause damages to the system! Manual control overrides all functional limitations, including pump/valve exercise, frost protection and monitoring and signal functions.

- **valve**: Control of valve (continuous): 0...100=0...100%, 101=AUTOMATIC (operated by program).
- **valve**: Control of the valve (2-point): 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).
- **valve:** Control of valve (3-point): 0=close, 1=open, 2=stop, 3=AUTOMATIC (operated by program).
- **pump**: Pump: 0=OFF, 1=ON, 3=AUTOMATIC (operated by program).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.5.1	6.5.805 v	6.5.806 0-100:man 101:automatic	6.5.80	6.5.80	6.5.80	6.5.810	
2.5.x.5.2	6.5.811 v	6.5.812 0:close 1:open 3:auto		6.5.81	6.5.81	6.5.815	
6.5.816 2.	6.5.817 v	6.5.818 0:close 1:open 2:stop 3:automatic		6.5.81	6.5.82	6.5.821	
6.5.822 2.	6.5.823 p	6.5.824 0:off 1:on 3:automatic		6.5.82	6.5.82	6.5.827	

6.6 Service functions (2.5.x.6.n)

The heating circuit service menu permits trained personnel to configure the basic parameters, assign input terminals and output terminals and to set individual parameters for additional functions. This menu also displays data of interest for maintenance and service personnel.

General values (2.5.x.6.1.n)

The menu “General values” offers data of interest to maintenance and service personnel.

- **pump:** The running time of the pump since installation.
- **curdegDNum**: Current degree day number. This value represents how warm/cold the year has been up to now. This value can be used to compare the heating consumption between this year and the previous year.
- **histAvDegrC**: Degree day number of the previous year.

All these parameters can also be set manually in case the controller or a pump has to be exchanged.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.1.1	pump		h	0	999999	0	
2.5.x.6.1.2	curdegDNum	cur.degDayNum					
2.5.x.6.1.3	histAvDegrC	last years degree-day numb.					

Caretaker (2.5.x.6.2.n)

The “Caretaker” function service menu lists the date and time when the heating circuit last switched from summer to winter and from winter to summer mode as a result of the caretaker function.

- **swOfDa, swOftime**: Last change from winter to summer mode. The heating circuit was switched off.

- **swOnDate, swOntime**: Last change from summer to winter mode. The heating system was switched on.
- **swOfDur**: This parameter displays the duration since the last switch-off.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.2.1	6.6.828 swO	6.6.829 switchOff-date					
2.5.x.6.2.2	6.6.830 swO	6.6.831 switchOff-time					
2.5.x.6.2.3	6.6.832 swO	6.6.833 switchOn-date					
2.5.x.6.2.4	6.6.834 swO	6.6.835 switchOn-time					
2.5.x.6.2.5	6.6.836 swO	6.6.837 switchOff-duration	6.6.8				

Switch-on optimization (2.5.x.6.3.n)

The service menu of the “switch-on optimization” function displays the curve for the heating-up time in relation to the outdoor temperature.

- **HeUpDuxx**: heatingup duration at xx°C outdorr temperature

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.3.1	HeUpDu+25	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	0.0	
2.5.x.6.3.2	HeUpDu+15	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	1.5	
2.5.x.6.3.3	HeUpDu+5	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	4.5	
2.5.x.6.3.4	HeUpDu-5	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	7.5	
2.5.x.6.3.5	HeUpDu-15	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	48.0	
2.5.x.6.3.6	HeUpDu-25	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	48.0	
2.5.x.6.3.7	HeUpDu-35	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	48.0	
2.5.x.6.3.8	HeUpDu-45	heating-up durat. At xx°C outdoortemp.	H	0.0	48.0	48.0	

Switch-off optimization (2.5.x.6.4.n)

The service menu for the function “switch-off optimization” offers values of interest concerning the function and the building:

- **begBlock**: The time when energy blocking last started.
- **endBlock**: The time when energy blocking last ended.
- **buildTiC**: The building time constant, generated automatically by experience over a long time.
- **multipl**: This is a multiplier used to calculate how much influence the new calculated building time constant should have on the currently valid building time constant. The multiplier is getting smaller the more often a calculation has taken place.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.4.1	begBlock	beg. Energy blocking		00:00	23:59		
2.5.x.6.4.2	endblock	end energy blocking		00:00	23:59		
2.5.x.6.4.3	buildTiC	building-timeconst.	H	0.0	48.0		
2.5.x.6.4.4	multipl	corr.-multiplicator		0.0	1.0		

Heating curve adaption (2.5.x.6.5.n)

If the function “heating curve adaption” is not activated (see chapter “AF – heating curve adaption”), the list below represents the heating curve which is used to calculate the values for the flow temperatures setpoint for occupation time 1..

If the function “heating curve adaption” is activated and the adaption is set to **automatic (type=1**, see menu), the below list displays the current optimized flow temperature values of the heating curve. When the function is activated it will work with the values previously set.

If the function “heating curve adaption” is activated and the adaption is set to **manual (type=2**, see chapter “AF – heating curve adaption”), the flow temperature values of the heating curve in the list below must be set manually.

- **HC(+/-)xx**: heating curve value at xx°C outdoor temperature

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.5.1	HC+25	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.2	HC+20	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.3	HC+15	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.4	HC+10	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.5	HC+5	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.6	HC+-0	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.7	HC-5	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.8	HC-10	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.9	HC-15	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.10	HC-20	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.11	HC-25	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		

2.5.x.6.5.12	HC-30	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.13	HC-35	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.14	HC-40	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		
2.5.x.6.5.15	HC-45	heating curve-value bei xx°C outdoortemp	°C	0.0	160.0		

Setpoint limitation (2.5.x.6.6.1)

The service menu of the “setpoint limitation” function displays the currently valid flow temperature setpoint limited by this function according to parameters set in the chapter “AF – setpoint limitation”.

- **curFlLim**: this parameter shows the currently valid flow setpoint after limitation

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.6.1	curFlLim	flow-SP according to fl-limitation	°C	2.0	160.0		

Universal limitation (2.5.x.6.7.n)

If the current limit (**curLim)** is violated by the value supplied by the limitation sensor (**limit**), the universal limitation assumes control of the actuator from the flow temperature controller. The control behaviour of the universal limitation can be adjusted to the characteristics of the control system by means of the following controller parameters:

- **Xp:** PID-proportional Band. An increase will reduce the step response of the P proportion.
- **Tn:** PID-integral action time. An increase will extend the I-rate in time and thus reduce the impact. Tn=121.0min will deactivate the I-rate.
- **switchD:** PID-switching difference for a 2-point valve (*open/close* if **limit** (see chapter 0) is < or > than **limit1** (see chapter 0), *inactive* if **limit** is < or > than (**limit1** +/- **switchD**).

The following parameters show the current status of the universal limitation:

- **curLim**: This parameter shows the currently valid limit of the universal limitation
- **curUniLim:** This parameter shows the current value coming from the limitation sensor.
- **Y-uniLim:** This parameter shows the current correcting variable (signal to the actuator).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.7.1	Xp	proportional band	K	0.0	99999.9	25.0	
2.5.x.6.7.2	Tn	integral action time	min	0.1	121.0	1.0	
2.5.x.6.7.3	switchD	switch difference	K	0.0	99999.9	5.0	
2.5.x.6.7.4	curLim	cur limit value universal limitation	°C				
2.5.x.6.7.5	curUniLim	actual value universal limitation	°C				
2.5.x.6.7.6	Y-uniLim	correcting variable universal limitation	%				

Room compensation (2.5.x.6.8.n)

The service menu of the “room compensation” function displays the parameters for the PI controller activated by this function.

- **Kp**: This parameter is the amplifier gain of the step response (P-rate). A higher factor will result in a bigger flow temperature correction.
- **Tn**: This parameter is the integral action time. An increase of this value will extend the I-rate in time and thus reduce the impact. Tn=121,0min will deactivate the I-rate.
- **curFlCorr**: This parameter shows the current flow temperature correction as a result of the room compensation.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.8.1	Kp	amplifier gain	K/K	0.0	50.0	5.0	
2.5.x.6.8.2	Tn	integral action time	min	0.5	121.0	121.0	
2.5.x.6.8.3	curFlCorr	correction SP-flow with compRoomTemp	K				

Delayed outdoor temperature (2.5.x.6.11.n)

The service menu of the function “delayed outdoor temperature” contains:

- **delOutdT**: The currently valid (calculated) delayed outdoor temperature.
- **virRoom**: The currently calculated room temperature.

This function uses the building time constant **buildTiC** (see chapter “AF – switch-off optimization”).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.11.1	6.6.839 de	6.6.840 delayed outdoortemperature	6.6.84				
2.5.x.6.11.2	6.6.842 vir	6.6.843 virtual roomTemp.	6.6.84				

Signal (2.5.x.6.14.n)

The last 10 errors detected by the heating circuit modules are recorded and displayed in the “Signal” service menu. The first parameter always shows the most recent error while the last parameter shows the oldest error. The parameter text displayed constitutes the short text of the detected error. When pressing ”OK”, the info text appears, providing a more detailed description of the error. The date and time at which the error was detected is documented. **Please note that a power failure or controller cold start will delete all recorded errors!**

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.14.1		6.6.845 --.--.-- --:--					
2.5.x.6.14.2		6.6.846 --.--.-- --:--					
2.5.x.6.14.3		6.6.847 --.--.-- --:--					
2.5.x.6.14.4		6.6.848 --.--.-- --:--					
2.5.x.6.14.5		6.6.849 --.--.-- --:--					
2.5.x.6.14.6		6.6.850 --.--.-- --:--					
2.5.x.6.14.7		6.6.851 --.--.-- --:--					

2.5.x.6.14.8		6.6.852 --.--.-- --:--					
2.5.x.6.14.9		6.6.853 --.--.-- --:--					
2.5.x.6.14.10		6.6.854 --.--.-- --:--					

Examples:

Parameter text:	6.6.855 FP system	6.6.856 Frost protection system triggered.
	6.6.857 GS-FrRoom	6.6.858 frost protection room triggered.
	6.6.859 Xw-Flow	6.6.860 Control deviation of flow temperature too high.
	6.6.861 Xw-room	6.6.862 Control deviation of room temperature too high.
	6.6.863 TI system	6.6.864 Trouble indication system detected (input TI-system = 1).
	6.6.865 Defective sensor	6.6.866 Sensor fault detected.
6.6.867 Info text	6.6.868 <date, time>	6.6.869 e.g.: The error was recorded on 18.02.06 at 13:57.

Pump (2.5.x.6.16.1)

The “pump” function service menu contains the following parameter:

- **timer**: The current extended running time of the pump in the heating circuit (as it happens).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.16.1	6.6.870 t	6.6.871 cur-extRun time	6.6.87				

Controller (2.5.x.6.18.n)

The control behaviour of the flow temperature control can be changed by adjusting the following parameters:

- **Xp:** PID-proportional band. An increase will reduce the step response of the P-proportion.
- **Tn:** PID-integral action time. An increase will extend the I-rate in time and thus reduce the impact. Tn=121. 0min will deactivate the I-rate.
- **switchD2pnt:** The switch difference of a two-point control can be set.
- **Tmot:** The actual running time of the actuator should be set accordingly.
- **Y-contr.:** The current calculated signal to the actuator is displayed.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.18.1	Xp	proportional band	K	0.0	500.0	25.0	
2.5.x.6.18.2	Tn	integral action time	min	0.1	121.0	1.0	
2.5.x.6.18.3	switchD2pnt	switchDiff. 2pnt.	K	0.0	50.0	1.0	
2.5.x.6.18.4	Tmot	actuator running time	s	10	600	120	
2.5.x.6.18.5	Y-contr.	Contr corr variable	%				

Capacity limitation (2.5.x.6.20.n)

The “Capacity limitation” function service menu contains the following parameters:

- **curSP-corr:** The current setpoint correction due to capacity limitation.
- **Tn:** The integral action time which has an influence on the signal to the actuator. An increase will extend the I-rate in time and thus reduce the impact. Tn=121.0min will deactivate the I-rate.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.20.1	curSP-corr	correction setpoint with capacity limitat	K				
2.5.x.6.20.2	Tn	integral action time	min	0.1	121.0	5.0	

Screed drying (2.5.x.6.22.n)

The “screed drying” function service menu contains the following parameters:

- **setpTemp**: This is the current flow temperature setpoint, calculated according to parameters in the “AF – screed drying” chapter.
- **status**: This parameter shows in which phase the function is currently working.
- **timer:** This parameter shows the remaining time of a timer, if the function needs a timer for the phase of the function it is currently in.
- **num-PowerFail**: This parameter shows the number of power failures during the active phase of the function.
- **trInd.Xw**: This parameter shows if there has been a trouble indication because of control deviation violation.
- **reset:** This parameter permits resetting (acknowledging) the trouble indication.
- **set:** This parameter permits setting a timer.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.22.1	6.6.873 setpTe	6.6.874	6.6.87				
2.5.x.6.22.2	6.6.876 status	6.6.877					
2.5.x.6.22.3	6.6.878 timer	6.6.879	6.6.88				
2.5.x.6.22.4	6.6.881 num-	6.6.882 nr of power failures					
2.5.x.6.22.5	6.6.883 trInd.X	6.6.884 trouble control deviation					
2.5.x.6.22.6	6.6.885 reset	6.6.886 reset error condition		6.6.88	6.6.888 1	6.6.889	
2.5.x.6.22.7	6.6.890 set	6.6.891 timer set	6.6.89	6.6.89	6.6.894 9	6.6.895	

Sensor correction (2.5.x.6.23.n)

If the temperature values displayed in the “current values” menu differ from the current system values, a correction factor can be entered to adjust the individual sensor values:

- **room:** Correction factor (offset) for the room temperature sensor.
- **outdoor:** Correction factor (offset) for the outdoor temperature sensor.
- **flow**: Correction factor (offset) for the flow temperature sensor.
- **return**: Correction factor (offset) for the return temperature sensor.
- **limit**: Correction factor (offset) for the limitation temperature sensor.
- **shift**: Correction factor (offset) for the shifting temperature sensor.
- **fl-corr**: Correction factor (offset) for the flow correction temperature sensor.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.23.1	room		K	-10.0	10.0	0.0	
2.5.x.6.23.2	outdoor		K	-10.0	10.0	0.0	
2.5.x.6.23.3	flow		K	-10.0	10.0	0.0	
2.5.x.6.23.4	return		K	-10.0	10.0	0.0	
2.5.x.6.23.5	limit		K	-10.0	10.0	0.0	
2.5.x.6.23.6	shift		K	-10.0	10.0	0.0	
2.5.x.6.23.7	fl-corr		K	-10.0	10.0	0.0	

Terminal assignment (2.5.x.6.24.n)

The “terminal assignment” service menu enables the user to assign the components used in the “heating circuit” module to an input or output terminal. Each input terminal has a substitute value, which is used by the module in case of a sensor error. The module will then signal the sensor error and use the substitute value. The substitute value will only be shown in the list if terminal number “99” is entered as the terminal for the sensor. If terminal number 99 is entered for the sensor, the module will keep working with the substitute value but will no longer display any error.

- **room:** The terminal number for the room temperature sensor.
- **^subst:** The substitute value for the above mentioned sensor (terminal number=99).
- **outdoor:** The terminal number for the outdoor sensor.
- **flow:** The terminal number for the flow temperature sensor.
- **return:** The terminal number for the return temperature sensor.
- **limit:** The terminal number for the limitation temperature sensor.
- **shift:** The terminal number for the shifting temperature sensor.
- **setp-poti:** The terminal number for the external setpoint potentiometer.
- **fl-corr:** The terminal number for the flow setpoint correction sensor.
- **volFlow**: The terminal number for the volumetric flow meter (pulse).
- **HeatCapa**: The terminal number for the heat capacity meter (pulse).
- **hAmount**: The terminal number for the heat amount.
- **system:** The terminal number for the system main switch. (DI)
- **key:** The terminal number for the overtime key. (DI)
- **oModRC:** The terminal number for the external operation mode switch at the remote control unit.
- **oModLoc:** The terminal number for the local operating mode switch.
- **ContrCont:** The terminal number for the actuator in the heating circuit (0..10 V continuous).
- **Contr2pnt:** The terminal number for the actuator in the heating circuit (2-point).
- **ContrOpen:** The terminal number for the actuator in the heating circuit (3-point, open).
- **ContrClos:** The terminal number for the actuator in the heating circuit (3-point, close).
- **pump:** The terminal number for the pump.
- **troublelnd:** The terminal number of the alarm signal (contact output).
- **Hea-dem:** The terminal number for external demand (0...10 V input).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.5.x.6.24.1	room	room temperature		0	255	0	
2.5.x.6.24.2	^-subst.		°C	-40.0	160.0	20.0	
2.5.x.6.24.3	outdoor	outdoor temperature		0	255	0	
2.5.x.6.24.4	^-subst.		°C	-40.0	160.0	0.0	
2.5.x.6.24.5	flow	flow temperature		0	255	0	
2.5.x.6.24.6	^-subst.		°C	-40.0	160.0	20.0	
2.5.x.6.24.7	return	return temperature		0	255	0	
2.5.x.6.24.8	^-subst.		°C	-40.0	160.0	20.0	
2.5.x.6.24.9	limit	limitation sensor		0	255	0	
2.5.x.6.24.10	^-subst.		°C	-40.0	160.0	20.0	
2.5.x.6.24.11	shift	shift sensor		0	255	0	
2.5.x.6.24.12	^-subst.		°C	-40.0	160.0	20.0	
2.5.x.6.24.13	setp-poti	setpoint poti		0	255	0	
2.5.x.6.24.14	^-subst.		°C	5.0	30.0	20.0	
2.5.x.6.24.15	fl-corr	flow setpoint corr.		0	255	0	
2.5.x.6.24.16	^-subst.		K	0.0	100.0	50.0	
2.5.x.6.24.17	volFlow	volumetric flow		0	255	0	
2.5.x.6.24.18	^-subst.		l/h	0.0	3200.0	0.0	
2.5.x.6.24.19	heatCapa	heat capacity		0	255	0	
2.5.x.6.24.20	^-subst.		kW	0.0	3200.0	0.0	
2.5.x.6.24.21	hAmount	amount of heat		0	255	0	
2.5.x.6.24.22	system	sys.-main switch		0	255	0	

2.5.x.6.24.23	^-subst.			0	1	0	
2.5.x.6.24.24	key	RC-key		0	255	142	
2.5.x.6.24.25	oModRC	RC-opMode-switch		0	255	0	
2.5.x.6.24.26	oModLoc	oMod-local switch		0	255	151	
2.5.x.6.24.27	^-subst.			0	9	0	
2.5.x.6.24.28	contr-cont	controller continuous		0	255	0	
2.5.x.6.24.29	contr2pnt	controller 2-point		0	255	0	
2.5.x.6.24.30	contrOpen	controller 3-pnt open		0	28	0	
2.5.x.6.24.31	contrClos	controller 3-pnt close		0	28	0	
2.5.x.6.24.32	pump	pump		0	255	0	
2.5.x.6.24.33	troubInd	trouble indication		0	255	207	
2.5.x.6.24.34	VDV	Hea-dem		0	255	0	

Chapter 7 **The Trend module**

7.1 Trend x

The trend function is used in order to record certain parameters during an extended period of time. By so doing, trend can simplify troubleshooting significantly. In addition, the collection of data provides confirmation that the system is operating properly.

A maximum of **10 trend modules** are available in HPM, enabling the following menu items to be found under parameter number 2.6.**1** (= Trend 1) to 2.6.**10** (= Trend 10). Every trend module records one data point. The recording interval is scaleable, ranging from 1 minute to 999.9 minutes. The recording memory is operated as a ring buffer (FIFO: first in first out).

The recorded values can be viewed directly in the controller display. Using the building management software **IRMA** or **EXOscada** enables the collected values to be read from the trend storage.

7.2 Current values (2.6.x.1.4.n)

The ”Current values” menu displays the recorded values. The recording date and time are stored in the info text, which is displayed by pressing “OK”. The parameter number where the next value will be stored is shown in the parameter **curNumReco.**

- **curNumReco:** The parameter number where the next value will be stored.
- **Vx:** Read value.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.6.x.1.4.1	7.2.896 curN	7.2.897 current numer recording					
2.6.x.1.4.2	7.2.898 V1	7.2.899 11.04.115 10:03					
2.6.x.1.4.3	7.2.900 V2	7.2.901 11.04.115 10:03					
…	7.2.902 …	7.2.903 11.04.115 10:03					
2.6.x.1.4.50	7.2.904 V49	7.2.905 11.04.115 10:03					
2.6.x.1.4.51	7.2.906 V50	7.2.907 11.04.115 10:03					

7.3 Additional function (2.6.x.3.n)

Recording (2.6.x.3.2.n)

In this menu, the recording interval is defined. The parameter **active = 1** starts the recording process.

- **active**: This parameter activates (1) or deactivates (0) the function.
- **intReco**: The recording interval (in minutes).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.6.x.3.2.1	7.3.908 ac	7.3.909		7.3.91	7.3.911	7.3.912	

7.3.913 2.	7.3.914 in	7.3.915 interval Recording	7.3.91	7.3.91	7.3.918	7.3.919	

Controller (2.6.x.3.10.1)

To allow clear allocation of the trend function to the assigned data point, a descriptive name of up to 19 characters in length can be entered here (e.g. “DH secondary flow”). Entering letters requires a PC/laptop and the building management software **IRMA** or **EXOscada**.

- **longDes**: Freely adjustable reference name for the trend program

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.6.x.3.10.1	longDes	free adjustab. Prog.-longRef.					

7.4 Status (2.6.x.4.1)

The Status menu provides an overview of the current mode of operation of the trend program. The operating status **opStatus** displays the current operating mode as plain text.

- **opStatus**: Operating status of the trend program.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.6.x.4.1	opStatus						

7.5 Service function (2.6.x.6.n)

Recording (2.6.x.6.2.n)

This menu item displays the latest, as well as the current, recording value.

- **recoValue**: Last recorded value.
- **curValue**: Current value of the parameter.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.6.x.6.2.1	recoValue	last recorded value					
2.6.x.6.2.2	curValue	cur value					

Terminal assignment (2.6.x.6.3.1)

The trend function can be assigned to any terminal (1…255) of the controller.

- **DPnt**: The terminal number which in- or output value should be recorded.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.6.x.6.3.1	DPnt	data point		0	255	0	

Reference/delete (2.6.x.6.4.1)

The “SF – reference/delete” menu permits deleting the contents of the recording memory.

- **cold start mem**: Cold start memory.
  =1: Clear memory.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
2.6.x.6.4.1	7.5.920 cold	7.5.921 cold start memory		7.5.92	7.5.92	7.5.924	

Part II Advanced configuration and settings

Table of contents

Part II Advanced configuration and settings

Chapter 8 **Global**

The “Global” menu is part of the controller operating system. General functions and parameters, such as for instance the controller software version, can be found here. This area is reserved for skilled service personnel, since erroneous parameter settings in this menu can lead to the controller and heating system being damaged.

The “Global” menu can only be accessed after entering **the level 4 access code**. Press and hold “OK” for 3 seconds and you will then be prompted to enter the access code. Please refer to the chapter “Access codes” for more information.

8.1 Signal (1.1.1.n)

The “Signal” menu is used only in combincation with the BMS software IRMA.

8.2 Error (1.2.n)

The “Error” menu is used only in combincation with the BMS software IRMA.

8.3 Service (1.3.n)

The “Service” menu displays important information to the service technician concerning the controller. This includes hardware type, software version and date, battery status etc. In this menu the service technician can also initiate warm or cold starts and set the access codes.

8.3.1 Hardware (1.3.1.n)

The “Service – hardware” menu displays important information to the service technician concerning the hardware of the controller.

- **Con_type**: Controller type (shown in display).
- **c.type**: Controller type (numeric number).
- **hw.type**: Hardware type.
- **serialnr**: Serial number.
- **functTest**: Shows if function test was performed.

- **dtOfManuf**: Manufacturing date.
- **dtOfServ**: Date of latest service. Can be set by service technician.
- **contr.Temp**: Controller internal temperature.
- **BattVolt**: Battery voltage
- **prodNr**: Production number.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.3.1.1	RU_type	controller type					
1.3.1.2	c.type						
1.3.1.3	hw.type			00	FF		
1.3.1.4	Serialnr			0	9999999	0	
1.3.1.5	functTest						
1.3.1.6	dtOfManuf	date of manufacturing					
1.3.1.7	dtOfServ	date of last service					
1.3.1.8	contr.Temp		°C				
1.3.1.9	BattVolt	Batterie voltage	V				
1.3.1.10	prodNr	poduction number AAYYMMDDNNNN					

8.3.2 Software (1.3.2.n)

The “Service – software” menu displays important information for the service technician concerning the software of the controller.

- **progDat:** Date of operating system.
- **version:** Version of operating system.
- **language:** Shows the language set currently used in the controller. This parameter can be adjusted by the service technician.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.3.2.1	progDat	program date					
1.3.2.2	version	version					
1.3.2.3	Language						

8.3.3 Cold-warm start (1.3.4.n)

The “Service – cold-warm start” menu offers the possibility to initiate cold or warm starts of the controller. It also offers information on the total number of cold and warm starts and when a warm start was last initiated.

- **warm start:** If the service technician enters “1”, a warm start is initiated. All parameters will retain their value.
- **coldStSys:** If the service technician enters “1”, a cold start is initiated. All parameters will be set to their default values. The system diagram will **not** be lost.
- **no.cold starts**: Number of cold starts.
- **no.warm res.:** Number of warm starts.
- **timeSWarmS**: Amount of time elapsed since last warm start.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.3.4.1	warm start			0	1	0	
1.3.4.6	coldStSys	cold start system		0	1	0	
1.3.4.8	no.cold starts	number of cold starts					
1.3.4.9	no.warm res.	number of warm starts					

1.3.4.10	timeSWarmS	time since last hot start	s				

8.3.4 Access codes (1.3.5.n)

The “Service – access codes” menu permits viewing and changing of access codes for each access level. It also permits setting at which level an access code should be required.

- **level x:** Access code at level x.
- **protect**: Access code needed at level x.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.3.5.1	level 1	code for chosen access level					
1.3.5.2	level 2	code for chosen access level					
1.3.5.3	level 3	code for chosen access level					
1.3.5.4	level 4	code for chosen access level					
1.3.5.7	protect	accesss code needed at level		1	5	1	

8.3.5 Project management (1.3.7.n)

The “project management” function is activated automatically after the system diagram number has been entered. The controller records the first 200 parameters that have been changed after the initial start-up. When the controller has been adjusted and a backup is being made, the service technician‘s project management software will read these parameters first.

- **active** : This parameter activates (1) or deactivates (0) the function.
- **delete**: This parameter will delete the list (1).
- **numOfPara**: This parameter shows the number of recorded parameters.
- **parNo.n**: This parameter shows the parameter number of the changed parameter.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.3.7.1	active			0	1	0	
1.3.7.2	delete			0	1	0	
1.3.7.3	numOfPara			0	200		
1.3.7.4	parNo.1						
1.3.7.5	parNo.2						
1.3.7.6	parNo.3						
…	…						
1.3.7.202	parNo.199						
1.3.7.203	parNo.200						

8.4 System clock (1.4.n)

The “System clock” menu permits the time and date of the system clock to be adjusted. This menu is also visible in the menu “Timer – service”, in which adjusting the system clock is also possible.

8.4.1 Status (1.4.1.n)

The “System clock – status” menu shows information for the daylight saving time and the current weekday.

- **Summer or winter:** current season of the clock (daylight saving or non-daylight saving time). Calculated based on the current date and current mode.
- **day:** Current weekday (calculated from current date).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.4.1.1	Sum/Win	current season summer / winter					
1.4.1.3	day:	calculated from current date					

8.4.2 Time (1.4.2.1)

The “System clock – time” menu permits viewing and adjusting the current time of day.

- **curTime:** Current time.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.4.2.1	curTime	current time		00:00	23:59		

8.4.3 Date (1.4.3.1)

The “System clock – date” menu permits viewing and adjusting the current date.

- **curDate:** Current date.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.4.3.1	curDate	current date		01.01.90	31.12.89		

8.4.4 Mode (1.4.4.n)

The “System clock – mode” menu permits adjusting the operation of the system clock:

- **typSuWiSwitch:** Type of summer/winter (daylight saving) switch:

  =0: Summer/winter switch according to dates given in dateSu and dateWi.

  =1: Automatic summer/winter switch according to statutory regulations in Europe.
- **dateSu:** Date when clock should switch to summer (daylight saving) time (typSuWiSwitch=1).
- **dateWi:** Date when clock should switch to winter time (typSuWiSwitch=1).
- **operating mode:** Operating mode of the system clock:

  *=0***:** Clock is running according to frequency from the power line.

  *=1***:** Clock is running according to internal crystal.
- **adjustment:** If the clock is running according to the internal crystal (operating mode=1) the frequency can be adjusted using this parameter.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
1.4.4.1	typSuWiSwitch	su/wi-switch at 0:date/time 1:by law		0	1	1	
1.4.4.2	dateSu	date summer		01.01.90	31.12.89	--.--.--	
1.4.4.4	dateWi	date winter		01.01.90	31.12.89	--.--.--	
1.4.4.7	operating mode	0:powerln. 1:crystal		0	1	1	
1.4.4.8	adjustment			-99	99		

8.5 Structure (1.5)

8.5.1 Display (1.5.6)

The “Structure – display” menu permits the look of the display to be adjusted. It is possible to show terminal values in the default display.

- **terminalRx**: Sensor terminal number to be displayed in row x.
- **shortDesRx**: Short description of terminal value displayed in row x.
- **textR1**: The text that should be displayed in row x.

Par. no.	Name	Info text	Un.	Min.	Max.	Default	Description
1.5.6.1	terminalR1	Number of terminal displayed in row 1		0	247	0	
1.5.6.2	shortDesR1	short Descry. For row 1					
1.5.6.3	textR1	Text for row 1					
1.5.6.4	terminalR2	Number of terminal displayed in row 2		0	247	0	
1.5.6.5	shortDesR2	short Descry. For row 2					
1.5.6.6	textR2	Text for row 2					
1.5.6.7	terminalR3	Number of terminal displayed in row 3		0	247	0	
1.5.6.8	shortDesR3	short Descry. For row 3					
1.5.6.9	textR3	Text for row 3					
1.5.6.10	terminalR4	Number of terminal displayed in row 4		0	247	0	
1.5.6.11	shortDesR4	short Descry. For row 4					
1.5.6.12	textR4	Text for row 4					

8.5.2 Web (1.5.7)

The “Structure – web” menu permits the adjustment of system relevant web graphics.

- **Contr.name**: Name of controller in the web graphic.
- **protect**: access to web graphic is protected by access level x.

  =0: unprotected

  =1: protected
- **username x**: username for level x.
- **level x**: password for level x

Par. no.	Name	Info text	Un.	Min.	Max.	Default	Description
1.5.7.1	contr.name	name of controller					
1.5.7.2	protect	access 0:unprotected 1:protected		0	1	0	
1.5.7.4	username 1					user	

1.5.7.5	level 1					11111111	
1.5.7.6	username 2					operator	
1.5.7.7	level 2					22222222	
1.5.7.8	username 3					service	
1.5.7.9	level 3					44444444	

Chapter 9 **Interfaces**

This section introduces the parameters of the various controller interfaces. HPM is always equipped with a standard USB interface but can also be equipped with a number of additional interfaces.

- **USB**: Service interface. Every HPM comes equipped with the USB interface for testing and uploading of configuration files or updates.
- **RS-485:** Serial interface. Can be used to connect the controller with the heat pump, in a network with other controllers or with a BMsystem.
- **Ethernet:** TCP/IP network. Can be used to communicate in an Ethernet network or via the Internet.

9.1 USB (3.1.1.n)

As a standard, every HPM comes equipped with a micro USB interface. This is a passive interface, meaning it needs a PC with a master USB interface in order to function.

The applications of the USB interface includes data backup, parameter setting, uploading updates or configuration files and communication via a remote control program (IRMA control).

- **monActive**: Monitor active (internal use)..
- **monActive**: USB interface active (1) or not active (0).

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
3.1.1.1	monActive			2400	38400	9600	
3.1.1.2	active			0	1	0	

9.2 RS-485 (3.2.n)

Interface – RS-485 (RJ11 or terminal connection)

The RS-485 interface with RJ11 plug can be used to connect remote displays or control units

The RS-485 interface with terminal connector has to be used to conect the HPM with the heat pump. (CAUTION! A special adapter cable is necessary!)

9.2.1 General values (3.2.1.n)

To establish a connection between a PC and the controller, or in combination with several controllers, all components (controller, interface adaptor, PC) must operate at the same transfer speed and communication has to be activated using “Active = 1”.

The transfer speed of the RS-485 interface set in this menu must match the baud rate defined by the communication partner (remote control unit or heat pump)

- **active** : This parameter activates (1) or deactivates (0) the function.
- **baudrate**: Baud rate of the RS-485 interface. The default value is 9600 baud. The following transmission speeds are possible: 2400, 4800, 9600, 19200 and 38400.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
3.2.1.1	active				1	0	
3.2.1.2	baudrate			2400	38400	9600	

9.3 Ethernet (3.3.n)

The ethernet interface provides the controller with a number of options for communication. Primarily, it enables connecting to a building management system, either by using an Ethernet network or via the Internet. It also enables use of the built-in web server and communication with the controller, either via a normal browser through the Internet or directly.

- **active** : This parameter activates (1) or deactivates(0) the ethernet.
- **host name**: Own host .name
- **MAC**-adr: Ethernet address in MAC format.
- **DHCPPC**:

  =0: Fixed IP-address.

  =1: IP addresses is provided by DHCP server.
- **IP-nr**: If DHCPPC=0, the fixed IP address should be entered here.
- **netMask**: Network mask.
- **defaultGW**: Default gateway address.
- **nameserver**: Nameserver.
- **Link**:Shows whether the connection is OK.
- **lowSpeed**: set fixed speed (10 Mbit/s) for ethernet connection

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
3.3.1	active	Activation interface		0	1	1	
3.3.2	host name	Own host name				HPM-800000	
3.3.3	MAC-adr	Ethernet address				00:1F:FC:80:00:00	
3.3.4	DHCPC	Receive IP address automatically		0	1	1	
3.3.5	IP-nr	Internet protocol address				192.168.178.131	
3.3.6	netMask	Network mask				255.255.255.0	
3.3.8	defaultGW	Default gateway				192.168.178.1	
3.3.9	nameserver	Nameserver				192.168.178.1	
3.3.10	Link						
3.3.11	lowSpeed	10 MBit/s fixed		0	1	0	

Chapter 10 Configuration

The “Configuration” menu permits adjustments to be made to the connection between the logical inputs/outputs of the programs and the physical input/output terminals of the controller. The functionality of the keys at the front of the controller can also be determined from here.

Specific settings can be made for the terminals, such as terminal type, smoothing time constant, unit and substitute value.

10.1 Input terminals (4.1.n)

Each input terminal can be adjusted for specific needs, as far as the hardware allows. The terminals can also be used as output terminals.

The input terminals can be used in the following way:

17		sensor	N		16
18		sensor	L		15
19		sensor	ground for 13		14
20		sensor	relay		13
21		sensor	not used		12
22		sensor	relay		11
23		sensor	ground for 9 + 11		10
24		sensor	relay		9
25	sensor ground	**contact / counter**	not used	earth	8
26		**contact / counter**	relay		7
27		0..10V	ground for 5 + 7		6
28		0..10V	relay		5
29		sensor ground	not used		4
30		not used	relay		3
31		not used	ground for 1 + 3		2
32		not used	relay		1

10.1.1 Sensors (terminals 17-24) (4.1.x.n)

The terminal programs converts the raw value (**rawVal**, the current input resistance) to the terminal value (**termVal**). The terminal value is transmitted to the assigned module.

The relation between raw value and terminal value is defined by a characteristic curve (see explanation and curve on page 25). This curve is itself defined by the parameters **EPSou1**, **OpntTI1** (start of the curve), **EPSou2** and **OpntTI2** (end of the curve).

- **termMod**: Terminal mode

  =*0*: Terminal is a sensor value (AI)

  =*1*: Terminal is a contact input (DI)

- **typeSensor**: Type of Sensor

  =*0*. PT1000

  =*1*: NI1000

  =*2*: NI1000LG (Landis & Gyr)

- **rawVal**: Raw value; resistance is measured directly at the terminal.

- **smTConst:** Smoothing time constant. The calculation of the terminal value from the raw value is delayed in order to reduce the influence of parasitic errors on the sensor cable.
- **curStat:** Using this parameter the incoming signal can be inverted if **termMod=1**
- **termVal:** The calculated terminal value that is passed on to the module.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.1.x.3	termMod	Terminal mode 0:none, 1:EK		0	1	0	
4.1.x.8	typeSensor	0=PT1000, 1=Ni1000, 2=Ni1000LG					
4.1.x.14	rawVal		kOhm	0.0	5.0		
4.1.x.115	smTConst	Smoothing time constant	s	0	100	1	
4.1.x.117	curStat	0:log 0>active 1:log 1>active		0	1	1	
4.1.x.210	termVal	Terminal value					

10.1.2 Counter/Contact (terminals 25-26) (4.1.9.n)

Terminals 25 and 26 can be used as pulse **counter** inputs or digital inputs, such as for an overtime key.

- **termType**: Terminal mode.

  =*3*: Terminal is used to receive a signal from a button (e.g. overtime)

  =*4*: Terminal is a contact input (DI).

  =*5*: Terminal is a counter, e.g. for reading pulses from a heat meter.

- **rawValue:** Raw value. Resistance is measured directly at the terminal.
- **unit**: Only for termType=5: The unit of one pulse:

  =*41*: kw/h

  =*9*: l (liter)

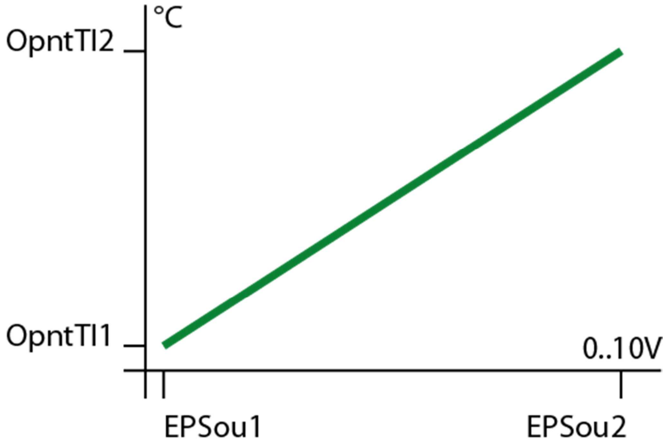
  =*10*: m3

- **standard:** Standardized. Only for termType=5. This parameter permits setting what significance a single pulse should have (e.g. 1.000 kWh, 1.000 l or 0.001 m3 per pulse).
- **curStat:** This parameter permits the incoming signal to be inverted. Only valid for **termType**=4.
- **termVal:** The calculated terminal value which is passed on to the module.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.1.9.2	termType	4:EK 5:counter 3:button		3	5	3	
4.1.9.14	rawValue	raw value		0	9999999	0	
4.1.9.105	unit	unit		0	255	41	
4.1.9.109	standard	Standardized	unit	0.000	9.999.000	1.000	
4.1.9.117	curStat	0:log 0>active 1:log 1>active		0	1	1	
4.1.9.210	termVal	Terminal value					

10.1.3 0…10 V (terminals 27-28) (4.1.11.n)

Terminals 27 and 28 are universal terminals and can therefore be used either as inputs or open collector (0...10V) outputs. **Terminal 27 and/or 28 are assigned as input terminals by default. When the terminals are activated in the menu "Configuration – output terminals – 0..10V (27-28)", they will become output terminals and will vanish from this list**.



- **rawValue** : Raw value; voltage measured directly at the terminal or sent out via this terminal.
- **EPSoux**: Entry point of the characteristic curve (see above graphic).
- **unit**: Unit of the terminal value, default is 150=%
- **OpntTlx**: Output point of the characteristic curve (see above graphic).
- **smTConst:** Smoothing time constant.
- **termVal:** The calculated terminal value which is passed on to the module or sent by it.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.1.11.14	rawValue		V	0.0	10.0		
4.1.11.103	EPSou1	entry point 1	V	0.0	10.0	0.0	
4.1.11.104	EPSou2	entry point 2	V	0.0	10.0	10.0	
4.1.11.105	unit			0	255	150	
4.1.11.106	OpntTl1	output pnt 1	°C	-999999.9	999999.9	0.0	
4.1.11.107	OpntTl2	output pos 2	°C	-999999.9	999999.9	100.0	
4.1.11.115	smTConst	smoothing time constant	s	0	100	1	
4.1.11.210	termVal	terminal value	°C				

10.1.4 Heat capacity/volumetric flow (terminals 35-36) (4.1.13.n)

Terminals 35 and 36 can be used to transmit values like heat capacity or volumetric flow to the modules. These can be either calculated by the controller from the pulse of an input terminal and parameter values, or entered by technical personnel during initial installation.

- **rawVal** : Raw value; the time measured between the last two pulses.
- **smTConst:** Smoothing time constant. The calculation of the terminal value from the raw value is delayed to reduce the influence of parasitic errors. 0 = no smoothing
- **termVal:** The calculated terminal value which is passed on to the module.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.1.13.14	rawValue	raw value	s				
4.1.13.115	smTConst	smoothing time const	s	0	100	0	

4.1.13.210	termVal	terminal value					

10.1.5 Free terminal (terminal 100) (4.1.15.n)

Terminal 100 is intended for free use with no assignment to a hardware terminal. It can provide either a configurable substitute value or a value written via the communication network to which the controller is connected. (e.g. outdoor temperature).

If no source address (**adrSrc** = 0) is entered, the default value **defValue** is transmitted as terminal value to the module.

If **adrSrc** = 1 or 2 is entered as a source address, the **rawvalue** is transmitted as a terminal value. It can be written to by master controllers (e.g. CLEVER master) connected to the same communication network as the controller.

- **active**: This parameter activates (1) or deactivates (0) the terminal.
- **termType**: Type of terminal

  =1: 0...10V input.

  =2: Sensor (PT1000).

  =3: Potentiometer.

  =4: Digital input (DI).

  =5: Stage input (0,1,2,3,4,5) can be used as operating mode switch (OFF, auto, summer, holiday, continuous operation or manual operation).
- **adrSrc**: Address of the source:

  =0: default value (4.1.15.125).

  =1: raw value (4.1.15.14) If communication is lost use last raw value.

  =2: raw value (4.1.15.14) if communication is functional; otherwise defValue (4.1.15.125) in case connection to master is lost.
- **rawVal** : Raw value. Value written by the master controller via the network.
- **unit:** Dimension of the terminal value.
- **defValue:** Default value.
- **termVal:** The calculated terminal value which is passed on to the module.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.1.15.1	active	active		0	1	0	
4.1.15.2	termType	0:MS 1:0-10V 3:pot. 2:Pt1000 4:EK 5:sta		0	5	0	
4.1.15.4	adrSrc	Address source		0	2	0	
4.1.15.14	rawValue	raw value		-40.0	160.0		
4.1.15.105	unit	unit		0	255	108	
4.1.15.125	defValue	Default value	°C	-999999.9	999999.9	0.0	
4.1.15.210	termVal	Terminal value	°C				

10.2 Output terminals (4.2.n)

The HPM has 7 relay outputs at terminals 1, 3, 5, 7, 9, 11 and 13.

10.2.1 Relay outputs (terminals 1, 3, 5, 7, 9, 11, 13) (4.2.x.n)

- **valSource** : Source value. The value coming from the module.
- **curStat:** This parameter permits inverting the value coming from the module.

  *=0***:** Value will be inverted.

  *=1***:** Value remains as sent by module.
- **termVal:** Current terminal value at the output terminal.

The ground for the relays on terminal 1+3 is on terminal 2.

The ground for the relays on terminal 5+7 is on terminal 4.

The ground for the relays on terminal 9+11 is on terminal 10.

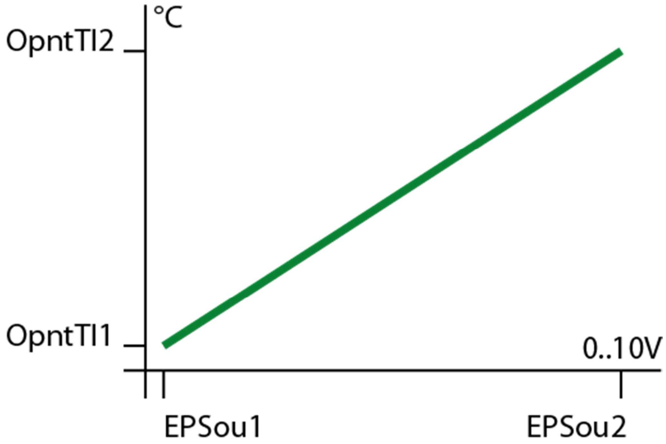
The ground for the relay on terminal 13 is on terminal 14.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.2.x.14	valSource	source value		0	1	0	
4.2.x.117	curStat	0:log 0>active 1:log 1>active		0	1	1	
4.2.x.210	termVal	terminal value					

10.2.2 0...10 V (terminals 27-28) (4.2.8.n)

Terminals 27 and 28 are universal terminals and can therefore be used either as inputs/outputs (0…10V) or open collector outputs. When used as outputs, they can for instance be used to control electronic pumps (START/STOP or MIN/MAX) or valve actuators with a continuous 0…10V signal. If terminal/s 27 and/or 28 has been assigned as output terminal(s), they will not be available in the list of input terminals.

curStat = 0 inverts the value **valSou** coming from the module. **EPSou1**, **EPSou2**, **OpntTl1** and **OpntTl2** are used to set the characteristic curve of the 0…10V output.



If the terminal is assigned as an output in a module, it is automatically configured accordingly (pump = open collector, valve continuous = 0…10V).

- **active:** This parameter activates (1) the terminal and defines it as an output terminal, in which case the terminal will vanish from the list of input terminals.
- **valSou:** The value coming from the module.
- **EPSoux:** The entry point of the characteristic curve (see above graphic).
- **OpntTlx:** The output point of the characteristic curve (see above graphic).
- **termVal:** Current terminal value at the output terminal.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.2.8.1	active	raw value		0	1	0	
4.2.8.14	valSou	sourceVal		0	1000		
4.2.8.103	EPSou1	entryPnt1 source		0	1500	0	
4.2.8.104	EPSou2	entryPnt2 source		0	1500	100	
4.2.8.106	OPTerm1	output pnt 1	V	0.0	10.0	0.0	
4.2.8.107	OPTerm2	output pnt 2	V	0.0	10.0	10.0	
4.2.8.210	termVal	terminal value	V				

10.3 Keys (4.4.n)

10.3.1 Maintenance (terminal 141) (4.4.1.n)

Pressing “UP” while in the standard menu will cause the menu for the maintenance personnel to appear. The heat pump can then be started or stopped by pressing “+” or “-”. This will set the terminal value from terminal 141 to “1”. This terminal can be assigned to the **maintenance** of the heat pump module. Activation of this terminal will result in the start of the heat pump.

If the maintenance function is activated, the display message “HEAT PUMP MAINTENANCE ACTIVE” is shown.

- **rawValue**: This parameter shows the raw value of the terminal.
- **termVal** : The value which is being sent to the module.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.4.1.14	rawValue	raw value		0	1		
4.4.1.210	termVal	terminal value					

10.3.2 Overtime keys (terminals 142-144) (4.4.2.n)

Pressing “DOWN” while in the standard menu will cause the menu for the overtime keys to appear.

It is possible to navigate to the desired overtime key by using “UP” and “DOWN”, followed by activating overtime for the desired module. The activation will result in the terminal value of the corresponding terminal being set to “1”. These terminals can be assigned to the input terminal key of the respective module. This configuration is performed automatically when selecting a system diagram:

- terminal 142 = The overtime key for heating circuit 1.
- terminal 143 = The overtime key for heating circuit 2.
- terminal 144 = The overtime key/forced loading for the domestic hot water circuit.
- **rawValue**: This parameter shows the raw value of the terminal.
- **termVal** : The value which is being sent to the module.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.4.2.14	rawValue	raw value		0	1		
4.4.2.210	termVal	terminal value					

10.4 Switch (4.5.1.n)

10.4.1 Local operating mode switch (terminal 151) (4.5.1.n)

Pressing “+” while in the standard menu will cause the menu for the controller’s local operating mode switch to appear. This terminal (151) is assigned to the **oModLoc** input terminals of all modules, making them react immediately to the position of the operating mode switch.

The operating mode switch has 6 positions:

0=OFF: All modules are set to the status “Not active/Off” and all controller outputs (relays and 0...10V) are switched off. Frost or building protection are not active.

1=AUTO: This is the normal operating mode, resulting in a timer-related operation and setpoint reduction in non-occupation mode (day/night).

2=SUMMER: The heating circuits of the controller are set to “switch-off operation”. All other modules remain in automatic mode.

3=HOLIDAY: The setpoints for non-occupation time apply to the heating circuits and domestic hot water circuit.

4=CONTINUOUS: The setpoints for occupation time 1 (OT1) apply and the heating circuits and domestic hot water circuit will operate continuously with reduction during the night.

5=MANUAL: The output values set in the menu “manual operation” in each module will apply. This allows manual operation of pumps and valves.

- **active**: This parameter permits activation (1) or deactivation (0) of the main operating switch.
- **rawValue**: This parameter shows the rawValue and how it was set by the customer. **← ???**
- **defValue**: The default value in case the main switch is deactivated (active=0).
- **termVal** : The operating mode that is being sent to the modules.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
4.5.1.1	active	active		0	1	1	
4.5.1.14	rawValue	raw value					
4.5.1.125	defValue	default-value		0	5	0	
4.5.1.210	termVal	terminal value					

Chapter 11 System diagram

The parameters in this menu are set automatically when the system diagram is first entered during initial installation. When the controller is started for the first time, you will be asked a series of questions which will result in the controller being configured automatically (see the instruction). Naturally, these parameters can also be set manually.

- **sysDiagram**: The number of the choosen system diagram.
- **typeSensor**: The type of sensor used:

  =0: Pt1000

  =1: NI1000

  =2: NI1000 Landis and Gyr
- **typeHPx**: Type of heat pump

  =1: contact
- **buffer stor**.: A buffer tank is included (1) or not included (0) in the system (only visible if **sysDiagram**=99999).
- **numOfDHW**: The number of domestic hot water circuits (only visible if **sysDiagram**=99999).
- **numOfHC**: The number of heating circuits (only visible if **sysDiagram**=99999).
- **cold start**: Cold start to load new system diagram.

Par. no.	Name	Info text	Un.	Min.	Max.	Def.	Description
5.1	sysDiagram	system diagram (look at Manual)		0	99999		
5.2	typeSensor	0:Pt1000 1:Ni1000 2:Ni1000LG		0	2		
5.3	type-HP1	Type of heat pump for HP1					Only visible if 99999
5.4	type-HP2	Type of heat pump for HP2					Only visible if 99999
5.5	type-HP3	Type of heat pump for HP3					Only visible if 99999
5.6	buffer	buffer storage		0	1		Only visible if 99999
5.7	no-DHW	number DHW circuits		0	1		Only visible if 99999
5.8	no-hCu	number heatCircuits		0	2		Only visible if 99999
5.100	cold start	system cold start load diagram		0	1		

Chapter 12 **System overview**

The “System overview” menu permits a quick overview to be obtained of the controller and the heating system. This function is especially useful for technical personnel when performing maintenance or attempting to find the reason behind a trouble indication or an alarm. The user can reach the system overview menu by pressing “-“ while in the standard menu.

12.1 Controller (6.1.n)

The “Controller” menu displays the most relevant parameters concerning the controller.

- **PAW-HPMx**: The controller type, f.i. PAW-HPM1
- **progDat**: The production date of the software inside the controller.
- **version**: The version of the software inside the controller.
- **prodNr**: The production number of the controller.
- **curTime**: The current system time.
- **curDat:** The current system date.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
6.1.1	PAW-HPMx	controller type					
6.1.4	progDat	program date					
6.1.5	version	version					
6.1.6	prodNr	production number AAYYMMDDNNNN					
6.1.7	curTime	current time		00:00	23:59		
	curDate	current date		01.01.90	31.12.89		

12.2 Heating circuit x (6.x.n)

The “Heating circuit x” menu displays the most relevant parameters concerning the heating circuit with the number x.

- **opStatus**: Current status of the heating circuit program.
- **trouble**: Displays if there are any pending trouble indications.
- **source**: Displays which source has influence on the setpoint and the amount of compensation.
- **outdoor**: Current outdoor temperature.
- **SP-room**: Current setpoint for the room temperature.
- **room**: Current room temperature.
- **SP-flowTemp**: Current calculated flow temperature setpoint.
- **flowTemp**: Current flow temperature.

- **return**: Current return temperature.
- **pump**: Current signal to the pump.
- **Y-contr**: Current signal to the valve.
- **roomOT1**: Room temperature setpoint for occupation time 1.
- **roomNO**: Room temperature setpoint for non-occupation time.
- **floRedNO**: Flow temperature reduction in non-occupation time.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
6.x.1	opStatus	OpStatus					
6.x.2	trouble						
6.x.3	source	setpoint source and –compensation					
6.x.4	outdoor	outdoor temperature	°C				
6.x.5	SP-room	setpoint room temp.	°C				
6.x.6	room	room temperature	°C				
6.x.7	SP-flow	setpoint flow temperature	°C				
6.x.8	flow	flow temperature	°C				
6.x.9	return	return temperature	°C				
6.x.10	pump	pump					
6.x.11	Y-contr.	controller output	%				
6.x.12	roomOT1	setpoint OT1	°C	2.0	50.0	20.0	
6.x.13	roomNO	setpoint NO	°C	2.0	50.0	15.0	
6.x.14	floRedNO	supp-reduct. NO	K	0.0	50.0	10.0	

12.3 Domestic hot water circuit (6.4.n)

The “Domestic hot water circuit” menu displays the most relevant domestic hot water parameters.

- **opStatus**: Current status of the DHW circuit program.
- **trouble**: Displays if there are any pending trouble indications.
- **source**: Displays which source has influence on the setpoint and the amount of compensation.
- **SP-stor**: Current setpoint for the storage tank temperature.
- **storage**: Current storage tank temperature (top).
- **storage2**: Current storage tank temperature (middle).
- **SP**-storFl: Current setpoint for the flowtemperature to the storage tank.
- **stoFlow**: Current storage flow temperature
- **SP**-loadFl: Current setpoint for the loading temperature to the storage tank
- **loadFlow**: current loading temperature
- **solStor**: Current solar storage tank temperature.
- **solColl**: Current solar collector temperature.
- **loadPu**: Current signal to the loading pump.
- **Y-stor:** Current signal to the valve (open-close).
- **exchPu**: Current signal to the heat exchanger pump.
- **Y-loadFl**: Current signal to valve for control of loading temperature.
- **Y-storFl**: Current signal to valve for control of storage tank temperature.
- **stoPu**: Current signal to storage tank pump.

- **solPu**: Current signal to solar pump.
- **circPu**: Current signal to circulation pump.
- **SP-OT1**: Occupation time 1 setpoint.
- **SP-NO**: Non-occupation time setpoint.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
6.4.1	opStatus	OpStatus					
6.4.2	trouble						
6.4.3	source	setpoint source and –compensation					
6.4.4	SP-stor	setpoint storage	°C				
6.4.5	storage	storage temperature	°C				
6.4.6	storage2	storage temperature2	°C				
6.4.7	SP-storFl	setpoint controller	°C	2.0	160.0		
6.4.8	stoFlow	storage flow temp	°C				
6.4.9	SP-loadFl	setpoint controller	°C	2.0	160.0		
6.4.10	loadFlow	loadFlow temp.	°C				
6.4.11	solStor	solar-storage temp.	°C				
6.4.12	solColl	solar-collector temp.	°C				
6.4.13	loadPu	loadPump					
6.4.14	Y-stor	contr corr variable	%				
6.4.15	exchPu	exchanger load pump					
6.4.16	Y-loadFl	contr corr variable	%				
6.4.17	Y-storFl	contr corr variable	%				
6.4.18	stoPu	storage load pump					
6.4.19	solPu	solarPump					
6.4.20	circPu	circulation pump					
6.4.21	SP-OT1	setpoint OT1	°C	2.0	160.0	50.0	
6.4.22	SP-NO	setpoint NO	°C	2.0	160.0	2.0	

12.4 Buffer tank (6.5.n)

The “Buffer Tank” menu displays the most relevant parameters for the buffer tank.

- **opStatus**: Current heat pump program status.
- **trouble**: Displays if there are pending trouble indications.
- **SP-zone1**: Current setpoint for buffer zone 1.
- **buffer 1**: Current temperature at the top of the buffer tank.
- **buffer 3**: Current temperature at the buttom of the buffer tank.
- **solarFL**: flow temperature from the solar collector
- **earth buffer**: flow temperature from the earth buffer tank
- **aHS-fl**: flow temperature from the additional heat source
- **Fl-T-He**: current flow temperature coming from buffer tank
- **loadPu**: status of buffer loading pump
- **solPu**: status of solar pump
- **earthBPu**: status of earth buffer tank pump

- **aHS**: status of additional heat source

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
6.5.1	opStatus	see explanation in user manual					
6.5.2	trouble	see explanation in user manual					
6.5.3	SP-zone1	current setpoint buffer zone 1	°C				
6.5.5	buffer1	buffer temp top					
6.5.7	buffer3	buffer temp bottom					
6.5.8	solarFl	flTemp.solar	°C				
6.5.9	earthBuf	earth buffer	°C				
6.5.10	aHS-fl	flTemp.aHS	°C				
6.5.11	Fl-T-He	flow temp. Heat	°C				
6.5.12	loadPu	buffer load.pump					
6.5.15	solPu	solarPump					
6.5.16	earthBPu	earth buffer pump					
6.5.17	aHS	add. heat source					

12.5 Heat pump x (6.6.n, 6.7.n, 6.8.n)

The "Heat pump" menu displays the most relevant parameters of the heat pump (HP1=6.6.n, HP2=6.7.n, HP3=6.8.n.)

- **opStatus**: Current heat pump program status.
- **trouble**: Displays if there are pending trouble indications.
- **source**: Displays which source has influence on the setpoint and the amount of compensation.
- **setpoint**: Current calculated setpoint for the flow temperature from the heat pump.
- **HP-rel**: Current release status of heat pump.

Par. no.	Name	Info text	Unit	Min.	Max.	Default	Description
6.x.1	opStatus	see explanation in user manual					
6.x.2	trouble	see explanation in user manual					
6.x.3	source	setpoint source and -compensation					
6.x.4	setpoint	temp.-setpoint	°C				
6.x.5		12.5.926 current release status of HP					

Part III Index